Dissertation

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Do Robots Lure Us to War?

The Impact of the Robotic Revolution on the Propensity for Armed Conflict

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Abstract

The second decade of the 21st century marked the return of great power competition, and Artificial intelligence (AI) is its new frontier. One of AI's most consequential applications could become autonomous weapons systems—military robots controlled by AI algorithms. Existing research suggests that unmanned capabilities might influence decision-makers' readiness to deploy lethal force, but what that impact might be, remains unclear. In this project, I developed a framework to define and understand the decision space for the use of force in the presence of emerging technology and applied it to autonomous weapons systems. I implemented the framework in a game-theoretic model of major power confrontation over a third country. Finally, in a survey experiment, I tested the impact of robotic warfare on the Russian public's support for military interventions.

Overall, my research suggests that innovations in autonomous weapons systems may increase the likelihood of war through three mechanisms. Namely, by (1) improving prospects of success, (2) decreasing economic costs of war, (3) increasing political benefits of fighting. None of these effects is guaranteed; however, each would marginally incentivize U.S. competitors to use force against a third country. The United States can mitigate this risk through technological advancement focused on improving its prospects of success in war and lowering mobilization costs (as opposed to reducing the economic costs of fighting) to increase the credibility of commitment and thus deter aggression. I found that robotic technology has no significant impact on the Russian public support for military intervention in a proposed scenario. At the same time, the results indicate that the support can be influenced the most through moral arguments. Casualties among Russian soldiers were not statistically significant predictor.

Acknowledgments

"If you set out on a journey let it be long
a wandering that seems to have no aim groping your way blindly
so you can learn the roughness of the earth not only with your eyes but by touch
so you can confront the world with your whole skin…"

A Journey by Zbigniew Herbert, translated by John and Bogdana Carpenter

This journey started long before I moved to Los Angeles, and it would not be possible without the countless people I met along the way. I will not be able to name all of them but I will name a few critical in last several years of this Ph.D. journey.

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Summary

Issue

This study is motivated by two overlapping trends. On the one hand, the second decade of the 21st century marked the return of great power competition, and there is a growing concern about the risk of direct confrontation between major powers, be it in Europe or Asia. On the other hand, we see rapid progress in military technologies, which change the battlefield before we can understand their broader implications. In particular, this might be the case with artificial intelligence (AI)—which itself becomes a new frontier of the great power competition—and its potential application in lethal autonomous weapons systems (LAWS).

Fully autonomous weapons systems do not exist yet, but potentially they could outperform human soldiers both physically and cognitively. On the physical level, robots are more resilient than the human body, so they could complete tasks that are too wary or too risky for human soldiers. On the cognitive level, AI operates faster than a human; it does not get tired, scared, or emotional. Consequently, LAWS could become relatively cheap enablers of missions that today are impossible, too costly, or too risky to undertake.

LAWS are already a topic of extensive debate, but much of it focuses on the legal and ethical ramifications of future wars if life and death decisions are delegated to machines. Meanwhile, augmenting or replacing human soldiers on the battlefield with AI-driven robots might have even broader implications that are poorly understood and remain understudied. Namely, it could change the very political calculation behind a decision to go to war.

Consequently, this study sought to answer three questions: Under what conditions could lethal autonomous weapons systems increase the likelihood of war? Could they increase Russia's propensity for war? And what should the United States do to mitigate such risk?

The war in Ukraine demonstrates that even if American troops do not fight against another major power, the cost of supporting the invaded country, defending the international order, and likely managing long-lasting ripple effects can be enormous. Thus, the ultimate goal of this study is to aid U.S. decision-makers and the defense community in understanding the broader implications of AI and military robotics to ensure that the development of capabilities designed to put fewer lives in harm's way does not create incentives to do it more frequently.

Approach

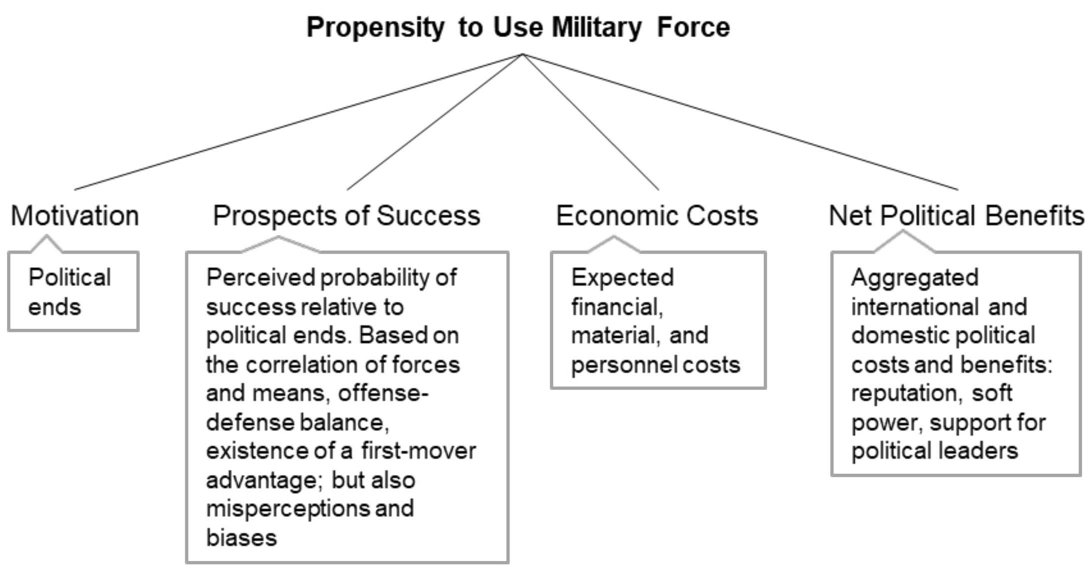
This study employs three methods: applied theory for framework development, game-theoretic model, and survey experiment. The framework defines the decision space. It provides a formalized and systematic way to reason, anticipate, and analyze the implications of different directions the policy and technological advancements could take. The model analyzes how

different factors interact in a decision scenario. In particular, studies the impact of emerging technologies on extended deterrence and explores mitigation options for the United States. The survey experiment drills down on the potential impact of robotic capabilities on Russian public support for military deployments.

Key Findings

The framework conceptualizes the decision to go to war as a function of four factors: motivation, prospects of success, economic costs, and net political benefits. Decision-maker's propensity for war marginally increases when emerging technology increases the motivation, prospects of success, and/or net political benefits and/or decreases the economic costs of war.

Figure S.1. Framework for the Impact of Emerging Technology on the Propensity to Use Military Force.



War, however, is hardly a unilateral act. Instead, it is a strategic interaction between two or more parties, where one player's increasing propensity for war affects the other player's decision calculus. In this work, I specifically explore how technological shifts can affect the stability of major power competition between Challenger and Defender over a third country (Protégé). This is predicated on the assumption that the impact of emerging conventional military technologies on extended deterrence might be stronger than on direct deterrence, as the latter is still governed by nuclear weapons. I found that:

- LAWS are unlikely to influence the primary motivation for war because algorithms—the main advantage of autonomous weapons systems—are unlikely to be destroyed with kinetic force.

- However, when stakes are high enough for a full-scale war to be possible, deterrence requires more Defender's troops than warfighting. It implies that piecemeal support for Protégé does not allow Challenger to adequately assess his prospects of success, and war seems overly appealing.
- There is a widespread expectation that LAWS will decrease the costs of wars: they are expected to save soldiers' lives, they might also lower labor and sustainment costs, and they might lessen the requirements on defensive systems today deployed to protect the crew of manned capabilities.
- The model suggests that the decrease in the Challenger's economic cost does not always lead to deterrence failure: there are cases where the decrease in costs leads to the emergence of deterrence. But in most cases, the model confirms the intuition that deterrence is likely to fail as war costs decrease. Importantly, this conclusion holds regardless of whose costs of fighting decrease: Challenger's or Defender's. On the other hand, lowering the Defender's mobilization costs has an unambiguously positive impact on deterrence.
- The expected decrease in Challenger's costs of fighting threatens the relationship between Defender and Protégé: it can lead to deterrence failure where Defender withholds the support for Protégé, which in real life would have much broader negative consequences for Defender's international standing.
- However, a decrease in warfighting costs is not guaranteed. The perceived decrease will depend on countries' current cost functions: how much they value human life and what is the price differential between capital and labor. It will also depend on whether LAWS augment or substitute the existing capabilities. Finally, the cost will be influenced by the new dependencies the demand for LAWS' components will produce.
- Deterrence failure is less likely if the technology advances in a way that Challenger's prospects of success (in the model represented as a function of quantity and quality of troops where the latter is influenced by technology) or his political benefit of fighting deteriorates. That is when there is a tradeoff between cost and other parameters (e.g., robotic warfare is cheaper but causes significant backlash or is less likely to succeed).
- If that is not the case, deterrence failure can be averted if Defender improves the quality of her troops. As long as the ratio between the Defender's and Challenger's troops' quality remains constant, the improvements in the Challenger's troops' quality will not affect deterrence.
- Meanwhile, the impact of LAWS on the prospects of success will likely be temporal and localized. Shifts in perceived prospects of success would stem from an unequal adoption rate where early adopters have the upper hand. Such windows of opportunity have a particularly destabilizing effect if the technology favors offense or creates a first-mover advantage.
- However, moving first might not be perceived as beneficial since it would provide a training set for the adversary's AI. Leaders might also be wary of deploying LAWS first if they are uncertain about their performance. Autonomous weapons might also be very useful in defense. Finally, the proliferation might be relatively fast given their low cost.
- Net political benefits are conceptualized as aggregated international and domestic political costs and benefits, where the international component comprises reputation and soft power, and the domestic component is approximated by support for countries' political leaders. Improving Defender's political benefits of fighting supports deterrence

because it incentivizes Defender to support Protégé. On the other hand, improving Challenger's political benefits is detrimental to deterrence because as those benefits increase, it takes more effort to deter Challenger.

- The impact of LAWS on net political benefits will depend on the systems' performance and the public's views on their legality and ethics. Political leaders might see them as a liability: the difficulty of assigning responsibility for their mistakes might shift said responsibility up the command chain to the political leaders who authorize the deployment.
- The survey experiment revealed no statistically significant effect of robotic capabilities on the Russian public support for military operations in a scenario where the Russian minority in a neighboring country is under threat. This finding, coupled with the fact that Russian robotization efforts seem to be aimed at improving the prospects of success, suggests that if Russia's robotization program succeeds, Moscow's propensity to use force might further increase.
- At the same time, regression analysis suggests that the morality of war, followed by the prospects of success, are the most promising avenues to influence the Russian public support for wars. On the other hand, messaging about casualties among Russian soldiers might not be persuasive.

Recommendations

The key overarching implication of this study is that the future of warfare in the presence of LAWS is not predetermined. There are possible futures where AI and robotic revolution could incentivize conflict and those where technological advances strengthen stability. While the agency to shape the future is not unlimited, it is up to the U.S. Government, its allies, and partners to influence the course of technological advancement for the benefit of peace. To that end, the U.S. Government should:

- To avoid detrimental shifts in offense-defense balance and the emergence of windows of opportunity, prioritize investment in defensive autonomous capabilities. Sharing such defensive capabilities with allies and partners would improve their capacity and resiliency while demonstrating the defensive intent behind technological innovation to potential adversaries.
- Focus on small autonomous drones with limited operational range to signal defensive intent of protecting critical infrastructure, communication nodes, and command posts instead of attacking targets deep inside the enemy's territory. The impact of such capabilities on the decision calculus could be twofold: they could ease potential adversaries' threat perception without leading them to overestimate the prospects of success in attack.
- Diplomatic efforts should be undertaken to promote the defensive LAWS. If such efforts fail, the U.S. national security community could use cyber, information, and psychological operations to work against adversaries' trust in their offensive autonomous capabilities.
- Strengthen or at least maintain the U.S. technological edge in algorithmic warfare. Strengthen Verification and Validation (V&V) and Testing and Evaluation (T&E) of

autonomous weapons to build the confidence of the U.S. leaders to deploy LAWS when necessary and weaken the willingness of U.S. adversaries to confront LAWS proven effective and provide them with opportunities to train and improve.

- Limit or slow down the expected decrease in these costs. This could, for example, entail export controls. Adversaries will likely develop elaborate schemes to evade them, but that alone will affect the price due to the risk premium.
- Adopt immigration policies that would lower the supply of high-skilled workers, aiding adversaries' scientific and technological innovation.
- Prioritize lowering mobilization costs over the costs of fighting. In practice, this can mean two paths for technology development and deployment: the United States could invest in large quantities of capabilities that are cheap, expendable, and easy to use and pre-deploy them in allied and partner countries. The second path would instead focus on quality: these would be sophisticated but expensive capabilities the U.S. military brings onto the theater when a crisis develops.
- Ensure there is no a priori backlash against autonomous weapons systems as this could impede the U.S. commitments to allies and partners and undermine extended deterrence. To that end, the U.S. Government should lead dialog on LAWS with the American people and the expert community. The same applies to the dialog with allies and partners, who might have even stronger concerns about the legality and ethics of LAWS than the U.S. public.
- Publicize the efforts to ensure LAWS safety and reliability, i.e., to the extent possible, use the V&V and T&E as a confidence-building measure, both domestically and internationally.
- Highlight LAWS' applications in defense and their advantage in saving the lives of American and allied soldiers, as well as the negative consequences of lagging behind the adversaries, including the risk of deterrence failure.
- Continue to engage in the debate on ethics, as there might be strong ethical arguments in favor of military robots, even if their performance is not error-free. Specifically, some war crimes and crimes against humanity are unlikely to be committed by LAWS, for example, torture and rape.

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Chapter 1. Introduction

The second decade of the 21st century marked the return of great power competition. With Russia's meddling in Western countries' elections and China's military build-up in South China, the challenge faced by the United States and its allies has been the measures short of war, often called gray zone activities, hybrid warfare, or subversion.[1] Yet, the 2022 Russian full-scale invasion of Ukraine reminded the international community that a full-scale, high-intensity war unleashed by one of U.S. competitors to establish a new strategic reality is not out of the picture.[2] On the contrary, there is a growing concern about the possibility of China's invasion of Taiwan.[3] While strategic deterrence still prevents nuclear powers from fighting each other, the risk of waging wars on each other's partners seems to increase in the face of the weakening of the liberal international order.[4]

War has accompanied humanity since the beginning. It is a political will to change the status quo embodied in military action, defined by means at fighters' disposal and ways of using them. But if the nature of war as "a continuation of political intercourse, carried on with other means"[5] remains constant, the means of war continue to change. Since the mid-twentieth century, weapons systems have become increasingly expensive and advanced, thus requiring more costly, professionalized manpower.[6] Yet, technological developments of the last few decades, especially in the fields of robotics, have enabled a shift in a different direction, namely, toward expandable and relatively cheap unmanned warfare.

Recent conflicts, most notably the war in Ukraine, demonstrate that even unsophisticated, civilian unmanned technologies—already ubiquitous on a battlefield—can be cheap force multipliers. Used for surveillance, targeting, or artillery fire correction, they democratize access to real-time situational awareness and precision fires, earlier available only to the most developed militaries. Despite their vulnerability to countermeasures, they continue to be useful

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[1] Andrew Radin, Alyssa Demus, and Krystyna Marcinek, *Understanding Russian Subversion: Patterns, Threats, and Responses,* RAND Corporation, PE-331-A, February 2020.

[2] Office of the Director of National Intelligence, *Annual Threat Assessment of the U.S. Intelligence Community*, PDF, February 6, 2023.

[3] Oriana Skylar Mastro, "The Taiwan Temptation," *Foreign Affairs*, June 3, 2021.

[4] Michael J. Gaines, *Whither the Liberal International Order? Does an Attenuated Liberal International Order Affect the Propensity, Incidence, and Intensity of Conflict?*, RAND Corporation, RGSD-A2685-1, 2023.

[5] Carl von Clausewitz, *On War*, edited and translated by Michael Howard and Peter Paret, Princeton, NJ: Princeton University Press, 1984, p. 87.

[6] Martin Van Creveld, *The Changing Face of War: Combat from the Marne to Iraq*, New York City, NY: Presidio Pres, 2008, p. 267.

even in a contested environment due to their low cost, justifying a high attrition rate.[7] Furthermore, whether targeting multimillion-dollar manned systems or being targeted by expensive air defenses, the cost-benefit analysis and comparison with more sophisticated equipment speak strongly in favor of unmanned capabilities. Some consider this shift so profound that they call it "the most important weapons development since the atomic bomb."[8]

So far, robotic capabilities used in combat are remotely operated. But that does not have to be the case in the future. Indeed, there is a widespread expectation that advances in artificial intelligence (AI) will enable unmanned platforms to perform many tasks autonomously.[9] Notably, AI itself is a frontier for great power competition. As Russian President Vladimir Putin famously said, "Artificial intelligence is the future, not only of Russia but of all of mankind. (…) Whoever becomes the leader in this sphere will become the ruler of the world."[10] The 2018 United States Department of Defense Artificial Intelligence Strategy recognizes that China's and Russia's investments in AI "threaten to erode our technological and operational advantages and destabilize the free and open international order."[11] The strategy posits that "failure to adopt AI will result in legacy systems irrelevant to the defense of our people, eroding cohesion among allies and partners."[12]

The lure of deploying AI as control systems of lethal robotic platforms, or lethal autonomous weapon systems (LAWS), is potentially twofold: LAWS could outperform humans both on a physical and cognitive level. On the physical level, robots are much more resilient than the human body, and they can complete tasks that might be too wary, too risky, or, for other reasons, impossible for human soldiers. Former Deputy Defense Secretary Robert Work has asserted that the spread of unmanned systems on the battlefield is inevitable due to the cost of human life.[13] On the cognitive level, a human cannot compete with AI in situation assessment and decision-making speed. AI does not get tired, scared, or emotional. Moreover, once the decision-making is delegated to AI, there is no need to maintain a communication link with operators, allowing

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7 Mykhaylo Zabrodskyi, Jack Watling, Oleksandr V Danylyuk and Nick Reynolds, "Preliminary Lessons in Conventional Warfighting from Russia's Invasion of Ukraine: February–July 2022," Royal United Services Institute for Defence and Security Studies, 2022, p. 37.

8 Peter Singer, *Wired for War. The Robotics Revolution and Conflict in the 21st Century*, Penguin Books, 2009, p. 10.

9 Some of the functions of unmanned capabilities are already automated. For example, the Global Hawk surveillance drone flies autonomously on a pre-loaded flight plan, which can be overridden by a human operator (Maj Jim Molinari, *Global Hawk Operational Overview*, Presentation, RAV/0032-01, undated).

10 President of the Russian Federation, "Open lesson 'Russia Focused on the Future' [Otkrytyj urok «Rossija, ustremljonnaja v budushhee»]", webpage.

11 Department of Defense, "Summary of the 2018 Department of Defense Artificial Intelligence Strategy. Harnessing AI to Advance Our Security and Prosperity," 2018, p. 5.

12 Department of Defense, 2018, p. 5.

13 Robert O. Work, "AI and Future Warfare: The Rise of Robotic Combat Operations," Mad Scientist Conference, University of Texas at Austin, April 24, 2019.

operations in a contested environment. That is, the type of environment we precisely expect from peer competitors. Thus, the United States and its competitors have indicated that they expect the proliferation of robotic capabilities with a growing degree of autonomy on future battlefields.[14]

However, the possibility of delegating life and death decisions to AI raises major practical, legal, and ethical concerns. The critics point out that "once developed, they will permit armed conflict to be fought at a scale greater than ever, and at timescales faster than humans can comprehend."[15] At the same time, while AI is excellent at performing a narrow set of tasks with clearly defined rules, the same will not necessarily be the case in a more complex and dynamically changing environment. Furthermore, the mistakes made with machine speed can be much more deadly than those made by individual soldiers. Finally, AI might be unable to distinguish between combatants and non-combatants or assure the proportionality of force, which would deem them illegal in light of international law.

Due to these concerns, the international community debates whether LAWS require additional regulations or should be banned altogether.[16] The formal discussion is held primarily by the Group of Governmental Experts (GGE) on emerging technologies in the area of lethal autonomous weapons systems under the auspices of the United Nations Convention on Certain Conventional Weapons (CCW). In 2019, the GGE adopted a set of "guiding principles" for LAWS, including ensuring human-machine interaction that maintains the human responsibility for decisions about the use of force and ensures that international humanitarian law is upheld.[17] The recommendations are largely aligned with the United States' policy on autonomous weapon

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[14] See, for example: John R. Hoehn, Kelley M. Sayler, Michael E. DeVine, *Unmanned Aircraft Systems: Roles, Missions, and Future Concepts*, Congressional Research Service, R47188, July 18, 2022; Ronald O'Rourke, Navy *Large Unmanned Surface and Undersea Vehicles: Background and Issues for Congress*, Congressional Research Service, R45757, April 17, 2023; Andrew Feickert, Jennifer K. Elsea, Lawrence Kapp, and Laurie A. Harris, *U.S. Ground Forces Robotics and Autonomous Systems (RAS) and Artificial Intelligence (AI): Considerations for Congress*, Congressional Research Service, R45392, Updated November 20, 2018; Jeffrey Edmonds, Samuel Bendett, Anya Fink, Mary Chesnut, Dmitry Gorenburg, Michael Kofman, Kasey Stricklin, and Julian Waller, *Artificial Intelligence and Autonomy in Russia*, Arlington, Va.: CNA, DRM-2021-U-029303-Final, May 2021; Krystyna Marcinek, and Eugeniu Han, *Russia's Asymmetric Response to 21st Century Strategic Competition: Robotization of the Armed Forces,* RAND Corporation, RR-A1233-5, 2023; Elsa B. Kania, "'AI weapons' in China's military innovation," The Brookings Institution, April 2020; Ryan Fedasiuk, "Chinese Perspectives on AI and Future Military Capabilities," *CSET Policy Brief*, Center for Security and Emerging Technology, August 2020; Guangyu Qiao-Franco, and Ingvild Bode, "Weaponised Artificial Intelligence and Chinese Practices of Human–Machine Interaction," *The Chinese Journal of International Politics*, Vol. 16, 2023, pp: 106–128.

[15] Future of Life Institute, "An Open Letter to the United Nations Convention on Certain Conventional Weapons," webpage, August 20, 2017.

[16] Kelley M. Sayler, *International Discussions Concerning Lethal Autonomous Weapon Systems*, Congressional Research Service, IF11294, Updated February 14, 2023.

[17] GIP Digital Watch, "GGE on lethal autonomous weapons systems," webpage, undated.

systems, which requires their design to allow "appropriate levels of human judgment over the use of force."[18]

The extensive use of drones in the war in Ukraine further exacerbates the debate on LAWS regulations.[19] Yet, it does not appear that any preemptive ban is on the horizon or that the guiding principles will dramatically limit the development of LAWS. The ban itself would be difficult to implement as the main building blocks of LAWS—hardware and software—are also being developed for civilian purposes. The U.S. policy on autonomy in weapon systems imposes a high standard for LAWS development but does not prohibit it. The permissible level of autonomy remains unclear as the meaning of "appropriate levels of human judgment" is not well defined, and, as Paul Scharre pointed out, it changes over time.[20] Furthermore, weapons systems might permit multiple levels of human control, and it might be difficult to verify how much autonomy they exercised in individual use cases.[21] Finally, according to Robert Work, the United States might have to rethink its position on autonomy if the competitors choose to do that first.[22] The North Atlantic Treaty Organization (NATO) recognizes the challenge of this competition in its AI Strategy and implementation plan:

> "In the current era of strategic competition, the Alliance's ability to maintain an edge in autonomous technologies, bolster deterrence and defence, and increase resilience also requires Allies and NATO to protect against interference and deception in our systems, protect the Alliance's armed forces, populations and territory from harmful use of autonomous systems, and protect against acquisition of our technologies by potential adversaries and strategic competitors. Allies and NATO recognise that competitors and potential adversaries are

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[18] Department of Defense Directive 3000.09, "Autonomy in Weapon Systems," PDF, January 25, 2023; for a summary of the U.S. policy on LAWS, see: Kelley M. Sayler, *Defense Primer: U.S. Policy on Lethal Autonomous Weapon Systems*, Congressional Research Service, IF11150, May 15, 2023.

[19] Stuart Russell, "AI weapons: Russia's war in Ukraine shows why the world must enact a ban," *Nature*, Vol. 614, 2023, pp: 620–623; Zachary Kallenborn, "Russia may have used a killer robot in Ukraine. Now what?," *Bulletin of the Atomic Scientists*, March 15, 2022. On the use of drones in Ukraine, see: Samuel Bendett, "The Ukraine war and its impact on Russian development of autonomous weapons," Atlantic Council, August 30, 2022.

[20] Paul Scharre, *Army of none: Autonomous weapons and the future of war*, WW Norton & Company, 2018.

[21] It is already difficult to establish how much autonomy existing drones exercise. Producers are not always reliable source of information because depending on the market they advertise to producers might have an incentive to exaggerate or downplay the level of autonomy. Most recently, that has been part of confusion about the true capabilities of systems used in Ukraine. See: Gregory C. Allen, "Russia Probably Has Not Used AI-Enabled Weapons in Ukraine, but That Could Change," Center for Strategic and International Studies, May 26, 2022.

[22] Atlantic Council, "Ending Keynote: Art, Narrative, and the Third Offset," YouTube video, May 2, 2016; For comparison of Chinese, Russian, and US autnoous weapons practices, see: Ingvild Bode, Hendrik Huelss, Anna Nadibaidze, Guangyu Qiao-Franco, and Tom F.A. Watts, "Prospects for the global governance of autonomous weapons: comparing Chinese, Russian, and US practices," *Ethics and Information Technology*, Vol. 25, No. 5, 2023.

> investing in and demonstrating autonomous systems with the goal of surpassing Allied capability development."[23]

Consequently, it seems that as the great power competition intensifies, the demand for LAWS will continue to rise.

Peter Singer argued that unmanned capabilities have changed who fights wars.[24] In turn, introducing AI into robots' control systems might drastically change how wars are fought, with all the consequences for the ethics of war. There is already a considerable body of literature on this topic, and it continues to grow. However, augmenting or replacing human soldiers on the battlefield with AI-driven robots might have even broader implications that are poorly understood and remain understudied. Namely, it could change the very political calculation behind a decision to go to war. Thus, while a large body of literature focuses on the characteristics of future robotic wars, this study seeks to understand whether new capabilities could change how often decision-makers resort to war in the first place.

Study Goals, Approach, and Structure

Research Questions

This study asks the following questions about the nexus between technology and the decision to use force:

1. Under what conditions could lethal autonomous weapon systems make the use of military force in interstate conflict more likely?
2. How could the availability of lethal autonomous weapon systems impact Russia's propensity to use force?
3. How can the United States mitigate the risk of increasing likelihood of war in the presence of lethal autonomous weapon systems?

Research Objectives

Technology advances rapidly and is often deployed before decision-makers fully understand its broader implications. The main reason is that formal reasoning about such technology is usually absent, and decision-makers must rely on heuristics. This research addressed this issue by developing a formalized and systematic way to reason about technological advances and their propensity to shape military conflict. The study does not determine whether or when LAWS

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[23] North Atlantic Treaty Organization, Summary of NATO's Autonomy Implementation Plan, webpage, October 12, 2022. See also: North Atlantic Treaty Organization, Summary of the NATO Artificial Intelligence Strategy, webpage, October 22, 2021.

[24] Singer, 2009, p. 10.

could become operational.[25] Instead, it provides tools to anticipate and analyze the implications of different directions the policy and technological advancements could take. As such, it can be applied to other emerging technologies at different stages of their development.

Specifically, this exploratory research project 1) provides a framework to define and understand a decision space of the use of force in the presence of LAWS specifically and other emerging technologies more broadly, 2) proposes a game-theoretic model to analyze how different factors interact in a decision scenario, 3) provides insights into the potential impact of robotic warfare on public support for Russia's military deployments, 4) offers policy recommendations aimed at mitigating the risk of increased propensity to resort to war in the presence of LAWS.

In its empirical component, the study focuses on Russia. This decision was motivated by the recognition that Moscow is determined to significantly increase the share of unmanned capabilities in its Armed Forces makeup,[26] and its impact on the country's already aggressive behavior does not receive enough attention. The decision to focus on Russia predated the 2022 war in Ukraine, but the conflict has provided additional incentives to continue the study. The war hindered Russia's ability to proceed with its quest for robotic force; however, it did not invalidate the motivations behind it. On the contrary, persistent problems with manpower might serve as an additional impulse to continue on the robotization path.[27] Similarly, the losses Russia has incurred over the course of war degraded the country's ability to engage in other conflicts in the foreseeable future, but they might also create resentment that will make Russia an even more aggressive actor.[28]

The war in Ukraine demonstrates that even if American troops do not fight against another major power, the cost of supporting the invaded country, defending the international order, and likely managing long-lasting ripple effects can be enormous. Thus, the ultimate goal of this study is to aid U.S. decision-makers and the defense community in understanding the broader implications of AI and military robotics to ensure that the development of capabilities designed to put fewer lives in harm's way does not create incentives to do it more frequently.

Approach and Structure

This study employs three methods: applied theory for framework development, game-theoretic model, and survey experiment. These three methods are the most established

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[25] Indeed, some experts argue that both the general public, and the policy makers then to overestimate the current state of affairs and underestimate the level of effort still necessary to achieve truly autonomous weapon systems. See: *The Impact of Artificial Intelligence on Strategic Stability and Nuclear Risk Volume I Euro-Atlantic Perspectives*, ed. Vincent Boulanin, Stockholm International Peace Research Institute, May 2019.

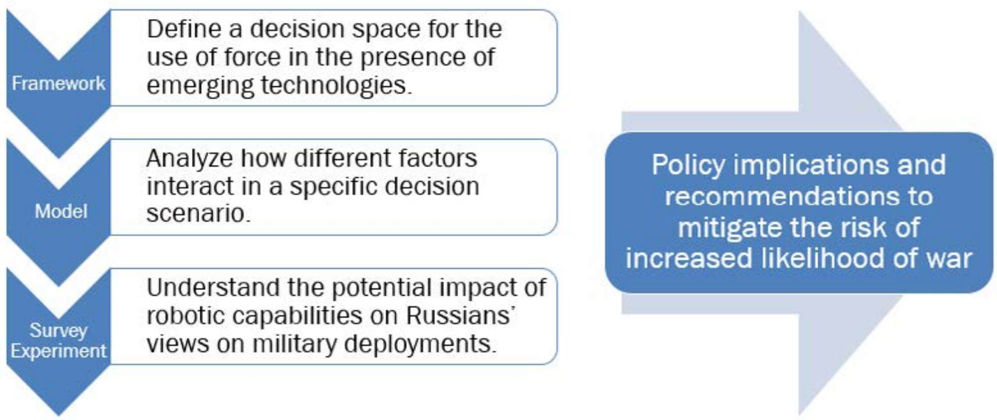
[26] Marcinek and Han, 2023.

[27] Marcinek and Han, 2023.

[28] Office of the Director of National Intelligence, 2023, p.12.

approaches to studying emerging technologies.[29] Each of them addresses all three research questions, albeit to various degrees. They are used in concert to gradually build insight from a general understanding of the nexus between emerging military technologies and the decision to use force to an empirical analysis of the impact of robotic warfare on Russian public support for military operations (Figure 1.1).

Figure 1.1. Research Design.



Following the introduction, Chapter Two begins with a review of existing literature on the potential impact of unmanned military capabilities on the likelihood of war. Next, Chapter Two proposes a framework for the impact of emerging military technologies on the decision to use force. The framework was used to analyze the potential effects of LAWS; however, future research can apply it to any emerging conventional technology.

Chapter Three presents the game-theoretic model studying a tripartite conflict scenario between two major powers and one weaker country. One example of such a scenario is Russia's war against Ukraine, a US partner. The specific purpose of this model is to understand how technological shifts can affect the stability of major power competition over third countries. To that end, framework factors are implemented as model parameters defining players' payoffs. The choice of a tripartite model focused on extended deterrence is motivated by the fact that LAWS are conceptualized as conventional technology. As such, they might have a stronger impact on competition over third countries than on bilateral relations between major powers because the later are still defined predominantly by nuclear deterrence.

Chapter Four discusses the results of a survey experiment conducted on a Russian convenience sample. The experiment tested the impact of robotic warfare on the support for a hypothetical military operation. The survey also elicited reasons for the support, or lack thereof.

[29] Michael C. Horowitz, "Do Emerging Military Technologies Matter for International Politics?," *Annual Review of Political Science*, Vol. 23, No. 1, 2020, pp: 386–7.

The survey was deployed two months before Russia's full-scale invasion of Ukraine, providing important context for understanding the war. As noted, the question of the impact of robotic warfare on Russians' support for military operations remains relevant even when Russia's ability to modernize its Armed Forces is degraded. It is the first such survey conducted on a Russian sample and, as such, can be a benchmark for future studies. Admittedly, in the ideal scenario, the empirical component would measure the impact of robotic warfare on Russian leadership's willingness to deploy military force, but such an approach was not feasible, and thus, this study focuses on the next best option, namely, the Russian public.

The study concludes with policy implications and recommendations for the United States and NATO allies. The recommendations focus on the avenues to hedge against the risk of increased propensity to resort to war in the presence of LAWS.

Chapter 2. The Impact of Emerging Military Technologies on the Decision to Use Force: Framework

In 1995, archaeological excavations in Schöningen, Germany, led by Dr. Hartmut Thieme, revealed what is known today as the world's oldest wooden spears.[30] Being at least 300,000 years old[31], they predate both *homo sapiens* and *homo neanderthalensis*. Bows and arrows are much later innovations. The oldest specimens, discovered in Sibudu Cave in South Africa, date back 60,000-70,000 years.[32] But no matter how long the progress took, from these early days of humanity until now, one of the main goals of major innovations in weapons development seems to have been to increase the harm inflicted on the enemy (be it an animal or another human), to decrease risks to own forces, or, preferably, both.

Over the course of history, new technologies have been shifting warfighting inputs from labor to capital. From fists to stones to spears, to the machine gun, to bombs, to nuclear weapons, the goal of increasing damage to the adversary while keeping own troops safe has often been achieved by increasing the magnitude of fire power going hand in hand with the growing distance between fighting sides. However, until recently, these improvements tent to come at the expense of precision. Intercontinental ballistic missiles carrying nuclear warheads might have been the culmination point of that trend—while they can reach targets thousands of miles away from the launch site, they cause fatalities and infrastructure damage of catastrophic proportions and without any discrimination.

The enormous scale of potential nuclear destruction has had a sobering effect on the second half of the 20th century. Despite huge military potentials and ideological hostilities between the two nuclear superpowers, the United States and the Soviet Union resisted a direct confrontation for fear of uncontrolled escalation into the nuclear apocalypse. This fear also accelerated research on the causes of war and the role of military technologies in the decision-making process.

Today, researchers resort to that toolkit when they analyze the shift away from indiscriminate force due to the emergence of precision-guided munitions followed by the advances in robotic technologies. If mutually assured destruction has been a cornerstone of global stability, it is only logical to ask what effects on that stability might have unmanned capabilities that have enabled

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[30] Hartmut Thieme, "Lower Palaeolithic hunting spears from Germany," *Nature*, Vol. 385, No. 6619, 1997, pp: 807–810.

[31] Daniel Richter, and Matthias Krbetschek, "The age of the Lower Paleolithic occupation at Schöningen," *Journal of human evolution*, Vol. 89, 2015, pp: 46–56.

[32] Marlize Lombard, and Laurel Phillipson, "Indications of Bow and Stone-Tipped Arrow Use 64 000 Years Ago in KwaZulu-Natal, South Africa," *Antiquity*, Vol. 84, No. 325, 2010, pp: 635–48.

tracking individual targets for hours to strike them with surgical precision while keeping operators safe thousands of miles away.

However, applying classic international relations concepts to the study of robotic warfare poses serious challenges. In this chapter, I will first demonstrate these challenges by briefly presenting the existing literature on the impact of LAWS on the onset of war. Next, I will propose a comprehensive framework to provide a structure for the analysis. The framework is adopted from existing literature and builds upon a broader decision theory and international relations research.

Literature Review

Lethal autonomous weapons systems as such do not exist yet. Thus, the literature on their impact on the decision to use force remains scarce, highly speculative, and produces conflicting conclusions.

Some of the earlier works used historical analogies to nuclear stability and lessons learned from remotely piloted drones to argue that an arms race for autonomous weapons systems is almost inevitable and will lead to crisis instability and escalation risks.[33] Researchers attributed that risk to machine decision-making speed and the unpredictability of the combat environment, which makes testing algorithms inherently difficult, if not impossible. As technology becomes more complex, errors should be expected as a rule, not rare exceptions. Faced with unpredictable situations, AI algorithms on both sides can respond with escalation, falling into "accidental war."[34] And even if that does not happen, and AI functions properly, its decision-making speed might impede the ability to fully utilize diplomatic means to de-escalate the conflict both before the crisis unfolds into war and after the hostilities start.[35]

LAWS can also impede diplomatic solutions by changing the incentive structure behind the negotiated resolution. A recent RAND study observed that manned systems might be better for deterrence because taking them down is perceived as more escalatory and thus incentivizes players to seek offramps.[36] Researchers argue that lowering the costs of war—both human and materiel—reduces domestic political risks of wars.[37] Consequently, LAWS could create a moral

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[33] Jürgen Altmann and Frank Sauer, "Autonomous Weapon Systems and Strategic Stability," *Survival*, Vol. 59, No. 5, 2017, pp: 117–142.

[34] Altmann and Sauer, 2017, pp: 128–130.

[35] Zachary Davis, "Artificial Intelligence on the Battlefield: Implications for Deterrence and Surprise," *Prism: A Journal of the Center for Complex Operations*, Vol. 8, No. 2, 2019, pp: 114–131; Similar conclusion about crisis instability and escalation, see: Yuna Huh Wong, John Yurchak, Robert W. Button, Aaron B. Frank, Burgess Laird, Osonde A. Osoba, Randall Steeb, Benjamin N. Harris, and Sebastian Joon Bae, *Deterrence in the Age of Thinking Machines,* RAND Corporation, RR-2797-RC, 2020.

[36] Wong, 2020.

[37] For example, Frank Sauer and Niklas Schörnig, "Killer Drones – The Silver Bullet of Democratic Warfare?," *Security Dialogue*, Vol. 43, No. 4, 2012, pp: 363–80; Altmann and Sauer, 2017.

hazard: if the risk of escalation and domestic political costs of operations are lower with unmanned capabilities, leaders might choose to execute missions that they would not approve of if manned capabilities were involved.[38]

Another related argument about the impact of machine-speed decision-making on crisis instability points to the perceived first-mover advantage. Specifically, leaders could form a belief that LAWS make decisions so fast that whoever deploys them first wins. Leaders could be wary that leaving time for consideration makes their country vulnerable to an attack that, once launched, would be impossible to defend against if the adversary's AI already gains an advantage.[39]

Yet, as Michael Horwitz argues, the uncertainty surrounding LAWS' performance might limit leaders' willingness to deploy them, albeit this hesitation might not be shared by all actors, especially autocratic ones.[40] Nevertheless, the author concludes that LAWS' impact on interstate war will be modest because, ultimately, "countries go to war for political reasons."[41]

Finally, others speculated that the complexity of autonomous weapons could have a stabilizing effect because they could provide "a face-saving alternative for leaders trying to de-escalate a crisis."[42] Specifically, leaders could plausibly blame machine malfunctions for incidents creating crises in the first place. According to that perspective, the difficulty of attribution of responsibility and the inherent unpredictability of these systems could become their political advantage. As Nathan Lyes succinctly put it, "if there is no one to blame, there is no one to bomb."[43] The author further speculated that losing an unmanned system creates less demand for retaliation, which may lower the risk of escalation. This would suggest a lower prevalence of war in the presence of LAWS.

Empirical research on drone warfare supports some of the intuitions discussed above. Specifically, evidence from survey experiments suggests that using drones instead of crewed aircraft increases the respondents' support for deploying an aircraft into a contested area.[44] The same study found that drones were associated with higher support for shooting down another country's aircraft in a contested area. On the other hand, when a drone was shot down, the

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[38] Nathan Leys, "Autonomous Weapon Systems and International Crises," *Strategic Studies Quarterly*, Vol. 12, No. 1, 2018, pp: 60–1.

[39] Scharre, 2018; Davis, 2019, p. 125; Michael C. Horowitz, "When Speed Kills: Lethal Autonomous Weapon Systems, Deterrence and Stability," *Journal of Strategic Studies*, Vol. 42, No. 6, 2019, pp: 764–88.

[40] Horowitz, 2019, pp: 773–6.

[41] Horowitz, 2019, p. 784.

[42] Leys, 2018, p. 60.

[43] Leys, 2018, p. 60.

[44] Michael C. Horowitz, Paul Scharre, and Ben FitzGerald, "Drone Proliferation and the Use of Force: An Experimental Approach," Center for a New American Security, 2017.

demand for retaliation was lower relative to that expressed when a manned aircraft was in question.

However, considerable differences exist between the U.S. and India public (non-expert) responses. Specifically, the Indian public sample exhibited higher overall support for military deployments in all three cases (deployment, shoot down, retaliation), and the difference between the drone and manned aircraft was significant only for the deployment of aircraft in a contested area.[45] The difference between the U.S. and India public suggests that participants' interpretation of a generic survey scenario might have been contextual, and so were the answers provided. Specifically, the survey scenario placed the choice in the context of "an important territorial dispute with another country." Contrary to the United States, India is actively involved in several territorial disputes, which might have impacted participants' responses.[46] Unfortunately, there is no indication of how much of the difference in support between the United States and India can be explained by this factor.

Similarly, the impact of costs can be ambiguous. If LAWS costs are lower than those associated with manned capabilities, and thus the former are perceived as more likely to be used, they might be a useful coercion or deterrence tool. Traditionally, strategic studies literature argued that only costly signals are credible, and thus, cheap unmanned capabilities should not be a credible deterrence. Yet, literature on drone warfare argues that low costs—coupled with the persistent threat of precision strikes—are precisely what makes drones credible because the use of force seems more assured.[47] Survey research on coercion dynamics with drones supports this conclusion.[48]

Similar findings yielded a RAND wargame concerning a hypothetical conflict scenario involving the United States, China, Japan, South Korea, and North Korea in the presence of LAWS. The authors observed that game participants did not perceive substituting U.S. troops with robots as a reduced commitment. Furthermore, the wargame demonstrated that by virtue of delegating decision-making authority to LAWS, the players saw LAWS as a credible signal of resolve and commitment.[49]

In conclusion, this brief survey of literature on the nexus between LAWS and the onset of war demonstrates that the existing research does not provide a consistent answer to the question about LAWS' impact on the propensity for war. The arguments presented in the literature span from the near certainty that LAWS will lead to crisis instability and inadvertent escalation to hesitation about whether they will even be deployed, given the inherent unpredictability of AI

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[45] Horowitz, Scharre, and FitzGerald, 2017, pp: 5–6.

[46] Iftikhar Gilani, "India's border dispute with neighbors," Anadolu Agency, May 31, 2020.

[47] Amy Zegart, "Cheap fights, credible threats: The future of armed drones and coercion," *Journal of Strategic Studies*, Vol. 43, No. 1, 2020, pp: 6–46.

[48] Zegart, 2020.

[49] Wong et al., 2020, p. x.

performance. Similarly, some believe that the lower the costs, the lower the threshold for military operations, while others claim that the lower the costs, the higher the credibility of deterrence.

This lack of consistent and conclusive answer in part stems from the fact that the technology in question does not exist yet, so the literature remains highly speculative, applies theory selectively, and sometimes lacks methodological rigor that would allow evaluation of the validity of presented claims. It also focuses on the potential risks (and sometimes promises) associated with LAWS without much consideration for the conditions under which those risks (or promises) would materialize. Finally, applying analogies and frameworks developed for strategic weapons to new conventional capabilities is challenging. Contrary to nuclear weapons, conventional technologies do not have a decisive effect or extreme cost attached. Consequently, the causal chain between the availability of conventional technology and the decision to use force is much longer, and therefore, the effect is difficult to isolate. However, I posit that available military technologies are always inherently present in the decision to use force as they define the boundaries of what is possible. However, tracing that causal link requires a new comprehensive framework.

Framework

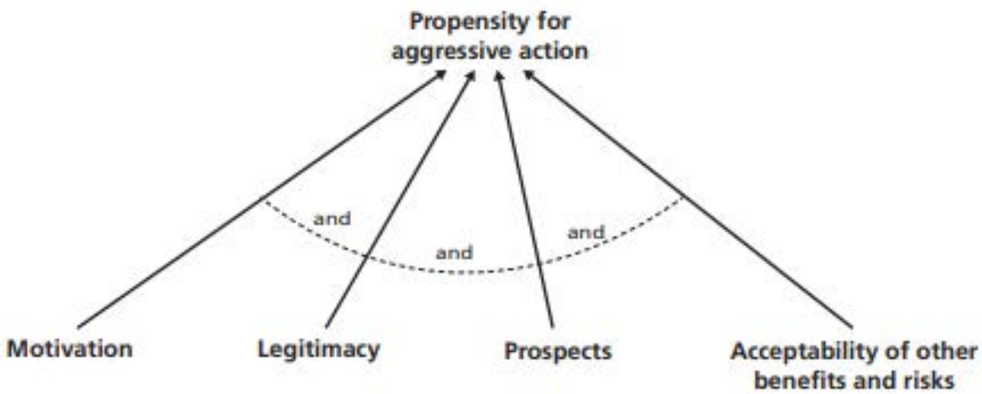
The problem of an elusive causal chain has already been recognized in the literature.[50] But while others analyze the impact of emerging technologies on deterrence strictly through the lenses of deterrence literature,[51] I use a more general approach. The framework I propose builds upon Davis et al.'s generic factor tree to explain the propensity for aggressive action (Figure 2.1).[52] The advantage of this factor-tree three is that, on the one hand, it includes factors corresponding to the classic decision framework—namely, the expected utility theory—while on the other, it introduces the notion of legitimacy as an important contributing factor.

[50] Michael J. Mazarr, Ashley L. Rhoades, Nathan Beauchamp-Mustafaga, Alexis A. Blanc, Derek Eaton, Katie Feistel, Edward Geist, Timothy R. Heath, Christian Johnson, Krista Langeland, Jasmin Léveillé, Dara Massicot, Samantha McBirney, Stephanie Pezard, Clint Reach, Padmaja Vedula, and Emily Yoder, *Disrupting Deterrence: Examining the Effects of Technologies on Strategic Deterrence in the 21st Century,* RAND Corporation, RR-A595-1, 2022, p. 6.

[51] Mazarr et al., 2022, pp: 10–14.

[52] Paul K. Davis, Angela O'Mahony, Christian Curriden, and Jonathan Lamb, *Influencing Adversary States: Quelling Perfect Storms*, Santa Monica, Calif.: RAND Corporation, RR-A161-1, 2021.

Figure 2.1. Generic Factor-Tree Considering an Aggressive Action.



SOURCE: Davis et al., 2021, p. 36.

The utility theory assumes that unitary rational actors weigh the expected benefits and costs of every possible course of action and select the one that maximizes their utility. Consequently, the utility framework in its classic form consists of three elements: desired benefits of action, probability of receiving those benefits, and costs of undertaking the action. Or, in the language of the factor tree above, expected utility is a function of motivation, prospects, and acceptability of other benefits and risks.

However, this classic rational choice framework does not address the most heated debate on LAWS, namely, the ethical, moral, and legal aspects of lethal machine autonomy. The prevalence of that debate in the English-language sources can be measured quantitatively—approximately a fifth of all journal articles on autonomous weapons indexed in the EBSCOhost database also mention ethics.[53] Some of the main concerns raised in this debate are the ability of computer algorithms to comply with international humanitarian law and laws of armed conflict, as well as the morality of killing humans when the decision to kill is delegated to a machine.[54] It seems unreasonable to assume that this debate has no bearing on future decisions to deploy LAWS. Consequently, "Legitimacy" as the fourth decision factor in Davis et al.'s factor tree is a good candidate to represent these considerations.

The specific issue of the impact of the legality and ethics of LAWS on the propensity to deploy them is part of a larger shortcoming of the expected utility framework. Namely, it does not account well for the political repercussions of the decision to use military force because it

53 As of June 6, 2023, the search term *autonomous weapons* produced 2,464 articles, whereas *autonomous weapons AND ethics* produced 529 articles.

54 For an overview of the concerns, see: Vincent Boulanin, Neil Davison, Netta Goussac, and Moa Peldán Carlsson, *Limits on Autonomy in Weapon Systems. Identifying Practical Elements of Human Control*, Stockholm International Peace Research Institute and the International Committee of the Red Cross, June 2020; Forrest E. Morgan, Benjamin Boudreaux, Andrew J. Lohn, Mark Ashby, Christian Curriden, Kelly Klima, and Derek Grossman, *Military Applications of Artificial Intelligence: Ethical Concerns in an Uncertain World,* RAND Corporation, RR-3139-1-AF, 2020.

aggregates all costs into one variable. Monetizing intangibles, while very useful in cost-benefit analysis, might blur the understanding of the impact of technologies on the decision to use force if said technology is expected to affect the politics of war. Importantly, while the expected utility always decreases in economic costs, it does not have to be the case with political repercussions. The "rally-round-the-flag effect" suggests that political leaders can gain additional benefits from war beyond the primary motivation for war. Thus, disaggregation of the economic and political costs can provide additional insights.

Consequently, my proposed framework for the impact of emerging technology on the propensity to use military force combines Davis et al.'s factor tree with utility theory to comprise four decision factors: motivation, prospects of success, economic costs, and net political benefits (Figure 2.2). Countries go to war then they (1) have political ends that cannot be achieved otherwise (motivation); (2) believe they have reasonable chances of reaching those goals through war; (3) the costs are acceptable given the stakes, and (4) there are some additional domestic or international political benefits or at least the political penalty for war is not prohibitively high. Under certain conditions, each of these factors can be influenced by technology, which in turn would marginally change the propensity for war. Specifically, the propensity to use military force marginally increases when emerging technology increases the motivation, prospects of success, and/or net political benefits and/or decreases the economic costs of war.

Figure 2.2. Framework for the Impact of Emerging Technology on the Propensity to Use Military Force.

There are two important caveats to the framework. First, the impact of conventional technology is marginal; that is, it depends on the benchmark comparison. For example, LAWS will likely have little impact on the propensity of the Russian President to attack the continental United States because the perceived benchmark for economic cost is the cost of nuclear war. Thus, regardless of how many soldiers can be saved by sending LAWS into battle instead of ground troops, the propensity to attack the United States will likely remain low.

Second, as decision-makers are influenced by their cognitive biases, misperceptions, and other constraints of human decision-making,[55] technology impacts the *perceived* prospects, costs,

[55] Paul K. Davis, Jonathan Kulick, and Michael Egner, *Implications of Modern Decision Science for Military Decision-Support Systems,* RAND Corporation, MG-360-AF, 2005.

and benefits. The consequence of that assumption is that, for example, two actors who possess the same technology and consider the same military action can have different perceptions of the prospects of success. It is an important distinction for emerging technologies because, initially, there is little empirical data to inform and update decision-makers' beliefs. At the same time, I will follow the rational choice theory and assume that even if actors' beliefs do not reflect objective reality, they act on those beliefs as if they were correct to maximize their utility.

I will now discuss how technology affects each decision factor in general and under what conditions LAWS specifically could impact the decision factors in a conflict-incentivizing or disincentivizing manner. Table 2.1 on page 24 summarizes these findings.

Motivation

The desired benefits—the primary political motivation to initiate a war and "the first, and in some ways decisive, variable for interstate deterrence outcomes"[56]—can be affected by emerging technology either when it creates new reasons for war (or strengthens the existing ones) or when it invalidates (weakens) the old ones.

The former could be the case if a given technology is perceived as a game-changer that shifts the balance of power and consequently incentivizes conflict.[57] Countries could conduct military operations to prevent others from developing a technology (thus preventing the change in the balance of power, i.e., preventive war[58]) or, although harder to conceive, to conquer the technology (thus acquiring more power themselves). The impact of technology can also be indirect if it requires input resources that are finite, scarce, and/or only temporarily available.[59]

One example of such preventive war was the Iraq war, supposedly launched to prevent Saddam Husain from developing weapons of mass destruction.[60] Similarly, the Israeli bombing of Iraqi nuclear facilities in 1981 was launched to ensure that Bagdad could not use a nuclear power plant to produce nuclear weapons.[61] Both cases could be seen as military operations aimed at preventing a shift in the regional balance of power.

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[56] Michael J. Mazarr, Arthur Chan, Alyssa Demus, Bryan Frederick, Alireza Nader, Stephanie Pezard, Julia A. Thompson, and Elina Treyger, *What Deters and Why: Exploring Requirements for Effective Deterrence of Interstate Aggression,* RAND Corporation, RR-2451-A, 2018, p. xiii.

[57] A.F.K. Organski, *World Politics*, New York, NY: Knopf, 1958. For the synthesis of hypotheses of how shift in balance of power can incentivize conflict, see: Stephen Van Evera, *Causes of War: Power and the Roots of Conflict*, Cornell University Press, 1999, pp: 73–104.

[58] Jack S. Levy, "Declining Power and the Preventive Motivation for War," *World Politics*, Vol. 40, No. 1, 1987, pp: 82–107.

[59] Jacquelyn Schneider, "The capability/vulnerability paradox and military revolutions: Implications for computing, cyber, and the onset of war," *Journal of Strategic Studies*, Vol. 42, No. 6, pp: 841–863.

[60] United Nations, "Briefing Security Council, US Secretary of States Powell Presents Evidence of Iraq's Failure to Disarm," Press Release, SC/7658, February 5, 2003.

[61] David K. Shipler, "Israeli Jets Destroy Iraqi Atomic Reactor; Attack Condemned by U.S. and Arab Nations," *The New York Times*, June 9, 1981.

However, there seem to be at least two conditions emerging technology must meet to become a candidate for a cause of war. First, the potential shift in the balance of power caused by technology has to be large enough to justify the risk of military operations. That is why this consideration seems to apply predominantly to weapons of mass destruction. Second, the nature of technology (or input resource) has to be such that it can be destroyed or conquered in combat operations.

Consequently, LAWS seem unlikely to be a cause for war. While AI might have the potential to shift the global distribution of power,[62] LAWS, conceptualized as conventional, relatively low-cost, more sophisticated successors of remotely operated weapons, seem less likely to have that much effect.[63] But even if they raise to that level of perceived threat through their qualitative or quantitative advantage, it would not change the fact that their strength lies in the algorithms controlling them. Meanwhile, algorithm development will likely take place in a distributed environment that is not conducive to elimination through combat operations. Algorithm development does not require as specialized equipment and highly controlled processes as, for example, nuclear weapons. And if bottom-up innovation can change drone hardware in real-time,[64] this can be no less the case with rapid bottom-up development of AI algorithms once the science and data behind them are already in place.

At the same time, it is conceivable that emerging technology can marginally change the propensity for war if it solves a problem that previously would be addressed with military operations. As discussed previously, LAWS could, for example, lower the demand for retaliation. They could also deflate threat perceptions if they prove particularly effective in defense. If a disarming attack is less likely, countries are less incentivized to contemplate a pre-emptive strike.[65]

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[62] Michael C. Horowitz, Shira Pindyck, and Casey Mahoney, "AI, the International Balance of Power, and National Security Strategy," in *The Oxford Handbook of AI Governance*, Justin B. Bullock and others (eds), Online edition, Oxford Academic, 2022.

[63] There is a growing body of literature on the risks LAWS can pose to nuclear capabilities, thus blurring the line between nuclear and conventional stability. However, for the purposes of this study, I assume LAWS will first have to establish themselves as reliable conventional weapons in relatively low-risk situation before they could be considered for deployment in more high-risk scenarios. See: Altmann and Sauer, 2017; Davis, 2019; James S. Johnson, "Artificial Intelligence: A Threat to Strategic Stability," *Strategic Studies Quarterly*, Vol. 14, No. 1, 2020, pp: 16–39; *The Impact of Artificial Intelligence on Strategic Stability and Nuclear Risk Volume I Euro-Atlantic Perspectives*, Vincent Boulanin, ed., Stockholm International Peace Research Institute, May 2019; *The Impact of Artificial Intelligence on Strategic Stability and Nuclear Risk Volume II East Asian Perspectives*, Lora Saalman, ed., Stockholm International Peace Research Institute, May 2019; *The Impact of Artificial Intelligence on Strategic Stability and Nuclear Risk Volume III South Asian Perspectives*, Petr Topychkonov, ed., Stockholm International Peace Research Institute, April 2020.

[64] "How Ukrainians modify civilian drones for military use," *The Economist*, May 8, 2023.

[65] Robert Jervis, "Cooperation under the Security Dilemma," *World Politics*, Vol. 30, no. 2, 1978, pp: 186–214

Prospects of Success

The impact of military technologies on the probability of success is more straightforward: the qualities and quantities of military capabilities available and expected to be used by each side aggregate to the correlation of forces and means and inform the assessment of the chances of winning the war. Suppose emerging technology is a qualitative force multiplier or enables a quantitative growth of the military force due to, for example, low cost, and the technology adoption rate is unequal. That would create a window of opportunity in which the perceived prospects of success of a country that has already adopted the technology increase vis-à-vis a country that has not. This would be especially the case if emerging technology favored offense over defense.[66]

In the case of LAWS, such windows of opportunity would likely be relatively narrow due to their low costs. We already see a rapid proliferation of drones, and their more sophisticated successors could follow the same path. Since military hardware innovation can be a capital-intensive endeavor, traditionally, it has been led by the United States and other high-income countries, China, Russia, as well as India, and Türkiye to a lesser extent.[67] However, the more the advantage of robotic capabilities shifts toward the quality of algorithms, the faster the potential proliferation might become, as algorithm development does not require as much capital as hardware development. Only very few countries can acquire aircraft carriers or nuclear weapons, but almost all can afford programmers. Consequently, the windows of opportunity might be too short to exploit.

Moreover, there is no reason to believe that LAWS would be inherently offensive capability. Today, we often think about them as such because of the legacy of drones. Remotely piloted unmanned aerial vehicles were developed under the U.S. technological and military superiority umbrella, which did not create any incentives for defensive drone capability. However, other countries—and the United States in the future—could choose to prioritize defensive applications. For example, a swarm of autonomous drones could serve as a defensive dome to protect the borders and critical infrastructure. In fact, LAWS could first be deployed as counter-drone weapons because shooting down an unmanned aircraft would arguably be the least controversial use case.

Even if countries adopt emerging technology at an equal rate, the very nature of technology still influences the perceived prospects of success. Specifically, the war is more likely if there is a reward for "jumping the gun," i.e., if the chances of winning increase for the first-mover.[68] Suppose decision-makers are wary that the other side might gain a decisive advantage by

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[66] Jervis, 1978.

[67] Jon Schmid, *The Determinants of Military Technology Innovation and Diffusion*, dissertation, Georgia Institute of Technology, 2018, pp: 180–2.

[68] Thomas Schelling, *Arms and Influence*, New Haven, Conn.: Yale University Press, 1966, p. 235.

deploying first. In that case, they might rush to secure the first-mover benefit for their country and, consequently, preclude a diplomatic solution.[69]

As discussed in the Literature Review section, LAWS' decision-making speed is a common argument in favor of their first-mover advantage. Supposedly, algorithms operate so much faster than human decision-makers that being caught off guard by adversaries' AI might lead to a decisive defeat. However, the fast pace of operations works both ways; that is, defeat can also be quick if the algorithm malfunctions or miscalculates. Given the inherent unpredictability of AI, the fear of such a humiliating defeat discrediting one's capabilities might limit decision-makers' eagerness to deploy LAWS. And even if trust in LAWS is high, decision-makers could still be wary of deploying first because such a move would provide a training set for adversaries' algorithms.

Emerging technology can also marginally change the propensity for armed conflict if it affects the risk of misperception of prospects of success. New technologies can either aid a more realistic assessment of one's chances of winning or instill more false optimism regarding the outcomes of a potential war, which is already prevalent in war decision-making.[70] Due to AI algorithms' inherent complexity and unpredictability, the accurate assessment of their effectiveness in combat will be difficult, if not impossible. However, its effect on the perception of prospects of success might not be uniform or constant over time.

Finally, emerging technologies could enable missions that were not feasible without them (i.e., whose prior probability of success was zero). Additional research is necessary to better understand what such missions could be for LAWS.

Importantly, the prospects of success are always relative to the political ends of military operations. Therefore, in practice, the impact of emerging technologies on the decision to use force will depend on the trends in the objectives of military operations. LAWS might be well aligned with some (e.g., protecting allies' critical infrastructure) and poorly aligned with others (e.g., "winning hearts and minds").

Economic Costs

Military technologies can also influence the costs of war, and as the costs decrease, so does the bargaining space necessary to reconcile conflicting interests without a fight.[71] That is,

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[69] For a review of literature and synthesis of hypotheses on the topic, see: Van Evera, 1999, pp: 73–104.

[70] Geoffrey Blainey, *The Causes of War*, New York: Free Press, 1988; Robert Jervis, *Perception and Misperception in International Politics*, Princeton, NJ: Princeton University Press, 1976; Robert Jervis, "War and Misperception," *Journal of Interdisciplinary History*, Vol. 18, No. 4, 1988, pp: 675–700; Jack S. Levy, "Misperception and the Causes of War: Theoretical Linkages and Analytical Problems," *World Politics*, Vol. 36, No. 1, 1983, pp: 76–99. For the synthesis of hypotheses see Van Evera, 1999, pp: 14–34. For the role of false optimism is selecting war strategies, see: Daniel Altman, "The Strategist's Curse: A Theory of False Optimism as a Cause of War," *Security Studies*, Vol. 24, No. 2, 2015, pp: 284–315.

[71] James D. Fearon, "Rationalist Explanations for War," *International Organization*, Vol. 49, No. 3, 1995, pp: 379–414.

decision-makers are less incentivized to compromise and find a peaceful solution when armed conflicts are perceived as more affordable. In other words, as the costs decrease, more causes could be worth the effort.

Since one of the primary motivations behind the robotic revolution is to save costly human lives, it seems logical that LAWS would decrease the costs of military operations and thus increase leaders' propensity to use force. The decrease in cost would not only stem from saving soldiers' lives but also from avoiding the costs of labor necessary to crew the manned—or even remotely piloted—capabilities. Additional savings could come from stripping military vehicles from expensive defensive systems developed predominantly to save the crew. Or from lowering sustainment costs if the expected duration of conflict decreases.

There are, however, several caveats. First, the decrease in costs due to saving soldiers' lives is only valid under the assumption that human life is worth more than a machine. Meanwhile, the value of soldiers' lives—for the purposes of war decision-making—is not the same across different countries. The cost of a soldier is not only represented by a well-established economic value of a statistical life,[72] but conceptually, it also depends on each country's investment in its soldiers and the scarcity of human resources. Countries that have to engage in costly competition for talent with the civilian sector and invest a lot in soldiers' training might perceive the benefit of substituting humans with machines as high. Conversely, countries with large populations or weak economies where military service is seen as a preferred career path and/or countries that invest little in training might see less economic benefit in substituting humans with machines. The same logic applies even when we consider only the cost of labor (as opposed to the cost of life). The benefit of the substitution will depend on the comparison between the price of labor and the price of capital.

Second, it is not at all clear who or what exactly LAWS would replace. For example, in the foreseeable future, the United States seems to aim for human-machine teaming and augmentation, not substitution.[73] Advances in machine autonomy will likely render some roles of human soldiers obsolete, but they will also create a demand for new ones. Relatedly, freeing soldiers from some responsibilities means that they can be shifted to other responsibilities. Decreasing the number of troops is a political decision with serious political and societal consequences, and as such, it goes beyond the technological or even warfighting necessities.[74]

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[72] Elena Keller, Jade E. Newman, Andreas Ortmann, Louisa R. Jorm, Georgina M. Chambers, "How Much Is a Human Life Worth? A Systematic Review," *Value in Health*, Vol. 24, No. 10, 2021, pp: 1531–1541.

[73] U.S. Department of Defense, *Unmanned Systems Integrated Roadmap FY 2017-2042*, Office of the Secretary of Defense, 2018; U.S. Army, *Robotic and Autonomous Systems Strategy*, Fort Eustis, VA: U.S. Army Training and Doctrine Command, 2017; U.S. Army, *Operationalizing Robotic and Autonomous Systems in Support of Multi-Domain Operations*, whitepaper, Fort Eustis, VA: U.S. Army Training and Doctrine Command, 2018; Kyle Rempfer, "Automation doesn't mean soldiers will be out of a job, chief says," Army Times, January 14, 2020.

[74] Erik Gartzke provides an additional argument against removing humans from battlefields: adversaries would be more incentivized to inflict harm on civilians to achieve political effect. Erik Gartzke, "Blood and Robots: How

If, instead, LAWS replace remotely piloted unmanned capabilities, then the unit cost will likely increase. Similarly to precision-guided bombs that are more expensive than "dumb" bombs, autonomous weapons will likely be more expensive than their less sophisticated predecessors. The analogy is admittedly flawed because there is no significant difference in labor costs between precision-guided and "dumb" bombs. However, more research is needed to understand how exactly the economic costs of military operations would change under LAWS, especially given that they can comprise a broad range of capabilities.

Net Political Benefits

Net political benefits are conceptualized as aggregated international and domestic political costs and benefits, where the international component comprises reputation and soft power, and the domestic component is approximated by support for countries' political leaders. The net characteristic of this factor stems from the fact that a decision to use force can generate political benefits of one kind and costs of another.[75] For example, a nationalistically-motivated attack on another country may increase domestic political support while creating backlash and generating political costs on the international stage.[76]

In the context of strategic studies, reputational costs are usually understood as the price of not following through with a commitment or threat. Deterrence theorists posit that if a country abandons its commitments, i.e., loses its reputation as an actor who follows through, then its future deterrence posture will not be credible, and thus deterrence will fail.[77] There is a continuous debate on the relationship between following through with the threat or commitment and the reputation for resolve;[78] however, for the purposes of this framework, we can abstract away from the future use of reputation. Given that in the past, U.S. Presidents have invoked reputation as a justification for military interventions,[79] I will assume that improving one's reputation provides an additional benefit of fighting, which in turn increases the propensity to

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Remotely Piloted Vehicles and Related Technologies Affect the Politics of Violence," *Journal of Strategic Studies*, Vol.. 44, No. 7, 2021, pp: 983–1013.

[75] Matthew A. Baum, "How public opinion constrains the use of force: The case of Operation Restore Hope," *Presidential Studies Quarterly*, Vol. 34, No. 2, 2004, pp: 187–226.

[76] I do not claim that countries wage wars only when those political considerations produce net benefits; I use the term "benefits" for a stronger distinction from economic costs.

[77] For example, Schelling, 1966; Robert Jervis, *The Logic of Images in International Relations*, Princeton, N.J.: Princeton University Press, 1970; Jervis, 1976.

[78] For a review of debate, see: Robert Jervis, Keren Yarhi-Milo, and Don Casler, "Redefining the debate over reputation and credibility in international security: Promises and Limits of New Scholarship," *World Politics*, Vol. 73, No. 1, 2021, pp: 167–203.

[79] Jonathan Mercer, *Reputation and International Politics*, Cornell University Press, 1996, p. 2.

use force, whereas damaging reputation decreases it.[80] It does not appear that this type of reputation can be influenced by technology.

However, for the purposes of the framework, reputation is conceptualized not only as a reputation for resolve but also as a reputation for upholding international law or for honesty in diplomacy, both shown to be motivating foreign policy decisions.[81] Regardless of the area of application, the basic premise is that "a reputation as an unreliable partner may prevent a government from being able to make beneficial agreements in the future."[82]

Finally, international political costs and benefits of war can be conceptualized in terms of soft power stemming from the country's ideology, culture, and values.[83] Even though the war itself, by definition, does not belong to a soft power resource, its ends, ways, or means—analogically to other instruments of foreign policy—might contradict or support the country's values, be legitimate or illegitimate, and therefore, strengthen or weaken country's soft power.[84]

Those international political costs and benefits might be influenced by emerging military technologies through legal, ethical, and moral concerns they evoke, which is particularly relevant for LAWS, given the ongoing debates on the subject.[85] The exact impact of LAWS on countries' soft power and international reputation will depend on how well the technology performs. LAWS' lack of emotional reaction and better optimization algorithms could limit violence to a necessary minimum and reduce war crimes. That is, LAWS might comply with international law and ethics better than human soldiers. On the other hand, LAWS could be unable to distinguish combatants from non-combatants, adhere to the proportionality rule, or they could attack targets that are legally legitimate but morally unacceptable.

The literature suggests that domestic costs and benefits change over the course of war: there tends to be an initial spike in wartime president's approval (the "rally-round-the-flag" effect[86]),

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[80] Reputation can also be the primary motivation for war: Allan Dafoe, Jonathan Renshon, Paul Huth, "Reputation and Status as Motives for War," *Annual Review of Political Science*, Vol. 17, No. 1, 2014, pp: 371–393.

[81] Rachel Brewster, "Unpacking the State's Reputation," *Harvard International Law Journal*, Vol. 50, No. 2, 2009, pp: 231–270; Andrew T. Guzman, *How international law works: a rational choice theory*, Oxford University Press, 2008; Anne E. Sartori, "The Might of the Pen: A Reputational Theory of Communication in International Disputes," *International Organization*, Vol. 56, No. 1, 2002, pp: 121–49; Anne E. Sartori, *Deterrence by Diplomacy*, Princeton, N.J.: Princeton University Press, 2005; Andrew C. Blandford, "Reputational Costs beyond Treaty Exclusion: International Law Violations as Security Threat Focal Points," *Washington University Global Studies Law Review*, Vol. 10, No. 4, 2011, pp: 669–726.

[82] Robert O. Keohane, *After Hegemony: Cooperation and Discord in the World Political Economy*, Princeton University Press, 1984, p. 258.

[83] Joseph S. Nye Jr, *Bound to lead: The changing nature of American power*, New York: Basic books, 1990.

[84] Joseph S. Nye Jr, "Soft Power and American Foreign Policy," *Political Science Quarterly*, Vol. 119, No. 2, 2004, pp: 255–270.

[85] On the state of debate, see: Steven Umbrello, Phil Torres, and Angelo F. De Bellis, "The future of war: could lethal autonomous weapons make conflict more ethical?," *AI and Society*, Vol. 35, 2020, pp: 273–282.

[86] As Yuval Feinstein shows, the occurrence and magnitude of the effect depends on "the characteristics of events, the historical circumstances and the reactions of national leaders." (Yuval Feinstein, *Rally'round the Flag: The*

but the support for military operations and, by extension, for their political leaders[87] decreases over time due to war casualties among own soldiers and civilians.[88] Another body of literature suggests that political leaders can be penalized for not following through with threats,[89] although there could also be a penalty for threatening in the first place.[90] Albeit to different degrees, domestic political costs of war occur both in democracies and autocracies.[91]

The effect of emerging technologies on domestic political benefits can, therefore, be threefold. The propensity for the use of force marginally increases when technology strengthens the "rally-round-the-flag" effect, minimizes casualties, and/or makes commitments and threats easier to uphold. Therefore, similarly to the international political costs and benefits, the impact of LAWS will depend on leaders' expectations regarding LAWS' performance in combat. Suppose political leaders believe LAWS will perform well. In that case, they might expect their approval ratings not to fall throughout the conflict.

Conversely, if they believe there is a high risk that LAWS will repeatedly attack and kill civilians, they will see LAWS as a political liability. Furthermore, the challenge of assigning responsibility to the algorithm, its developer, or a commander who ordered deployment[92] might

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Search for National Honor and Respect in Times of Crisis, Oxford University Press, 2022, p. 6). In many cases it does not not occur at all (Bradley Lian and John R. Oneal, "Presidents, the use of military force, and public opinion," *Journal of Conflict Resolution*, Vol. 37, No. 2, 1993, pp: 277–300; TaeJun Seo and Yusaku Horiuchi, "Natural Experiments of the Rally 'Round the Flag Effects Using Worldwide Surveys," *Journal of Conflict Resolution*, online edition, April 23, 2023). Importantly for the generality of the framework, the "rally-round-the-flag" effect is not uniquely American phenomenon (Dieuwertje Kuijpers, "Rally around all the flags: the effect of military casualties on incumbent popularity in ten countries 1990–2014," *Foreign Policy Analysis*, Vol. 15, No. 3, 2019, pp: 392–412).

[87] Michael Tomz, Jessica L.P. Weeks, and Keren Yarhi-Milo, "Public Opinion and Decisions About Military Force in Democracies," *International Organization*, Vol. 74, No. 1, 2020, pp: 119–43.

[88] John E. Mueller, *War, presidents, and public opinion*, New York: Wiley, 1973; Eric V. Larson and Bogdan Savych, *Misfortunes of War: Press and Public Reactions to Civilian Deaths in Wartime,* RAND Corporation, MG-441-AF, 2006; Scott Sigmund Gartner, "The multiple effects of casualties on public support for war: An experimental approach," *American political science review*, Vol. 102, No. 1, 2008, pp: 95–106; Scott Sigmund Gartner and Gary M. Segura, "War, casualties, and public opinion," *Journal of conflict resolution*, Vol. 42, No. 3, 1998, pp: 278–300.

[89] James D. Fearon, "Domestic Political Audiences and the Escalation of International Disputes," *American Political Science Review*, Vol. 88, No. 3, 1994, pp: 577–92; Michael Tomz, "Domestic audience costs in international relations: An experimental approach," *International Organization*, Vol. 61, No. 4, 2007, pp: 821–840; Erik Gartzke and Yonatan Lupu, "Still looking for audience costs," *Security Studies*, Vol. 21, No. 3, 2012, pp: 391–397.

[90] Joshua D. Kertzer, and Ryan Brutger, "Decomposing audience costs: Bringing the audience back into audience cost theory," *American Journal of Political Science*, Vol, 60, No. 1, 2016, pp: 234–249.

[91] Bruce Bueno De Mesquita and Randolph M. Siverson, "War and the survival of political leaders: A comparative study of regime types and political accountability," *American Political Science Review*, Vol. 89, No. 4, 1995, pp: 841–855; Jessica L. Weeks, "Autocratic Audience Costs: Regime Type and Signaling Resolve," *International Organization*, Vol. 62, No. 1, 2008, pp: 35–64; Mark S. Bell and Kai Quek, "Authoritarian public opinion and the democratic peace," *International Organization*, Vol. 72, No. 1, 2018, pp: 227–242.

[92] For a summary of the debate, see: Ann-Katrien Oimann, "The Responsibility Gap and LAWS: a Critical Mapping of the Debate," *Philosophy and Technology*, Vol. 36, No. 1, 2023, pp: 1–22.

lead to a situation where there is no "bad apple" to blame for AI's misconduct, and thus, political leaders could worry that they will be personally blamed. So far, there is no indication that the U.S. public would be against LAWS in principle,[93] but deployment might update these priors in either direction. Finally, different societies might put different weights on war casualties and the characteristics of warfighting, creating a disadvantage for countries whose public is more concerned with ethical, moral, and legal aspects of war.

Experimental research shows that even holding the number of casualties constant, support for military operations is higher when there is no draft.[94] By the same token, LAWS might be associated with a higher approval for military operations simply because fewer soldiers are sent into battle (if that indeed is the case).

Relatedly, the "rally-round-the-flag" effect might be stronger in the presence of LAWS if there is a belief that military operations can accomplish their goals at lower human and financial costs. On the other hand, it is also possible that with fewer soldiers' lives at stake, general interest in military operations would fade, and therefore, the initial premium for wartime leaders would be lower.

Table 2.1 summarizes the conditions under which LAWS could lead to increased or decreased propensity for armed conflict.

Table 2.1. Potential Impact of LAWS on Propensity for Armed Conflict.

Framework elements and their definition	Assumptions about LAWS under which the use of force becomes more likely	Assumptions about LAWS under which the use of force becomes less likely
Motivation		
The political ends of considered military action; *rarely influenced by technology*.	LAWS are perceived as game-changer shifting the balance of power, while combat operations are a feasible mean to deny access to LAWS.	LAWS invalidate or weaken past causes for war (e.g., decreased demand for retaliation; alleviate adversaries' threat perception).
Prospects of success		
The perceived probability of success based on the correlation of forces and means between two sides, offense-defense balance, trust in the technology, and a first-mover advantage. The prospects of success are always relative to the political ends of considered military action.	Technical specs of LAWS (speed of decision-making, endurance, mobility, size, weight, footprint, lack of need for communication) enable operations previously impossible. Technical specifications of LAWS are better aligned with the political goals of the considered action. LAWS favor offense over defense. There is a	Technical specifications of LAWS are poorly aligned with the political ends of considered action (e.g., "winning hearts and minds"). The error rate is high, and leaders' confidence in technology is low. Consequently, leaders avoid conflict to avoid revealing the weaknesses of their LAWS. LAWS prove to be very

[93] Morgan et al., 2020; Koki Arai and Masakazu Matsumoto, "Public perceptions of autonomous lethal weapons systems," *AI Ethics*, 2023.

[94] Michael C. Horowitz and Matthew S. Levendusky, "Drafting support for war: Conscription and mass support for warfare," *The Journal of Politics*, Vol. 73, No. 2, 2011, pp: 524–534.

Framework elements and their definition	Assumptions about LAWS under which the use of force becomes more likely	Assumptions about LAWS under which the use of force becomes less likely
	window of opportunity due to the uneven proliferation of LAWS. There is a first-mover advantage (e.g., due to the tempo of AI operations), and the confidence in own algorithm is high. LAWS are force multipliers for other military technologies (e.g., in providing better situational awareness). LAWS are perceived as more likely to be actually deployed than their alternatives.	reliable and efficient in defensive applications and unreliable in offensive applications. Leaders limit deployments to avoid proving a training set to adversaries LAWS.
Economic costs		
Financial, material, and personnel costs of considered military action.	Costs of LAWS replacement are lower than for the predecessors. Labor costs are lower. Due to LAWS capabilities, expected duration of conflict decreases enabling lower sustainment costs. LAWS replace humans, not only augment	Training costs for support personnel for LAWS are higher than for the predecessors (e.g., due to wage competition with tech companies). The component prices are drastically higher than for predecessors.
Net political benefit		
The international reputation effects and domestic public support for considered military action; depends on the technology through casualties on both sides, perceived “fairness” of considered action, and confidence in the performance of new technology.	LAWS better comply with international humanitarian law and law of armed conflict (distinction, proportionality, and military necessity principles) due to a lack of negative emotions against the opposing forces. The error rate is low, so the number of civilian casualties decreases. LAWS enable reducing casualties among own soldiers. The performance of LAWS in combat improves the country’s international reputation.	The error rate is high, civilian deaths are increasing, and they are harder to explain. Notorious cases of legal but unethical kills (e.g., child soldiers). The lack of accountability of AI shifts the entire political responsibility up the command chain to the political level.

Conclusions

The framework proposed in this chapter was designed to reason through the four decision factors and explicitly assess under what condition and assumption an emerging technology can increase or decrease the propensity for military action. If the technology increases the motivation, prospects of success, and net political benefits but decreases the economic costs, the decision-makers’ propensity to use military force will increase.

Application of the framework to LAWS demonstrated that their impact on the propensity for armed conflict is far from certain. For example, they can have a stabilizing effect if we decide to prioritize capabilities more suitable for defense or when there emerges a belief that warfighting provides a training set for the adversary’s AI and thus hinders future prospects of success. Of course, not all situations in which the propensity to use force decreases are desirable. For example, a high error rate would lower the confidence in LAWS, marginally decreasing the propensity to deploy. However, a high error rate itself would be a highly undesirable phenomenon, given its potential human costs. Consequently, the framework sheds light on the tradeoffs of LAWS development. As much as we want LAWS to act efficiently, ethically, and within the legal bounds, the better they perform, the more incentives to use them.

There are two caveats to these findings. First, the framework factors are treated as independent, but this is unlikely to be the case. In particular, political costs and benefits—international and domestic—likely depend on motivation, prospects of success, economic costs, and who the adversary is.[95]

Second, the framework does not determine what happens if the directions of effects on different framework factors diverge. For example, the shift in economic costs incentivizes war, yet the change in prospects of success disincentivizes it. Therefore, the framework helps to understand the ramifications of different design, acquisition, and policy decisions, but it cannot answer the question of whether overall emerging technology will lead to an increase or decrease in a leader's propensity for war. These dynamics will be further explored in a game-theoretic model in Chapter Three.

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[95] See, for example: Eric V. Larson, *Casualties and Consensus: The Historical Role of Casualties in Domestic Support for U.S. Military Operations,* RAND Corporation, MR-726-RC, 1996; Joe Clare and Vesna Danilovic, "Reputation for Resolve, Interests, and Conflict," *Conflict Management and Peace Science*, Vol. 29, No. 1, 2012, pp: 3–27; Benny Geys, "Wars, Presidents, and Popularity the Political Cost(S) of War Re-Examined," *The Public Opinion Quarterly*, Vol. 74, No. 2, 2010, pp: 357–374; George C. Edwards, William Mitchell, and Reed Welch, "Explaining Presidential Approval: The Significance of Issue Salience." *American Journal of Political Science*, Vol. 39, No. 1, 1995, pp: 108–34; Michael R. Tomz and Jessica L.P. Weeks, "Public opinion and the democratic peace," *American political science review*, Vol. 107, No. 4, 2013, pp: 849–865.

Chapter 3. The Impact of Emerging Technologies on Extended Deterrence: Game-Theoretic Approach

The framework indicates that leaders' propensity to resort to war increases when technological advances lead to a decrease in the economic costs of war or an increase in prospects of success and political benefits of fighting. However, war is hardly a unilateral act. Instead, it is a strategic interaction between two or more parties. In this work, I specifically explore how technological shifts can affect the stability of major power competition over third countries. The focus on extended deterrence, which the existing literature singles out as one of the key understudied areas in the context of LAWS,[96] seems particularly relevant given that this study conceptualizes autonomous weapons as a conventional capability. If that is the case, the impact of LAWS on direct deterrence might be negligible as the bilateral interaction between major powers is still governed by nuclear deterrence.[97] However, while nuclear weapons deter major powers from waging wars against each other, they might not be sufficient to deter an attack of one major power against a partner of another major power. One example of such a war is Russia's full-scale assault on Ukraine, a U.S. partner. Another such scenario could be China's invasion of Taiwan.

To understand how such scenarios can play out when U.S. competitors introduce autonomous weapons systems into the mix, I turn to formal modeling using a game-theoretic approach. I first established the conditions of the status quo (here defined as a situation before the notional technological shift) in which war is absent, i.e., the challenging major power is deterred from aggression against the other major power's partner. I found two deterrence mechanisms, both hinging on the defender's support for its partner. First, when the challenger was to challenge but then back down in the face of the defender's involvement level. Second, when the challenger and defender were to fight, but the fight would be very costly for the challenger. Both cases come down to the defender committing enough troops for the challenger not to challenge. I focus the analysis on the former case when the challenger does not challenge due to the political costs of backing down.

Next, I examined how deterrence is affected by changes in the aggressor's military technologies that impact troops' quality, economic costs, and political benefits of fighting. Since there is a widespread expectation that the robotic revolution will decrease the costs of wars, I specifically focus on how the decrease in challenger costs would impact deterrence dynamics. I found that there always exists such a low value of the challenger's unit cost of fighting that the

[96] Wong et al., 2020, p. x.

[97] For an overview of deterrence types, see: Michael J. Mazarr, *Understanding Deterrence,* RAND Corporation, PE-295-RC, April 2018.

deterrence fails. However, there are special cases of no deterrence in the status quo, where the decrease in costs leads to the emergence of deterrence.

Focusing on the cases of precarious deterrence where a small decrease in challenger's costs leads to deterrence failure, I explored how changes in the challenger's quality of troops and political benefits brought about by the robotic revolution could affect that state. Overall, the model confirms the intuition that if there is a tradeoff between the costs and at least one other parameter (for example, robotic warfare is cheaper but causes significant backlash or is less likely to succeed), then the net effect of technological shift on deterrence might be negligible. However, finding the deterrence-preserving relative rate of change in each decision factor might be difficult in real life. Meanwhile, influencing adversaries' economic costs, political benefits, and troops' quality is difficult in and of itself. Consequently, seeking to preserve deterrence in the face of technological advances might be better served by changes in the defender's decision factors.

Thus, in search of recommendations for the U.S. Government, I examined the defender's mitigation options when the challenger's technology develops in a war-incentivizing manner. Suppose the U.S. competitors' pursuit of autonomous weapons systems (or other emerging technologies) has a destabilizing effect. What can the United States do to preserve deterrence? How should the United States think about its pursuit of robotic capabilities to disincentivize conflict? Should the United States mirror competitors' efforts? In other words, is keeping up sufficient to maintain the status quo? If not, which technology characteristics should the United States focus on in its technology development to disincentivize conflict?

The results suggest that to preserve deterrence, the defender should focus on technology and policy solutions that decrease mobilization costs. This is the only choice that has an unambiguously positive effect on deterrence. Investing in technologies that can be force multipliers should also pay off, even though it can have a destabilizing effect in some very specific and unlikely cases. But as long as the ratio between the defender's and challenger's troops' quality remains constant, the net effect of the challenger's innovation in quality will be zero. The results also suggest that making sure that there is no backlash against LAWS, or even better, that they are perceived as a more ethical option and thus yield higher political benefits of war, would have a positive effect on upholding commitments to allies and as such, it would strengthen deterrence. Last but not least, the model suggests that lowering the costs of warfighting to mirror the challenger's decreasing costs should not be the priority because, in fact, it incentivizes war: the cheaper the fight for the defender, the fewer reasons to avoid the war.

Modeling extended deterrence in the game-theoretic approach is not new, but my model makes several new contributions. Existing models of extended deterrence usually focus on the conditions under which allies fulfill their formal commitments.[98] My model, on the other hand,

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[98] See for example, Gerald Sorokin, "Alliance Formation and General Deterrence: A Game-Theoretic Model and the Case of Israel," *The Journal of Conflict Resolution*, Vol. 38, No. 2, 1994, pp: 298–325; Catherine C. Langlois,

focuses on the deterrence itself: whether the challenger attacks the third country or not, regardless of whether the defender supports its partner. Furthermore, I model technology-impacted inputs of this decision-making, disaggregating economic costs and political benefits. Also, the existing models often treat the challenger's and defender's decision to fight as binary: either they engage in war or not. Meanwhile, my model assumes that when major powers decide to engage in war, the force size selected from continuous distribution is part of their strategic choice. In turn, the fighting force of both sides combined with the troops' quality determine the prospects of success, in other models often treated as given.[99]

Model Overview

I model a three-player strategic interaction, where two players, Challenger (he/him/his) and Defender (she/her), are near-peers major powers, and the third player is an ally or a partner of Defender and a foe of Challenger. I call this third player Protégé (they/them/their), following the convention introduced by Zagare and Kilgour.[100] One pertinent example would be the United States as the Defender and China or Russia as the Challenger. In such a scenario, the Protégé could be any U.S. partner or ally threatened by any of these two major powers, e.g., Taiwan or Ukraine. The fact that Defender and Challenger are near-peers is represented by the efficiency of their troops—the ratio between their quality and cost—which I assume to be comparable. Furthermore, I assume that if Challenger and Defender committed the exact same number of troops, but Defender fought alongside Protégé, then those pooled troops have a higher probability of winning than Challenger fighting alone. I also assume that Protégé is a weaker country than Defender and Challenger, which means that the quality of Protégé's troops is lower than the quality of Challenger's and Defender's troops. While real-life representatives of the Protégé figure can develop pockets of excellence where their tactics or will to fight can be superior, their weapons systems are, on average, less effective than those of Challenger or Defender figure representatives. In real life, players could be both individual states or alliances, but in the game, they make decisions as if they were risk-neutral unitary actors who aim to maximize expected utility.

I model this interaction as a sequential-move game of perfect information. That is, all utility-relevant parameters are known, and the actions are observed. In the real world, players might have uncertainty about their own parameters or the parameters of other players. For example,

"Power and Deterrence in Alliance Relationships: The Ally's Decision to Renege" *Conflict Management and Peace Science*, Vol. 29, No. 2, 2012, pp: 148–169; Renato Corbetta, "Determinants of Third Parties' Intervention and Alignment Choices in Ongoing Conflicts, 1946–2001," *Foreign Policy Analysis*, Vol. 6, No. 1, 2010, pp: 61–85; Frank Zagare, and Marc Kilgour, "Alignment patterns, crisis bargaining, and extended deterrence: A game-theoretic analysis," *International Studies Quarterly*, Vol. 47, No. 4, 2003, pp: 587–615.

[99] For example, Zagare and Kilgour, 2003.

[100] Zagare and Kilgour, 2003.

they may not know how their adversary assesses the quality of their troops and, thus, the probability of winning. It especially might be the case when players enter the first crisis since the deployment of new military technology. However, my model is the first game of extended deterrence to define payoffs as functions of parameters impacted by technologies and to use contest success functions where Defender and Challenger choose their level of effort to increase the probability of winning. Consequently, the model yields new, interesting results even without formally introducing uncertainty. Adding the complexity of imperfect information can be a valuable extension of this model.

At the beginning of the game, Challenger decides whether to threaten Protégé with war if they do not concede to Challenger's political demands. These demands can be interpreted as a territorial dispute, dispute over natural resources, or conflict over a broader foreign policy course chosen by Protégé (e.g., Ukraine's choice of Transatlantic integration opposed by Russia). The game does not account for the possibility of a negotiated settlement, and thus, the motivation for war can be both a divisible or indivisible issue.[101] Intuitively, the game starts when the possibility of a negotiated solution fails, and Challenger has to either forego the demands or threaten Protégé with military aggression. If Challenger decides not to threaten, the game ends, and the peaceful status quo is preserved.

If Challenger threatens the Protégé, Defender decides whether to provide military support to Protégé. The decision to support the Protégé requires from Defender to specify the magnitude of support. That is, Defender commits to Protégé's defense a specific number of troops. Equipped with that knowledge, Protégé decides whether to concede to Challenger's demands or not. This order of steps is a reverse of the sequence proposed by Zagare and Kilgour (2003). In real life, receiving the Defender's support is an iterative process that continues long after an outbreak of war (e.g., Western support for Ukraine drastically grew over the course of the full-scale Russian invasion of 2022). For the purposes of this model, where the force size is the players' strategic decision, it is reasonable to assume that Defender's decision is input to Protégé's ultimate decision to concede or fight. I assume that Protégé can fight even without Defender's support, which is an option not accounted for by Zagare and Kilgour's model. This choice is motivated by the example of Ukraine in 2014, which initially decided to fight for Donbas without any tangible support from its Western partners.

If Protégé decides to concede, the game ends. Otherwise, Challenger makes the final decision whether to back down or attack. Similar to Defender's, Challenger's decision to fight requires him to determine how much force to commit to fighting. Protégé, on the other hand, as a smaller and weaker country threatened by a major power, is assumed to commit their entire military capability if they choose to fight. This is motivated by the fact that in real life, the correlation of

[101] Fearon, 1995, discusses how bargaining can fail if the issue at stake is indivisible. In my model, players calculate the expected value of the issue at stake when they consider fighting over it, but if the fight does not take place, the winner takes all.

forces between any major power and a potential target of attack (e.g., Russia and Ukraine or China and Taiwan) favors the major power to the extent that the potentially attacked country is likely to perceive war as an existential threat, which precludes keeping any troops idle, both for operational and political reasons.

The payoffs in each outcome are defined by how much the players value the political stakes of potential war, how many troops they are willing to commit, what the quality of these troops and the costs of their mobilization and employment are, and finally, what are the political benefits of the fight, other than stakes themselves (e.g., rally around the flag effect). Conceptually, the payoffs are modeled as the expected utility of the value of war motivation weighted by their respective probability of winning plus the additional political benefit of fighting minus the cost of force employment. However, different outcomes are defined by different probabilities of winning and costs.

If the Challenger chooses not to threaten, the Challenger cannot win, while Protégé and Defender are certain to keep whatever is at stake (although Defender can value it less than Protégé). This is also the case when Challenger threatens but subsequently backs down. If Protégé concedes, the situation reverses: Challenger receives—with certainty—whatever motivated him to threaten Protégé, while Defender and Protégé lose it. Finally, if the war breaks out, then the respective probabilities of winning are Contest Success Functions of the quantity and quality of troops committed by each side.[102]

Importantly, as Defender and Protégé fight on one side, they have the same probability of winning. If Defender committed troops to defend Protégé, they would fight together, sharing the quality of Defender's troops. The intuition behind this shared quality is that when they fight together, the effects of Defender's capabilities (e.g., intelligence, targeting, logistics) spill over to Protégé's troops. If Defender did not commit any troops, then Defender's and Protégé's probability of winning is defined by Protégé's force size and quality. It might be counterintuitive that Defender can still incur benefits from a Protégé's win despite not joining the war. However, this is motivated by the case of 2014 Ukraine's fight against Russia for its pro-Western choice without any substantial U.S. support. In that case, the United States got the benefits of Ukraine's fight (for example, strengthening the partnership over the longer time horizon) without incurring the costs of military support.

Costs also differ between outcomes. When Challenger does not threaten, no one incurs any costs. Now, suppose the status quo is challenged, but Defender decides not to support Protégé. In that case, she will not incur the costs of mobilization or fighting. Yet, in every outcome following her decision, she will suffer the opportunity cost of the lost political benefit of fighting and the cost of weakening the partnership with Protégé. This political opportunity cost is imposed on every player who gives up on the fight, i.e., on Protégé when they concede and on

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[102] First introduced in Gordon Tullock, "Efficient rent-seeking," in J.M. Buchanan, R.D. Tollison, G. Tullock (Eds.), *Toward a Theory of Rent-Seeking Society*, Texas AM University Press, College Station, 1980, pp: 97–112.

Challenger when he backs down. Furthermore, if Defender decides not to support Protégé, Protégé also suffers the cost of weakening the partnership, while Challenger will see it as an additional benefit.

When players commit troops to fight, the cost of that commitment will depend on whether the fight takes place. If it does not, players who committed troops incur only the mobilization costs, which are lower than the cost of fighting. For the Defender, this happens when she commits to support Protégé, but Protégé concedes, or Protégé does not concede, but Challenger backs down. In the latter case, also Protégé incurs only mobilization costs.

Model Specification

The game has three players, C, D, and P. Players fight over political stakes, which is their motivation $m > 0$. Challenger and Protégé value them the same. Defender can value these political stakes of war differently, i.e., she weights them by the factor of α such that, for Defender, the value of stakes is $0 < \alpha m \leq m$. In real life, Challenger's and Protégé's valuation of stakes could be different, but it is not essential to my focus on autonomous weapons systems, which I found unlikely to change primary motivations for wars (see Chapter Two). Players seek to maximize their respective payoffs $U_i, i \in [C, D, P]$.

Stage 1

At the beginning of the game, Challenger chooses whether to threaten Protégé. If he does not threaten, he receives 0, and the other players retain their valuation of stakes. That is, players' payoffs are $U_C = 0, U_D = \alpha m, U_P = m$.

Stage 2

If Challenger threatens the Protégé, Defender decides whether to provide military support to Protégé or not. That is, she decides how many troops t_D she commits to fighting. When Defender does not support Protégé $t_D = 0$. Otherwise, she chooses some $t_D \geq t_L > 0$, where t_L is the minimum number of troops that Defender has to commit when she joins the fight. This lower bound prevents a superficial situation when Defender commits almost no troops only to avoid the penalty for avoiding the fight altogether. Intuitively, support below this threshold is indistinguishable from no support in terms of the political benefit of fighting: for example, involvement this small would not warrant the "rally-round-the-flag" effect because it would not capture the attention of the domestic audience. Furthermore, it would also not improve Defender's reputation on the international stage because it would be clear that her support for Protégé is purely symbolic. Finally, it would also not merit higher prospects of success: there would be no spillover effects of the Defender's troops' quality, and the additional quantity would not make up for arising coordination problems.

Stage 3

Next, Protégé decides whether to concede to Challenger's demands or not. That is, they decide whether to commit 0 or t_P. Contrary to Defender, Protégé's choice of force size is binary. It is motivated by the assumption that an attacked country will base its calculation on the maximum force it can employ. I assume that Protégé's military potential is greater than the minimum force size Defender or Challenger can commit to fighting, i.e., $t_P > t_L$. I normalize $t_P = 1$.

If Protégé concedes, players' payoffs depend on Defender's earlier decision to support or not. In both cases, Challenger receives m with certainty, and the other two players lose their respective weighted value of stakes. When Protégé concedes after receiving Defender's support, Defender incurs mobilization cost $d_D t_D$, which is the cost of committing troops to a war that did not take place.[103] Protégé loses the political benefit of fighting b_P. Consequently, if Protégé concedes after receiving Defender's support, players' payoffs are $U_C = m, U_D = -d_D t_D, U_P = -b_P$.

The parameter $b_i > 0, i \in [C, D, P]$ is the political benefit of fighting other than the stakes themselves, in practice modeled as the political costs of not fighting. It represents a domestic and international reaction to players' actions and, as such, is influenced by technology (e.g., there can be a backlash against some technologies but not others) but independent of force size. This can be justified by real-life examples of backlash against military operations based on individual stories of soldiers' wrongdoings. These "bad apples" seem to taint audiences' perception of militaries at war, regardless of the number of soldiers committing crimes. It is framed as a political benefit (i.e., $b_i > 0$). In real life, war can become political costs as opposed to benefits, but I exclude this possibility in my model because it usually happens long after the decision to launch an attack is made. Additionally, if $b_i < 0$, then $-b_i > 0$, so Challenger could threaten Protégé only to receive the benefit of public support for backing down and avoiding the fight. Note that permitting $b_D, b_P < 0$ would not change the results.

If Defender does not support Protégé, and Protégé concedes, then Challenger receives an additional benefit β_C of weakening the partnership between Defender and Protégé. Defender incurs the costs of avoiding the fight and of weakening the partnership, collapsed into one parameter b_D. Protégé suffers the same fate, but for them, the costs are disaggregated into b_P and β_P, to capture the payoff difference between conceding with and without Defender's support. Consequently, if Defender does not support Protégé and Protégé concedes, then players' payoffs are $U_C = m + \beta_C, U_D = -b_D, U_P = -(b_P + \beta_P)$.

[103] Note that the word "mobilization" is here used to describe the process of bringing forces to readiness, not the legal status of that process.

Stage 4

When Protégé does not concede and Challenger backs down, then Defender and Protégé win the stakes with certainty, whereas Challenger incurs the political cost of not following through with the threat. Overall, payoffs still depend on Defender's earlier decision. And so, if Defender supports Protégé, both Defender and Protégé incur mobilization costs, and so players' payoffs are: $U_C = -b_C, U_D = \alpha m - d_D t_D, U_P = m - d_P$.

If Defender does not support Protégé, then Challenger's loss is offset by the benefit of weakening Defender's and Protégé's partnership. Defender again incurs the costs of avoiding the fight, whereas Protégé incurs mobilization costs and the cost of weakening the partnership with Defender. Consequently, the payoffs are $U_C = -b_C + \beta_C, U_D = \alpha m - b_D, U_P = m - (d_P + \beta_P)$.

If Challenger attacks Protégé, he needs to decide how many troops t_C to commit to war. Same as for Defender, there is the minimum number of troops t_L that Challenger has to deploy when he chooses to attack Protégé. That is, $t_C \geq t_L > 0$. When Challenger chooses to fight with $t_C = t_L$, I will call it a low-intensity conflict, which can also be conceptualized as hybrid war or high-end gray-zone aggression. Conversely, when $t_C > t_L$, I will call it a full-scale war. Conceptually, the illegal annexation of Crimea and the subsequent conflict in the Donbas region in Ukraine between 2014 and 2022 would be the case of $t_C = t_L$, whereas Russia's war against Ukraine launched in 2022 would be the case of $t_C > t_L$.

When Challenger attacks Protégé, each player's payoff is the value of stakes weighted by their probability of winning $p_i \in (0,1)$ minus the cost of force employment $c_i > 0$. Note that the cost of force employment is higher than the mobilization cost ($c_i > d_i$) because conceptually, fighting requires conducting mobilization first. Respective probabilities of winning $p_i(t_C, t_D, t_P)$ are contest success functions combining the number of troops t_i with their quality q_i, which represents the effectiveness of converting resources to the probability of winning. q_i is the component of the probability of winning that can be influenced by technology. Protégé, as a country weaker than Challenger and Defender, has troops of lower quality than the other two players, i.e., $q_P < q_D, q_C$.

In real life, the quality and costs associated with different elements of force greatly vary within that force. For example, precision-guided munitions have higher effectiveness and higher costs than unguided munitions. Thus, the parameters should be interpreted as average values of these characteristics.

If Defender supports Protégé, the probability of winning for both of them is:

$$p_D(t_C, t_D, 1) = p_P(t_C, t_D, 1) = p(t_C, t_D, 1) = \frac{q_D(t_D + t_P)}{q_C t_C + q_D(t_D + t_P)} = \frac{q_D(t_D + 1)}{q_C t_C + q_D(t_D + 1)}$$

The Challenger's probability of winning is:

$$p_C(t_C, t_D, 1) = 1 - p(t_C, t_D, 1)$$

Thus, Challenger's, Defender's, and Protégé's payoffs are

$$U_C = \left(1 - p(t_C, t_D, 1)\right)m - c_C t_C,\ U_D = p(t_C, t_D, 1)\alpha m - c_D t_D,\ U_P = p(t_C, t_D, 1)m - c_P$$

If Defender does not provide support to Protégé, but Protégé and Challenger decide to fight, then the probability of winning for Defender and Protégé is

$$p_D(t_C, 0,1) = p_P(t_C, 0,1) = p(t_C, 0,1) = \frac{q_P}{q_C t_C + q_P}$$

The probability of winning for Challenger is

$$p_C(t_C, 0,1) = 1 - p(t_C, 0,1)$$

Again, the fact that Defender does not join the fight does not mean that she loses political stakes. For example, in 2014, the European Union did not provide meaningful military support to Ukraine, but Ukraine nevertheless signed the Association Agreement and the Deep and Comprehensive Free Trade Area Agreement.

Thus, Challenger's, Defender's, and Protégé's payoffs are

$$U_C = \left(1 - p(t_C, 0,1)\right)m - c_C t_C + \beta_C; U_D = p(t_C, 0,1)\alpha m - b_D; U_P = p(t_C, 0,1)m - (c_P + \beta_P)$$

Note that all else being equal, Protégé always prefers Defender to support, whereas Challenger prefers Defender not to support Protégé. Challenger always prefers Protégé to concede over all other possible outcomes. Protégé and Defender prefer Challenger not to threaten.

The extensive form of the game is presented in Figure 3.1.

Parameters summary

To summarize, players fight over stakes m, which Defender weight by $\alpha, \alpha \in (0,1]$. Each player $i, i \in [C, D, P]$ can choose to fight with military force t_i or 0. Defender and Challenger select the value $t_i \geq t_L$, whereas Protégé's force is fixed at $t_P = 1$. Their military is characterized by four parameters:

- The quality (effectiveness) q_i which, combined with the quantity of troops in the contest success function, determines players' probability of winning (also called here prospects of success) when Challenger decides to attack Protégé;
- The unit cost of employment c_i incurred only when Challenger attacks Protégé;
- The unit cost of mobilization d_i incurred when troops have been committed to war which does not take place, i.e., for Defender, when she supports Protégé but they concede or they do not concede, but Challenger backs down, and for Protégé, when they do not concede, and Challenger backs down.

- The political benefit of fighting b_i modeled as the cost of avoiding the fight, i.e., incurred by Defender when she does not support, Protégé when they concede, and Challenger when he backs down.

There is also a parameter β_i signifying the cost (for Protégé) or benefit (for Challenger) of weakening the partnership between Defender and Protégé. The parameters are summarized in Table 3.1.

Table 3.1. Model Parameters.

Parameter description	Parameter symbol	Parametric assumptions
Motivation, i.e., the value of war stakes	m	$0 < m$
Defender's weight on the value of war stakes	α	$0 < \alpha \leq 1$
Protégé's number of troops	t_P	$t_P = 1$
Minimum number of troops Defender and Challenger have to commit if they choose to fight	t_L	$0 < t_L < t_P$
Quality of troops	q_i	$0 < q_P < q_D, q_C$
Unit cost of force employment	c_i	$0 < c_i$
Unit cost of force mobilization	d_i	$0 < d_i < c_i$
Political benefit of fighting	b_i	$0 < b_i$
Value of partnership between Challenger and Protégé	β_i	$0 < \beta_i$

Recall that Challenger and Defender are both major powers and near-peers. One such case would be when Defender represents the United States, whereas Challenger represents the People's Republic of China. Given the U.S. technological superiority, in this specific case $q_D > q_C$. However, for the sake of the model's generality, I allow $q_C > q_D$. Yet, to represent the near-peer assumption, I assume that if Challenger and Defender both committed the minimum number of troops t_L, but Defender fought alongside Protégé, then those pooled troops would have a higher probability of winning than Challenger fighting alone. That is,

$$\frac{q_D(t_L + 1)}{q_C t_L + q_D(t_L + 1)} > \frac{q_C t_L}{q_C t_L + q_D(t_L + 1)} \Leftrightarrow q_D(t_L + 1) > q_C t_L \Leftrightarrow \frac{t_L + 1}{t_L} > \frac{q_C}{q_D}$$

For simplicity, I will also assume that $b_D < d_D t_L$ to exclude a parameter regime where Defender could commit troops knowing that Protégé will concede anyway. Such a parameter regime does not provide any additional insights.

Definitions

Throughout this chapter, I will use several terms defined below.

Definition 1: *Efficiency (s_i)* of player i's troops, $i \in [C, D, P]$, is a ratio between the quality q_i and unit cost of force employment c_i.

$$s_i = \frac{q_i}{c_i}$$

Definition 2: *Relative efficiency (r_i)* of player i, $i \in [D, P]$, is a ratio between the player i's and Challenger's troops' efficiency.

$$r_i = \frac{s_i}{s_C} = \frac{q_i c_C}{c_i q_C}$$

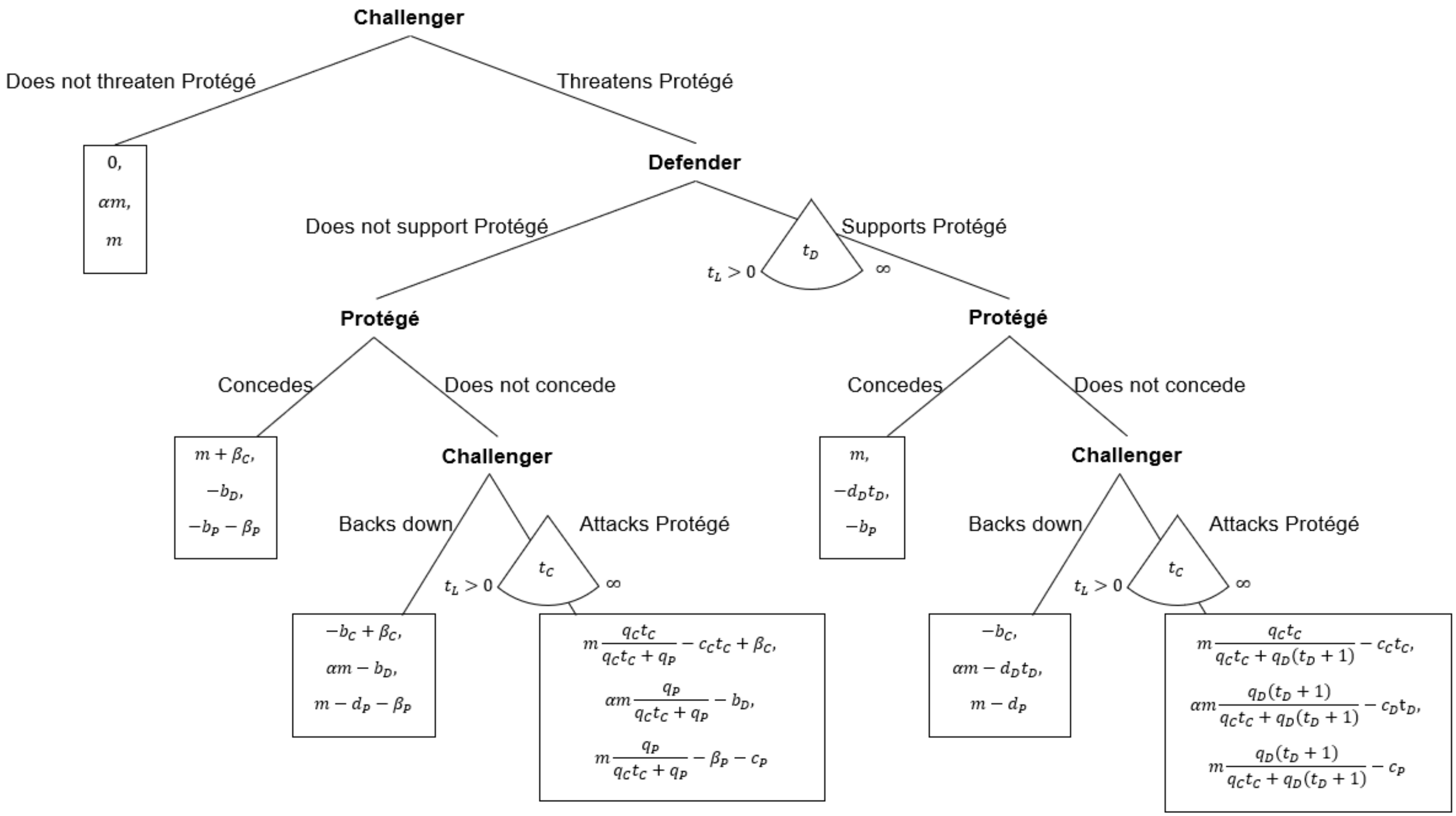
Definition 3: Defender's and Protégé's fighting force $K(t_D)$ is a combined force size times the quality of troops, where the quality of troops depends on whether Defender supports Protégé.

$$K(t_D) = \begin{cases} q_D(t_D + 1), & \text{if } t_D > 0 \\ q_P(t_D + 1) = q_P, & \text{if } t_D = 0 \end{cases}$$

Definition 4: *Challenger's relative probability of winning $w(t_C, t_D, t_P)$* is a ratio between Challenger's and Protégé's (Defender's) probability of winning. That is,

$$w(t_C, t_D, 1) = \begin{cases} \dfrac{q_C t_C}{q_D(t_D + 1)} \text{ if } t_D > 0 \\ \dfrac{q_C t_C}{q_P} \text{ if } t_D = 0 \end{cases}$$

Figure 3.1. Tripartite Game of Extended Deterrence.



Results

In this section, I will first define the conditions in which deterrence exists. Next, I will discuss whether full-scale war deterrence can survive when Challenger's economic costs of fighting decrease and how it depends on Challenger's troops' quality and political benefits of fighting. Finally, I will examine whether Challenger's technology's negative impact on large-scale war deterrence can be offset by improvements in the Defender's economic costs, mobilization cost, quality of troops, or political benefit. Specifically, I will discuss whether mirroring Challenger's technological advances ("keeping up" with Challenger) is sufficient to preserve deterrence.

Definition 5. *Full deterrence* is an equilibrium in which Challenger does not threaten Protégé. *Partial deterrence* is an equilibrium in which Challenger fights with t_L. *Deterrence failure* is an equilibrium in which Protégé concedes or Challenger attacks Protégé with $t_C^* > t_L$.

When does deterrence exist?

Let $\overline{t_D}$ be the value of t_D where Challenger is indifferent between backing down and fighting. Let t_D^0 be the value of t_D where Challenger earns 0 from fighting. Let t_D^L be the lowest value of t_D where Challenger fights with t_L.[104] Let t_D^* be Defender's optimal value of t_D conditional on Challenger threatening but not backing down (even if it was optimal for him to do so).

Proposition 1.

1. If $t_D^* < t_D^L$, then there is full deterrence if $U_D(t_D^*) \leq U_D(\overline{t_D})$ but if $U_D(t_D^*) > U_D(\overline{t_D})$, there is no deterrence.
2. If $t_D^L \leq t_D^*$, then there is partial deterrence if $t_D^* < t_D^0$ and $U_D(t_D^*) > U_D(\overline{t_D})$. Otherwise, there is full deterrence.

The proposition states that it is possible to achieve deterrence in two ways. In the first case, when Defender's optimal fighting force is such that Challenger's response would be to attack with $t_C^* > t_L$, the only way Challenger can be deterred from threatening Protégé is if Defender

[104] It might be counterintuitive that t_D^L is the *lowest* value of t_D where Challenger fights with t_L. In real life, we would expect the opposite. Namely, we would expect that low-intensity war ($t_C = t_L$) takes place only when Defender provides very little to no support. However, the proof of Lemma X in Appendix A demonstrates that Challenger's best response $t_C(t_D)$ is a concave function, where $t_C = t_L$ either when t_D is very small or very large. The latter is the case when Challenger is still motivated to fight but Defender and Protégé commit so many troops that Challenger's probability of winning is so small that he commits only t_L because suffering costs of additional troops would not be sufficiently offset by the expected benefit of fighting. Furthermore, Lemma X also shows that if $q_D(t_L + 1) > q_C t_L$, then $t_C = t_L$ only in the latter scenario.

instead commits enough troops for Challenger to back down. In other words, if Challenger threatens Protégé and the Defender's mobilization costs are so high that she is not willing to commit enough troops for Challenger to back down, then Challenger would always be better off going to war than not challenging. Therefore, the only way Challenger would not threaten is if Defender could commit enough troops for Challenger to back down, in which case he is better off not threatening in the first place.

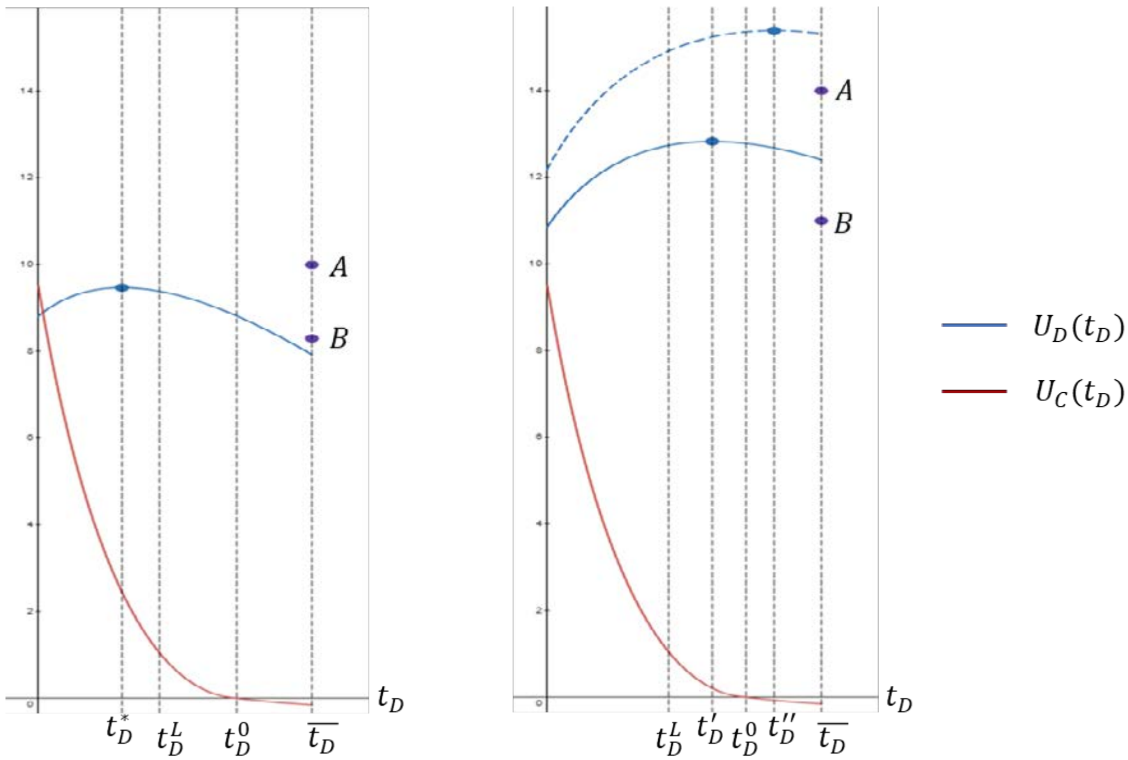
The second case is when Defender's optimal fighting force is such that Challenger's response would be to attack with t_L. In that case, there is partial deterrence if Defender would not commit enough troops for Challenger to back down and Challenger still earns a positive payoff from fighting with t_L. However, if Defender commits at least enough troops so that Challenger's payoff of fighting with t_L is negative, then there is full deterrence. Note that in this case, there are two mechanisms of full deterrence: either the Defender's optimal fighting force is less than t_D^0 but Defender is willing to commit enough troops for Challenger to back down (as in the first case) or $t_D^* > t_D^0$ in which case Challenger's fighting payoff is negative (thus, he prefers not to threaten). In the latter case, Defender does not need to commit enough troops for Challenger to back down. Instead, she just needs to commit enough troops such that if there is a war, Challenger is worse off than if he did not challenge.

These deterrence options are represented in Figure 3.2, where the horizontal axis is t_D, the blue curve represents notional $U_D(t_D)$ and the red curve represents $U_C(t_D)$ when there is war ($t_D < \overline{t_D}$) and Challenger fights with t_C^*.[105] Blue dots represent the Defender's maximum payoff of fighting (i.e., the value of $U_D(t_D)$ at t_D^*). The left plot illustrates case one, where (A, B) are Defender's two notional utility values at $\overline{t_D}$. In that case, there is full deterrence if $U_D(\overline{t_D})$ is at A (i.e., $U_D(\overline{t_D}) > U_D(t_D^*)$) and no deterrence if $U_D(\overline{t_D})$ is at B (i.e., $U_D(\overline{t_D}) < U_D(t_D^*)$).

The right plot illustrates case two for two Defender's notional utility functions (solid blue and dashed blue lines). If the utility happens to be the dashed line, the full deterrence exists regardless of the value of A and B because at $t_D^* = t_D''$ Challenger's payoff of fighting with t_L is negative, and thus, he would not threaten. If the Defender's utility happens to be the solid line and $t_D^* = t_D'$, there is partial deterrence if $U_D(\overline{t_D})$ is at B (i.e., $U_D(\overline{t_D}) < U_D(t_D')$) and full deterrence if $U_D(\overline{t_D})$ is at A (i.e., $U_D(\overline{t_D}) > U_D(t_D')$).

[105] For simplicity, the plot starts at $t_D = 0$, instead of $t_D = t_L$.

Figure 3.2. Full and Partial Deterrence (Notional Representation).



To limit the scope of analysis, I will focus on case one in the above proposition, where Defender chooses between full deterrence and no deterrence. A better understanding of when marginal technology changes might cause jumps between full and partial deterrence would also be beneficial, but second order to jumps between full and no deterrence.

Furthermore, when $t_D^* < t_D^L$ Challenger's optimal force size when he attacks Protégé is $t_C^* > t_L$. For simplicity, I will also assume that Protégé's force size is such that, should they fight without Defender's support, Challenger's optimal fighting force is $t_C^* > t_L$. Thus, for the remainder of the analysis, I will focus on the parameter space where, should the Challenger attack Protégé, he attacks with more troops than t_L. In other words, a low-intensity conflict (or high-end gray-zone aggression) is excluded from this analysis. This is an important distinction given that the literature indicates different criteria for deterring interstate war and aggression that falls below the full-scale war threshold.[106]

I will call the parameter space of analysis a *Potential Full-Scale War* (PFSW) regime. Deterrence in the PFSW regime is then *full-scale war deterrence*. The details of the PFSW regime are provided in Lemma A.3 in Appendix A. Broadly speaking, the PFSW regime occurs

[106] See: Michael J. Mazarr et al., 2018, and Michael J. Mazarr, Joe Cheravitch, Jeffrey W. Hornung, and Stephanie Pezard, *What Deters and Why: Applying a Framework to Assess Deterrence of Gray Zone Aggression,* RAND Corporation, RR-3142-A, 2021.

when the stakes for Challenger are sufficiently high relative to costs and the efficiency of Defender's troops s_D weighted by α is less than twice the Challenger's troops' efficiency s_C.

In PFSW, $\overline{t_D} > t_D^*$. That is, credible commitment to deterrence requires mobilizing more troops than the Defender's optimal fighting force. The tradeoff Defender faces when she chooses her level of support for Protégé is between costs and prospects of success. When Challenger backs down, Defender keeps with certainty whatever is at stake (thus, the probability of winning is higher than in other scenarios). However, to force Challenger to back down, Defender has to bear the costs of committing more troops than in the alternative courses of action (i.e., supporting Protégé in war or not supporting at all). And so, Defender commits enough support to ensure deterrence when the gain in expected benefit is greater than the loss from mobilization costs of this large number of troops.

The fact that $\overline{t_D} > t_D^*$ has an important real-life implication that ambiguous support can undermine deterrence. The model suggests that piecemeal, gradually growing support would not allow Challenger to adequately assess his prospects of success at the outbreak of the crisis. Meanwhile, overestimated prospects of success make war seem to be an option more appealing than not threatening in the first place.[107] This could have been, for example, the case of war in Ukraine, where Russia overestimated its prospects due to the underestimation of the support provided to Kyiv by the West.

Somewhat counterintuitively, in PFSW, Defender must commit strictly more troops than the amount required to give Challenger a negative payoff from going to war ($\overline{t_D} > t_D^0$). The reason has to do with the sequence of events. Since Challenger moves first (he threatens Protégé or not), there is a political price of backing down ($-b_C$). Therefore, *after* observing the threat, deterrence requires Defender to commit enough troops so that Challenger would rather incur this political cost of backing down instead of going to war. However, suppose the actions were reversed, i.e., Defender could commit troops *before* Challenger had the opportunity to threaten. Then Defender would only have to commit enough troops for Challenger to earn a zero payoff of going to war to achieve deterrence.

In real-world terms, this order of moves can be conceptualized as a difference between well-established alliances and less stable partnerships. Deterring aggression against a long-term ally may require less convincing the adversary of the seriousness of the commitment than deterring aggression against a partner to whom the commitment is less obvious. It can also be conceptualized as declaring support before and during a crisis. Once the crisis starts to unfold, it becomes harder and more costly to deter war because the adversary has already staked its political capital on the outcome of the crisis.

[107] In the confines of the model, Defender determines her level of support for Protégé only once, and based on that declaration, Challenger calculates his prospects of success. However, due to the anchoring bias, these observations are also valid in the real world, where countries can and do revisit their commitments.

Can full-scale war deterrence survive when Challenger's economic costs of fighting decrease, and how does it depend on Challenger's troops' quality and political benefits of fighting?

Saving soldiers' lives is one of the primary promises of the robotic revolution. Yet, many experts argue that lowering the costs of war lowers the threshold for it. This model largely confirms that intuition, but it also provides additional more nuanced insights.

Let t_C^* be Challenger's optimal fighting force. Let t_D' be Defender's optimal fighting force if there were no constraints on her fighting payoff maximization problem. Let $\widehat{K}$ be the minimum $K(t_D)$ for Protégé not to concede. Let $\widehat{t_D}$ be the minimum number of Defender's troops necessary for Protégé not to concede if $K(0) < \widehat{K}$. Let $g(t_D) = U_D(\overline{t_D}) - U_D(t_D^*)$. Then there is full deterrence when $g(t_D) > 0$.

Lemma 1. Challenger would never back down regardless of troops committed by other players when $c_C t_L < b_C$. Thus, full-scale war deterrence can exist only when $\frac{b_C}{t_L} < c_C$.

Proposition 2. Reducing Challenger's costs reduces deterrence.

$$\lim_{c_C \to \frac{b_C}{t_L}^+} g(t_D) = -\infty$$

To understand the mechanism behind Proposition 1, I will reference the following lemmas.

Lemma 2. In the PFSW regime,

1. t_C^* decreases in c_C
2. t_D' increases in c_C and q_D, and decreases in c_D and q_C
3. $\widehat{K}$ decreases in c_C, and increases in q_C
4. $\widehat{t_D}$ decreases in c_C and q_D, and increases in q_C
5. t_D^L decreases in c_C and q_D, and increases in q_C
6. $\overline{t_D}$ decreases in c_C and q_D, and increases in q_C and b_C

Lemma 3. In the PFSW regime,

1. $U_D(\overline{t_D})$ increases in c_C and q_D, and decreases in q_C, b_C, and d_D
2. If $t_D^* > 0$ then $U_D(t_D^*)$ increases in c_C and q_D, and decreases in q_C and c_D
3. If $t_D^* = 0$ and $K(0) \geq \widehat{K}$ then $U_D(t_D^*)$ increases in c_C, and decreases in q_C and b_D
4. If $t_D^* = 0$ and $K(0) < \widehat{K}$ then $U_D(t_D^*)$ decreases in b_D

The Proposition 2 states that as Challenger's unit cost of fighting approaches its lowest value where deterrence is still possible (Lemma 1), deterrence is bound to fail because Defender becomes better off supporting Protégé in war or not supporting at all. That is because as the unit cost of fighting decreases, Challenger is willing to commit more troops to fight (Lemma 2.1), and thus, the number of troops necessary for Challenger to back down, $\overline{t_D}$, increases (Lemma 2.6). When Challenger's unit cost of fighting approaches its lowest value where deterrence is still possible, $\overline{t_D}$ becomes so high that Defender's payoff of Challenger backing down becomes infinitely small.

Defender's payoffs of most alternative outcomes also decrease as Challenger's unit cost of fighting decreases (Lemma 3.2 and 3.3).[108] That is because the relative prospects of success change: the decrease in Challenger's fighting unit cost increases his optimal fighting force (Lemma 2.1), which in turn improves his prospects of success and worsens Defender's and Protégé's prospects of success. Thus, the Defender's expected benefit of fighting decreases. Still, as Challenger's unit cost of fighting approaches its lowest value where deterrence is still possible, $U_D(t_D^*)$ would not fall below $-b_D$.[109]

However, that is not to say that the decrease in c_C always makes deterrence failure more likely.

Proposition 3. $\frac{\partial g}{\partial c_C}$ is ambiguous

Corollary 1. There exist such parameter regimes where decreasing c_C leads to the emergence of full deterrence.

While the Defender's payoff of Challenger backing down and almost all alternative outcomes decrease as Challenger's unit cost of fighting decreases (Lemma 3.1–3.3), whether $U_D\left(t_D^*(c_C)\right)$ or $U_D\left(\overline{t_D}(c_C)\right)$ decreases faster, and therefore whether $\frac{\partial g}{\partial c_C}$ is positive or negative depends on what c_C was in the status quo. In some cases, reducing c_C can lead to deterrence because it makes war less attrictive for Defender and thus makes her more willing to commit enough troops to force Challenger to back down.

This intuition is presented in Figure 3.3, where the horizontal axis is c_C and two notional $g(c_C)$ are depicted by the red curves. The PFSW regime is defined for $c_C \in \left(\frac{b_C}{t_L}, D\right)$. If $g(c_C)$ happens to be as represented by the solid red line, there is full-scale war deterrence when $c_C > A$.

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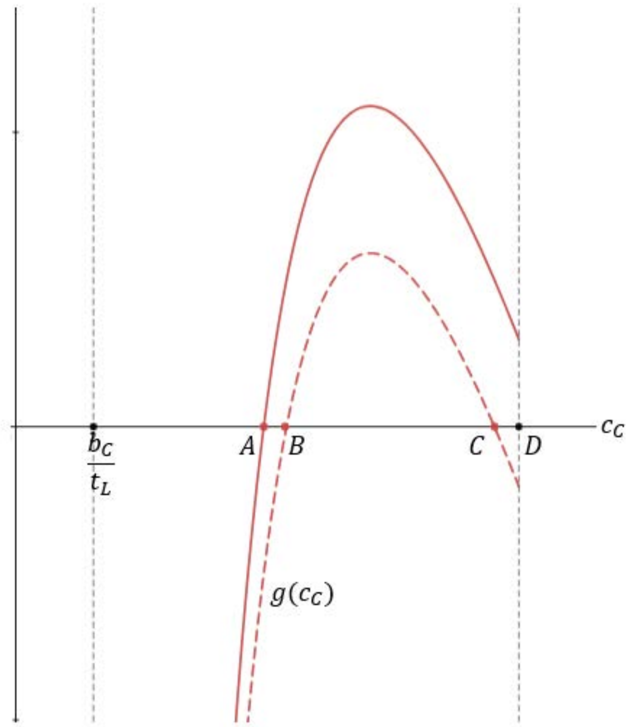
[108] Except for the case when the alternative outcome is Protégé conceding, where Defender's payoff does not depend on c_C (Lemma 2).

[109] Recall $b_D < d_D t_L$ and $U_D(\overline{t_D})$ decreases in d_D. Thus, there could be no deterrence if d_D was approaching infinity and the consideration of deterrence failure due to the decrease in c_C would be immaterial.

However, if $g(c_C)$ happens to be as represented by the dashed red line, then there is full-scale war deterrence when $c_C \in (B, C)$ and there is no deterrence when $c_C \in \left(\frac{b_C}{t_L}, B\right)$ or $c_C \in (C, D)$.[110]

Figure 3.3. Full Deterrence and No Deterrence as a Function of c_C (Notional Representation).



Numerical analysis indicates that the existence of the range (C, D) is rare because—except for specific combinations of parameters, including high c_D, q_D—usually $D < C$ (as the solid line). In other words, there are only narrow parameter regions where the decrease in Challenger's unit cost can lead to the emergence of deterrence. In those regions, there is no deterrence when Challenger's unit cost approaches the right limit because t_C^* is so low that the Defender's prospects of success in war are too high to warrant committing $\overline{t_D}$ so that Challenger backs down. As c_C decreases and t_C^* increases, the Defender's prospects of success in war decrease and thus Defender becomes better off committing $\overline{t_D}$ and forcing Challenger to back down.

In real-life terms, this phenomenon can be conceptualized as a situation where weak adversary can be largely ignored because a potential war would not take much resources. But as he gets more technologically advanced, he is treated more seriously in deterrence considerations and resources are employed to prevent an armed conflict. And finally, with even further technological progress such adversary becomes hard to contain and deter.

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[110] There also exist such parameter regimes where deterrence is impossible, i.e., where $g(c_C) < 0$ for all c_C for which the PFSW regime is defined. I exclude those trival parameter regimes from the analysis.

Proposition 4.

1. There exist parameter regimes where decreasing c_C leads to deterrence failure where Defender withdraws all the support for Protégé.
2. In a specific parameter regime, where Defender's cost of fighting is sufficiently high, there exist two values of Challenger's cost, $c_{C\_low} < c_{C\_high}$, such that Defender does not support Protégé and there is no deterrence at c_{C_high} but at c_{C_low} Defender supports Protégé with $\overline{t_D}$ and there is full deterrence.

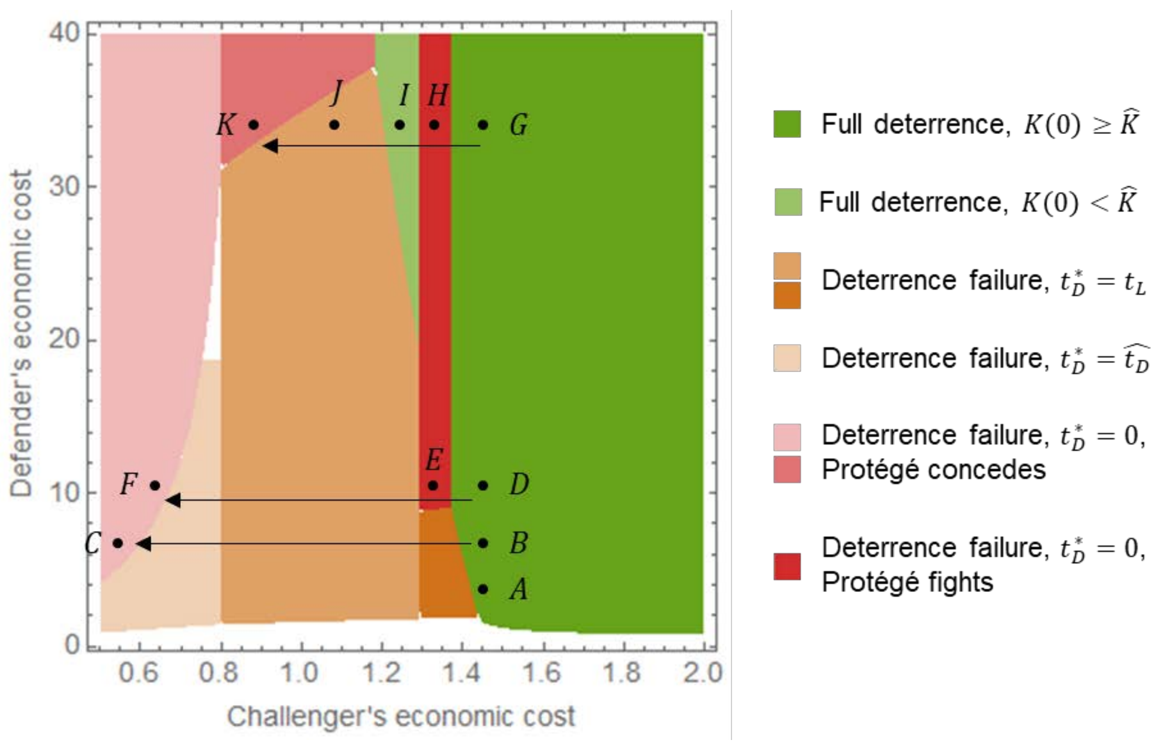
Lemmas 2 and 3 imply that the decrease in Challenger's economic cost of war might lead deterrence failure that undermines the partnership between Defender and Protégé. That is, instead of supporting Protégé with enough troops for Challenger to back down, Defender might not support Protégé at all.

Suppose that before the decrease in Challenger's costs, there was full deterrence. Once Challenger develops technology that decreases c_C, not only committing $\overline{t_D}$ might be disadvantageous but also t_D' might decrease to the point where Defender fights only with $\widehat{t_D}$, or t_L, or she is better off withholding the support altogether ($t_D^* = 0$). That is because, when Challenger's fighting cost decreases, his optimal fighting force, t_C^*, increases (Lemma 2.1), which in turn improves his prospects of success and worsens Defender's and Protégé's prospects of success. However, their prospects of success determine the expected benefit, which in turn dictates how much costs they are willing to bear, i.e., what Defender's optimal fighting force, t_D', is. Consequently, as Challenger's economic costs decrease, t_C^* increases, and t_D' decreases (Lemma 2.2). At the same time, Protégé becomes less willing to face Challenger alone, and the number of troops they require not to concede increases (Lemma 2.3–2.4). And so, as c_C decreases, not only might deterrence fail, but it might lead to a situation where Defender withholds support, and Protégé concedes. In the real world, such a situation could have broader negative consequences for a major power's standing on the international stage and the credibility of commitments to other partners.

Suppose, however, that in status quo, Defender's fighting costs are very high and Protégé was willing to fight alone. Then there could exist such a decrease in c_C that leads to the emergence of deterrence. That is because once Challenger develops technology that decreases c_C, Protégé requires Defender's support not to concede. Therefore, Defender faces different choices than in the status quo. Instead of choosing between deterrence and Protégé fighting alone, now Defender faces the risk of Protégé conceding. Consequently, it is possible that for c_{C_high} Defender was better off not supporting Protégé whereas for c_{C_low} Defender is better off committing to deterrence. The parameter regions where this can happen seems narrow, but they do exist.

These intuitions are represented in the region plot in Figure 3.4. The existence of such shift in c_C that Proposition 4 is true and necessary magnitude of that shift depends on other parameters, including Defender's economic costs. Suppose the status quo (c_C, c_D) were in point A. Then there exists no such shift in c_C that could lead to an equilibrium where Defender does not support Protégé (although I can still lead to deterrence failure and war with Defender's participation). However, if the status quo (c_C, c_D) were in point B, the shift to point C would lead to equilibrium where Protégé concedes without Defender's support, although such shift would have to be large. If the status quo (c_C, c_D) were in point D, a small shift to point E would lead to Protégé fighting alone, whereas a large shift to point F would be lead to Protégé conceding. Finally, if the status quo (c_C, c_D) were in point G then small shift (to point H) would lead to deterrence failure where Protégé fights alone, whereas slightly bigger shift (to point I) would lead to reemergence of deterrence because now Protégé requires defender's support. Even further decrease of c_C would (to point J) would lead to war were Protégé is supported by Defender but if c_C reached point K, Defender is better off not supporting Protégé even though Protégé concedes. The higher c_D, the smaller the shift in c_C to lead to the equilibrium where Defender does not support Protégé.

Figure 3.4. Deterrence as a Function of c_C and c_D for a Fixed Set of Other Parameters.



Finally, the impact of changes in c_C on the PFSW regime is also ambiguous. Lemma 2.5 implies that as c_C decreases, the PFSW space expands. In other words, as Challenger's costs decrease, some causes for conflict, previously not warranting full-scale war, now meet the threshold. At the same time, Lemma 1 implies that as c_C approaches $\frac{b_C}{t_L}$, Challenger has less incentive to back down, and thus, full deterrence becomes unlikely.

Proposition 2 implies that there always exists such a potential decrease in c_C that deterrence fails. However, this deterrence failure can be either exacerbated or prevented if technological innovation leads to changes in other Challenger parameters.

Proposition 5.

1. $\lim_{q_C \to +\infty} g(t_D) = -\infty$
2. $\frac{\partial g}{\partial q_C}$ is ambiguous
3. $\frac{\partial g}{\partial b_C} < 0$

Proposition 5.1. states that sufficiently high increase in q_C leads to deterrence failure. When the quality of Challenger's troops increases, his prospects of success in war increase, so the number of troops Defender needs to commit to deter Challenger increases (Lemma 2.6). Hence, her payoff of Challenger backing down decreases (Lemma 3.1). Even though the payoff of every alternative—except for Protégé conceding—also decreases (Lemma 3.2–3.4), the negative impact on $U_D(\overline{t_D})$ is larger as q_C grows. As q_C approaches infinity, deterrence is bound to fail because Defender would have to commit infinite troops for Challenger to back down, thus making the deterrence payoff infinitely negative. Meanwhile, $U_D(t_D^*) = -b_D$.

Similarly to the dynamics in c_C, there are also narrow regions where the increase in q_C leads to the emergence of deterrence, if the status quo q_C was very low, and thus, there was no deterrence. The mechanism is similar to the analogical phenomena in c_C. Namely, at a very low q_C, Defender's prospects of success in war are too high to warrant committing $\overline{t_D}$ so that Challenger backs down. As q_C increases, the Defender's prospects of success in war decrease, and thus, Defender becomes better off committing $\overline{t_D}$ and forcing Challenger to back down.

The impact of changes in political benefits on deterrence is more straightforward and unambiguous. The higher the political benefit of fighting, the more troops Defender has to commit to for Challenger to back down (Lemma 2.6), so Defender's payoff of that outcome decrease, and so does the value of g. In other words, at a sufficiently high b_C deterrence fails.

Let $\overline{c_C}$ be the lowest value of c_C where full deterrence exists and, therefore, at that point, $\frac{\partial g}{\partial c_C} > 0$. Let $\overline{q_C}$ be the highest value of q_C where full deterrence exists and, therefore, at that point, $\frac{\partial g}{\partial q_C} < 0$. Let $\overline{b_C}$ be the highest value of b_C where full deterrence exists. Putting aside narrow parameter regions where the decrease in Challenger's fighting cost or increase in his troops' quality leads to the emergence of deterrence, I will now focus on the case where there is full-scale war deterrence when $c_C \geq \overline{c_C}$ and $q_C \leq \overline{q_C}$, and there is no deterrence when $c_C < \overline{c_C}$ (i.e., solid red line in Figure 3.3, where $A = \overline{c_C}$) and $q_C > \overline{q_C}$. Specifically, I will focus on the case of precarious deterrence, where, in the status quo, $c_C = \overline{c_C}$, $q_C = \overline{q_C}$, and $b_C = \overline{b_C}$.

Let Δ_{j_i} be a technological shift in parameter j of player i, $j \in \{c_i, q_i, b_i, d_i\}, i \in \{C, D, P\}$. For sufficiently small Δ_{j_i}, the total change in the value of g caused by shifts in Challenger's economic costs, political benefits of fighting and the quality of his troops can be approximated by $\partial g = \frac{\partial g}{\partial c_C}\Delta_{c_C} + \frac{\partial g}{\partial q_C}\Delta_{q_C} + \frac{\partial g}{\partial b_C}\Delta_{b_C}$.

Proposition 6. Suppose in the status quo $c_C = \overline{c_C}$, $q_C = \overline{q_C}$, and $b_C = \overline{b_C}$. For sufficiently small Δ_{j_i}, if $\Delta_{c_C} < 0$ then there is deterrence when

$$\left(\Delta_{q_C} < 0 \vee \Delta_{b_C} < 0\right) \wedge \frac{\frac{\partial g}{\partial q_C}}{\frac{\partial g}{\partial c_C}}\Delta_{q_C} + \frac{\frac{\partial g}{\partial b_C}}{\frac{\partial g}{\partial c_C}}\Delta_{b_C} > -\Delta_{c_C}$$

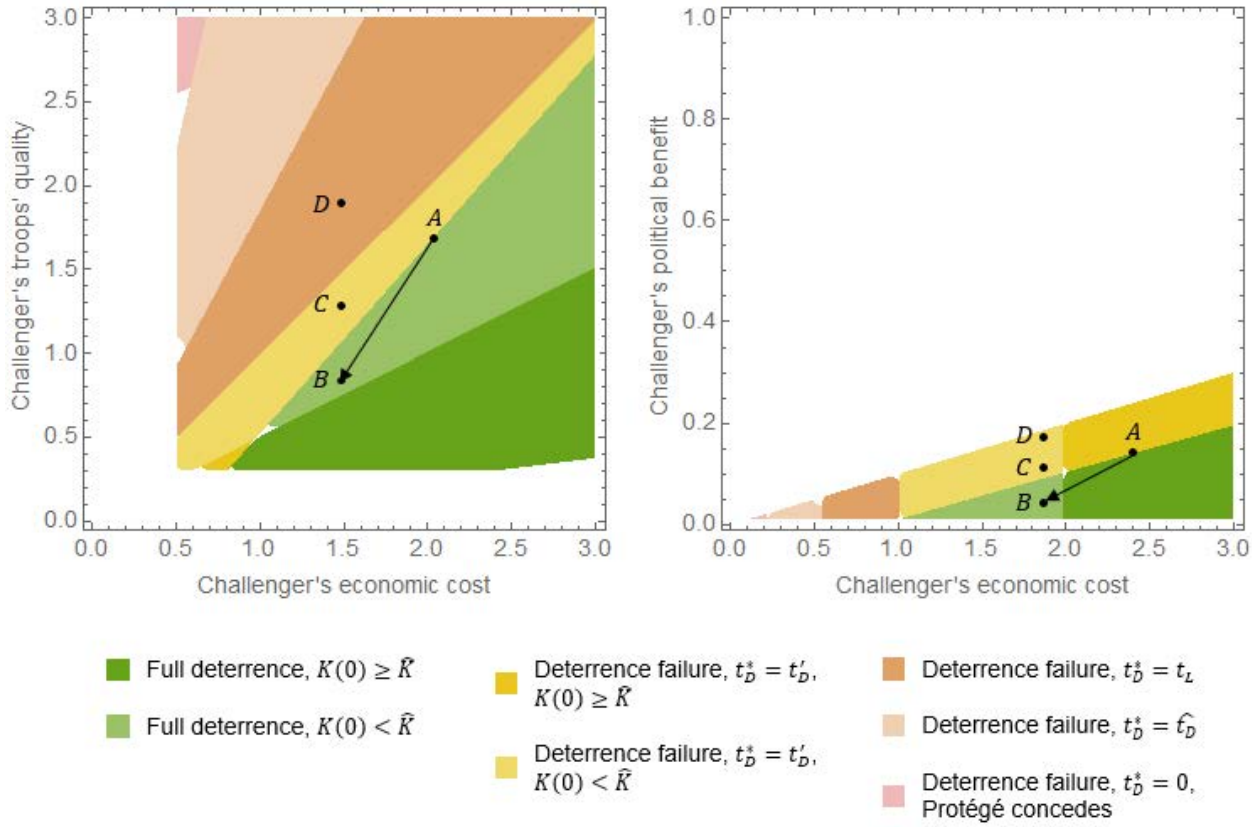
The proposition says that even if an emerging technology decreases the costs of war, deterrence can still be preserved if there is a sufficiently big deterioration in either the quality of Challenger's troops or his political benefits of fighting (or both). However, deterrence will then depend on the marginal rates of substitution between the costs and the other two parameters. In other words, even if cutting costs goes hand in hand with the degradation of Challenger's troops' quality or decrease in his political benefit of fighting, deterrence still may fail if Defender's losses from cutting costs are higher than gains from changes in the other two parameters. Deterrence failure is even more likely if the cost decrease is coupled with improvements in the other two parameters.

This intuition is represented in Figure 3.5. The left graph plots equilibria as a function of Challenger's economic costs of fighting and his troops' quality, where $A = (\overline{c_C}, \overline{q_C})$. The right graph plots equilibria as a function of Challenger's economic costs and political benefits of fighting, where $A = (\overline{c_C}, \overline{b_C})$. Thus, for clarity of presentation, each plot varies two parameters while holding all other parameters constant. In both cases, points B, C, D represent different changes in q_C and b_C, respectively, while holding the change in Challenger's economic costs constant. Point D represents the case when deterrence fails because $\Delta_{q_C} > 0$ and $\Delta_{b_C} > 0$, respectively. Point C is the case when $\Delta_{q_C} < 0$ and $\Delta_{b_C} < 0$, respectively, but deterrence fails because the marginal rate of substitution is too low. Finally, only shift to point B strengthens deterrence because

$$\frac{\frac{\partial g}{\partial q_C}}{\frac{\partial g}{\partial c_C}}\Delta_{q_C} > -\Delta_{c_C}, \frac{\frac{\partial g}{\partial b_C}}{\frac{\partial g}{\partial c_C}}\Delta_{b_C} > -\Delta_{c_C},$$

respectively.

Figure 3.5. The Impact of Changes in q_C and b_C on Deterrence.



Can Challenger's technology's negative impact on large-scale war deterrence be offset by an improvement in the Defender's economic costs, mobilization cost, quality of troops or political benefit? Is "keeping up" enough?

As discussed in the previous section, the decrease in Challenger's economic costs of fighting does not have to automatically lead to deterrence failure. Specifically, it can be offset by changes in the quality of troops or political benefits of fighting. That could be, for example, the case if the deployment of LAWS was cheaper than operations conducted with their predecessors, but LAWS' have a high error rate, and thus, the confidence in performance is low. Alternatively, there could be a strong domestic pushback or international norm against their use.

Suppose that is not the case, and the advances in robotic capabilities lead to such a change in Challenger's the quality of troops, economic costs and political benefits of fighting that deterrence failure becomes more likely. In that case, Defender can seek to preserve deterrence through technological innovation. Then the key question becomes, what should Defender prioritize in her research, development, and acquisition to incentivize deterrence and disincentivize war.

Public debate about the ramifications of innovation in military technologies and their proliferation sometimes reflects the argument that stability requires “keeping up” with the pace of adversaries’ progress. Intuitively, deterrence should be preserved when changes in Defender’s military technologies mirror Challenger’s technological progress. Suppose the worst-case scenario of precarious deterrence in the status quo (i.e., $c_C = \bar{c_C}$, $q_C = \overline{q_C}$, and $b_C = \overline{b_C}$) while Challenger’s technological innovation is such that $\Delta_{c_C} < 0, \Delta_{q_C} > 0, \Delta_{b_C} > 0$. According to that argument, deterrence should be preserved if $\Delta_{c_D} < 0, \Delta_{q_D} > 0, \Delta_{b_D} > 0$. But is that really the case?

Proposition 7.

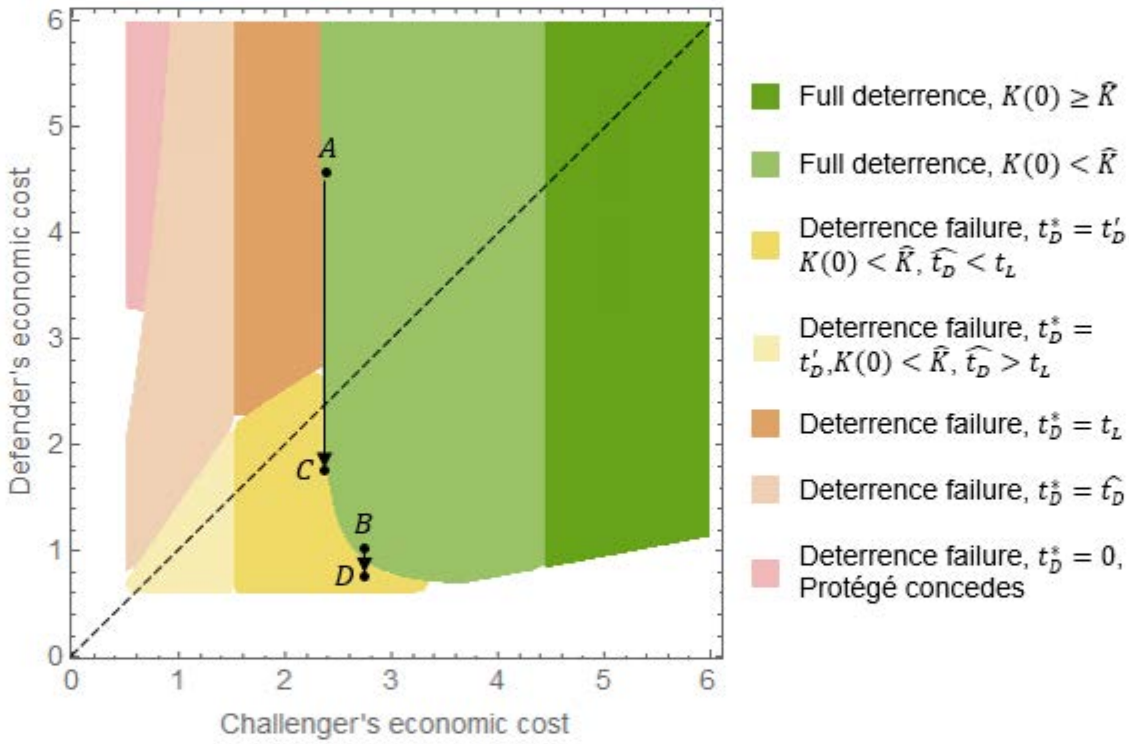
1. Suppose in the status quo, $t_D^* > 0$. Then $\frac{\partial g}{\partial c_D} > 0$ and $\frac{\partial g}{\partial b_D} = 0$.
2. Suppose in the status quo, $t_D^* = 0$. Then $\frac{\partial g}{\partial c_D} = 0$ and $\frac{\partial g}{\partial b_D} > 0$.

The proposition states that lowering the Defender’s costs of fighting cannot strengthen the deterrence; it can only weaken it or have no impact at best. That is because as the economic costs of fighting decrease, the payoff of fighting increases, while the payoff of Challenger backing down remains unchanged. Thus, war becomes a more appealing option. At the same time, numerical analysis indicates that it might be very difficult to predict just how bad the impact of lowering c_D might be. Figure 3.6 exemplifies that challenge. Specifically, if the status quo $(c_C, c_D) = A$ then deterrence will only fail when the decrease in the Defender’s costs of fighting is large (shift to C). However, if the status quo $(c_C, c_D) = B$ then even a very small decrease of c_D can lead to deterrence failure (shift to D).

A corollary of Proposition 6 is that even if emerging technology does not disturb the relative efficiency of Defender’s troops (i.e., Defender keeps up with Challenger’s technological progress so that r_D remains constant), the decrease in economic costs would ultimately lead to deterrence failure. The diagonal in Figure 3.6. represents a special case when not only r_D remains constant, but also Challenger and Defender have the same economic costs and quality of troops ($c_C = c_D, q_C = q_D$). Yet, as the diagonal approaches 0, deterrence fails.

On the other hand, improving Defender’s political benefits of fighting is indeed beneficial for deterrence, which might be counterintuitive at first. One could expect that the higher the political benefits of war, the more incentivized players would be to fight. However, that is not the case because the political benefits of fighting are modeled as a penalty for withdrawing from war, i.e., it is the opportunity cost imposed on Defender when she chooses not to support Protégé. Consequently, as b_D increases, Defender becomes better off supporting Protégé. Thus, in parameter regimes where she chooses between not supporting Protégé and supporting with the number of troops sufficient for Challenger to back down, Defender now chooses the latter.

Figure 3.6. Deterrence as a Function of c_C and c_D for a Fixed Set of Other Parameters.



Proposition 8.

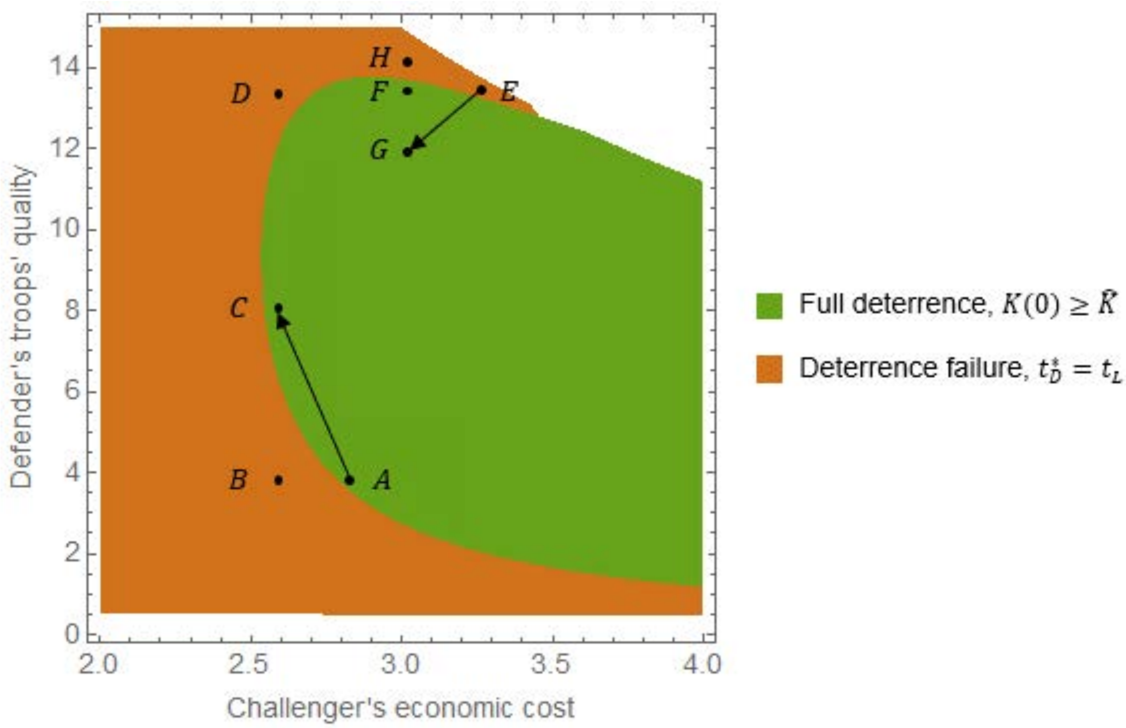
1. $\lim_{q_D \to 0} g(t_D) = -\infty$
2. $\frac{\partial g}{\partial q_D}$ is ambiguous
3. $-\frac{q_D}{q_C}\frac{\partial g}{\partial q_D} = \frac{\partial g}{\partial q_C}$, unless $t_D^* = 0$ and $K(0) \geq \hat{K}$. For any change in q_C and q_D, changes in the status quo depend on the ratio $\frac{q_D}{q_C}$.

Defender is incentivized to commit enough troops to deter Challenger when the gain in expected benefit is greater than the loss from additional costs. The gain in expected benefit is a function of the quality of troops. Thus, improvement in that area can help preserve deterrence. But the Proposition 8.3. states that improvements in Challenger's troops' quality Δ_{q_C} are neutral for deterrence if Defender improves her troops' quality so that the ratio between q_D and q_C remains constant. This implies that if the quality of Defender's troops was higher than Challenger's to begin with, then to preserve deterrence, she needs to improve the said quality faster than Challenger. More research is needed to examine whether this finding is robust to different ways of aggregating quality and quantity to contest success functions.

Furthermore, because of those narrow parameter regions where changes in troops quality has a non-monotonic effect on deterrence, there is no guarantee that improving troops quality will always remedy decreasing Challenger's costs of fighting. The region plot in Figure 3.7. shows an interesting case of those peculiar non-monotonic effects in both Challenger's costs and Defender's troops' quality. Suppose in the status quo $(c_C, q_D) = A$. Point B represents a shift in Challenger's costs that, holding Defender's quality constant, would lead to deterrence failure. However, if Defender develops own technology that shifts the quality of her troops to point C, deterrence is preserved. Yet, if she improves her technology up to point D, deterrence fails.

Alternatively, suppose the status quo $(c_C, q_D) = E$ where there is no deterrence. Point F represents a shift in Challenger's costs that, holding Defender's quality constant, would lead to the emergence of deterrence. However, suppose Defender develops own technology that shifts the quality of her troops to point H instead. In that case, deterrence does not emerge, and only degrading own quality to point G can strengthen deterrence.

Figure 3.7. Deterrence as a Function of c_C and q_D for a Fixed Set of Other Parameters.



These paradoxical effects in the narrow regions where deterrence is unstable speak to the fact that Challenger's costs and players' quality of troops have qualitatively the same effect on Defender's payoff of fighting and Challenger backing down. Both those payoffs decrease as Challenger's costs of Defender's troops' quality decrease or Challenger's troops' quality increases.

The non-monotonic effects in Challenger costs take effect in particular when Defender's costs are very high, and the only reason why she is supporting Protégé is that the quality of her troops is also high relative to Challenger's costs and quality of troops. However, in war, she is only willing to provide Protégé with the minimum number of troops still considered a support. When Challenger's costs are low (point D in Figure 3.7), the number of troops necessary for him to back down is too high for Defender; thus, they fight. When the costs are higher (point F in Figure 3.7), $\overline{t_D}$ is lower, and so, since Defender is not eager to fight in the first place, she is willing to commit to deterrence. But when the costs are at point E (Figure 3.7), now Challenger's optimal response is barely above t_L, so Defender is again better off fighting.

The non-monotonic effect of Defender's troops' quality works similarly. At point B, Defender's Defender would have to commit to deterrence $\overline{t_D}$ so high that she is better off fighting. Then at point C, she is at the right spot, where $\overline{t_D}$ is acceptable, while Challenger's fighting force is still relatively high, so Defender is better off committing to deterrence. But if the quality of her troops is very high (point D), Challenger's optimal fighting force is low, and Defender's prospects of success in war make committing to deterrence unwarranted.

Noteworthy, these peculiar effects require very specific conditions (i.e., they happen only at specific values of all other parameters), which makes them highly unlikely to occur in the real world. That is especially the case with the non-monotonic effects of changes in Challenger's economic costs, which seem to occur close to the low-intensity conflict threshold. More research, including simulations, might be useful to understand these dynamics better.

Finally, we will turn to Defender's mobilization costs.

Proposition 9.

$$\frac{\partial g}{\partial d_D} < 0$$

Since mobilization cost affects Defender's payoff only when Challenger backs down, their impact on deterrence is unambiguous: decreasing mobilization costs always incentivizes committing more troops to deterrence, and thus, it strengthens deterrence.

Conclusions

The model yields several interesting results. First, I found that there are two deterrence mechanisms. Defender could be maximizing her payoff with such force that Challenger responds with an attack on Protégé that yields him a negative payoff. That, however, can only happen in low-intensity war scenarios. The second deterrence mechanism is for Defender to commit enough troops that the Challenger's best response is to back down. Importantly, the number of Defender's troops necessary for Challenger to back down is higher than her optimal fighting force. This implies that halfhearted or piecemeal, gradually growing support leads to deterrence failure because Challenger overestimates his prospects of success. This is an important lesson for cases like Ukraine or Taiwan, where the United States is wary of committing substantial support up front not to aggravate a potential crisis. Yet, the model suggests that overcommitting might, in fact, be a better choice.

The second finding pertains to the impact on deterrence of decreasing Challenge's costs of fighting. Unfortunately, regardless of some very specific—and thus, in the real world, unlikely—cases where the decrease in costs leads to the emergence of deterrence, overall, the model confirms the intuition that as the costs of war decrease, deterrence is likely to fail. However, this can still be averted if the technology advances in a way that at least one other parameter deteriorates. Potentially, this could be an area of Defender's intervention, although finding the deterrence-preserving rate of change might be difficult.

If that is not possible, the model suggests that deterrence can be aided by changes in the Defender's technological parameters. However, mirroring Challenger's technological progress might not be beneficial. In particular, the model suggests that lowering the Defender's costs of fighting cannot strengthen deterrence; it can only weaken it or have no impact at best. On the other hand, lowering the mobilization costs has an unambiguously positive impact on deterrence. This implies that it might be beneficial to simultaneously pursue two different paths for the development of robotic capabilities. One focused on cheap capabilities for prepositioning in third countries (cheap mobilization capabilities), and another focused on high-quality, more expensive capabilities for warfighting when the Defender joins a war (expensive but high-quality fighting capabilities).

Relatedly, the model suggests that, in most cases, investing in technologies that can be force multipliers pays off. Specifically, if the ratio between the Defender's and Challenger's troops' quality remains constant, the net effect of the improvements in the Challenger's troops' quality will be zero. At the same time, this implies that if the quality of Defender's troops was higher than Challenger's to begin with, then she needs to improve faster than Challenger to preserve deterrence. Thus, it might be advantageous to focus efforts and investment in this area instead of decreasing the costs.

Finally, improving Defender's political benefits of fighting is also beneficial for deterrence because it incentivizes Defender to support Protégé. This finding highlights the need to avoid backlash against LAWS. If new capabilities were perceived as more ethical than their predecessors, they could have a positive impact on deterrence.

Chapter 4. The Impact of Robotic Warfare on Public Support for War: Online Survey Experiment on the Russian Public

In the past, Russian political and military leaders have expressed the belief that the future of war is unmanned and increasingly autonomous, and therefore, robotic capabilities should be one of the priorities of military investment.[111] Military planning documents reportedly set the target for military robotization—as the program is called in Russian-language military literature—to 30 percent of robotic platforms' share in total weapons structure.[112] Many of the tasks foreseen for those robotic capabilities require high levels of autonomy.[113]

Russia's war against Ukraine with subsequent sanctions, depletion of legacy military equipment and weapons, and brain drain undermine Russia's ability to deliver on its robotization vision. Yet, it does not invalidate the motivation standing behind it. Given the below-expectations performance of Russian troops,[114] Moscow will likely still seek efficient force multipliers that can decrease reliance on manpower. Therefore, the robotization timelines will likely be delayed, but there is no indication that Russia will abandon the quest for the robotic revolution altogether.

Despite common concern that Russia might be less constrained by a pushback against LAWS, as of June 2023, there has been no attempt to understand the impact of this revolution on the Russian public's support for military operations, even though the literature discussed in Chapter Two suggests that authoritarian countries are not impervious to political costs of wars. Meanwhile, there have been several studies on the U.S. public's support for drone operations, as well as studies of public opinion of LAWS.[115] This study aims to bridge this gap. Specifically, it seeks to understand whether there will be pushback against LAWS or, conversely, whether the support for military operations will increase when the public expects fewer soldiers to be sent to combat. To that end, the study uses an experimental design to measure the impact of technology on the public support for Russian military interventions on neighbor's territory. Given the

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[111] Vladimir Putin, "Extended meeting of the collegium of the Ministry of Defense of the Russian Federation [Rasshirennoe zasedanie kollegii Ministerstva oborony Rossiiskoi Federatsii]," *Army Collection* [*Armeiskii sbornik*], No. 1, 2021, p. 8; Valerii Gerasimov, "The value of' science is in foresight [Tsennost' nauki v predvidenii]," *Military-Industrial Courier* [*Voenno-promyshlennyi kur'er*], No. 8, 2013, p. 1.

[112] Vladislav Gordeev, "The Ministry of Defense promised to bring the share of robots in the armament structure to 30% [Minoborony poobeshchalo dovesti dolyu robotov v strukture vooruzhenii do 30%]," *RBK*, November 6, 2014.

[113] Marcinek and Han, 2023, p. vi.

[114] Dara Massicot, "Russia's Repeat Failures," *Foreign Affairs*, August 15, 2022

[115] Horowitz, Scharre, and FitzGerald, 2017; Morgan et al, 2020; Walsh and Schulzke, 2018; Michael C. Horowitz, "Public opinion and the politics of the killer robots debate," *Research and Politics*, Vol. 3, No. 1, 2016, pp: 1–8.

disparities between the U.S. and India public support in the same study settings,[116] I also measure the impact of strategic context on the support for military operations.

Furthermore, the study seeks to understand the factors underlying the support, or lack thereof. Where other studies only implied the reasons for support from treatment scenarios, I offer potential reasons for supporting the proposed operation based on the framework adopted for this study. This is an important distinction between the framework and the model, where net political benefits, informed by the public's views on military operations, are an independent factor/parameter. However, as the literature discussed in Chapter Two suggests, those views are influenced by the stakes of war, casualties on both sides, costs, and prospects of success. I posit here that they might also be influenced by some additional moral considerations. Therefore, I propose seven decision factors and elicit participants' views on the proposed operation through the lenses of those factors.

Survey Design

To measure the effect of robotic warfare on the Russian public's support for military operations and examine whether this effect depends on the strategic context, I used a two-by-two factorial design. After signing a consent form and responding to a few screening questions (for Survey Instrument, see Appendix B), the participants will be randomly assigned to one of four scenarios. Technology was signified by the type of unit conducting a proposed operation. Half of the sample was randomly assigned a scenario with airborne forces and the other half with a robotic unit. The strategic context was signified by adversaries' membership in NATO. For half of the sample, an adversary would be a NATO member, while for the other half, not a NATO member (Figure 4.1).

Figure 4.2. Factorial Design.

		Technology	
		Airborne Forces	**Robotic Unit**
Strategic Context	**NATO**	NATO, Airborne	NATO, Robot
	Non-NATO	Non-NATO, Airborne	Non-NATO, Robot

The scenario read as follows (with the treatment in bold only highlighted here for presentation purposes):

[116] Horowitz, Scharre, and FitzGerald, 2017.

> "Russian minority living in country X is in danger of ethnic cleansing. Country X is Russia's neighbor and **is/is not** a member of NATO. The Russian government is considering a limited armed intervention to protect the minority. The plan calls for a deployment of **airborne forces/ a robotic unit comprising ground and aerial autonomous robots with minimal support from military personnel**."[117]

After presenting the scenario, the participants were asked whether they supported the proposal. Next, they were asked how they viewed the operation in terms of motivation, prospects of success, international risks, civilian casualties, casualties among Russian soldiers, costs of operation, and its moral acceptability. Specifically, the participants were presented with seven factor statements in positive framing, as presented in Figure 4.2. The order of factor statements was randomized.

Figure 4.2. Elicitation of Views on the Hypothetical Plan.

Do you agree or disagree with the following statements when you think about the government's hypothetical plan?

	Strongly agree	Somewhat agree	Somewhat disagree	Strongly disagree
I believe that the safety of the Russian minority is a very important issue, and it would call for military intervention.	○	○	○	○
I believe the intervention would be successful.	○	○	○	○
I believe that the international risks associated with this intervention would be low. I would not expect any major negative consequences.	○	○	○	○
I believe there would be few casualties among civilians.	○	○	○	○
I believe there would be few casualties among our soldiers.	○	○	○	○
I believe that the costs associated with the intervention would be low.	○	○	○	○
I believe that such an operation would be morally acceptable.	○	○	○	○

Next, the participants were asked whether these factors spoke in favor or against the proposed intervention. Specifically, if a participant agreed with a statement in the positive framing (for example, "I believe that the costs associated with the intervention would be low"), they were asked whether a given factor spoke in favor of the hypothetical government's plan. If the participant did not agree with the statement, then the statement was reframed as negative (e.g., "I believe that the costs associated with the intervention would be high"), and the participant was asked whether the factor spoke against the government's plan (Figure 4.3.; for details, see Appendix B).

117 The survey was conducted in Russian; throughout this chapter I use English translations of the instrument.

Figure 3.3. Example of the Elicitation of Importance of Decision Factors

What, in your opinion, speaks in favor of the government's hypothetical plan?

	Does not speak in favor	Speaks somewhat in favor	Speaks strongly in favor
I believe the intervention would be successful.	○	○	○

What, in your opinion, speaks against the government's hypothetical plan?

	Does not speak against	Speaks somewhat against	Speaks strongly against
I do not believe the intervention would be successful.	○	○	○

This design accounted for a situation when, for example, someone believed that the proposed intervention would be successful but did not see it as an argument in favor of the hypothetical plan. Importantly, the design rests on two logical assumptions. First, I assumed that positive framing is not an argument against the hypothetical plan and vice versa. That is, if a respondent thinks that, for example, the number of casualties among Russian soldiers would be low, this could be an argument in favor of the hypothetical plan or not be an argument at all, but it cannot be an argument against it. Second, I assumed that if a respondent disagrees with a statement in positive framing (e.g., "the costs of the intervention would be low"), they would agree with the opposite (in this case, "the costs of the intervention would be high").

The questionnaire also included an open-ended question asking about any other contributing factors. Next, the participants were asked whether their views had changed and how, and the support question was repeated. This design aimed at a more informed response to the support question. It also allowed screening for inconsistency between the first and second support elicitation, given the response to the question about the change in views. The survey concluded with several demographic questions. The survey was conducted in Russian.

Data Collection

The survey was active between December 14 and December 21, 2021, that is, two months before Russia's full-scale invasion of Ukraine launched on February 24, 2022. The Russian public was alerted to the escalation of the situation in eastern Ukraine, but the majority did not believe that the tensions would evolve into a fully-fledged war.[118] This might have eased the

[118] Levada-Center, "Escalation in Donbas [Obostrenie v Donbasse]," webpage, December 14, 2021. Admittedly, that situation persisted until the very outbreak of the war. See: Levada-Center, "Ukraine and Donbas," webpage, March 4, 2022.

cognitive load on the respondents to think about potential military operations. At the same time, it might have primed the participants to consider Ukraine specifically as "country X," which will be discussed further in the Findings section.

To minimize potential risks for the participants, I used a data collection method that guaranteed no party had access to both the personal data needed for the payment and the responses. The online panel provider (TGM Research) responsible for the payment to participants distributed a link to the survey hosted on a separate online platform (Alchemer Survey). The survey platform supports integration with online panels through automatic redirects when certain conditions are met. Thanks to that functionality, TGM Research received information when the survey was complete without having access to responses.

This study was approved by RAND's Human Subjects Protection Committee (HSPC). Per HSPC's request, I paid for all submitted responses, regardless of the number of questions skipped. The cost of each completed survey was $1.40.

Sample

Due to cost and feasibility considerations, I used a convenience sample. The participants were recruited through a paid online panel maintained by TGM Research.[119] I collected N=2,031 complete, internally consistent responses with a response time of at least 2 minutes. Pearson's chi-squared tests did not reveal statistically significant differences in drop-out and exclusion rates between the interventions. The number of drop-outs and responses excluded in each study group are presented in Table 4.1. The study was sufficiently powered to detect the smallest meaningful effect of a ten percentage points difference in support at $\alpha=0.05$.

Table 4.1. Drop-out and Exclusion Rates.

Intervention	Assigned Intervention	Did not answer the second support question	Disqualified for inconsistency	Disqualified on response time below two minutes	Sample
NATO, Airborne	581	20	32	19	510
NATO, Robot	570	32	28	15	495
Non-NATO, Airborne	611	19	31	20	541
Non-NATO, Robot	548	26	28	9	485
Total	**2,310**	**97**	**119**	**63**	**2,031**

The participants were recruited using gender population quotas, and age quotas shifted toward younger age groups. Recruiting a relatively young sample was motivated by the fact that

[119] Initially, the panel was supposed to be provided by a Russian company. However, the company ultimately canceled the project (along with other politically charged surveys), invoking "political situation in Russia."

the scenario calling for the deployment of a robotic unit is futuristic in nature. Therefore, I shifted the sample towards younger age groups to elicit the opinion of those more likely to be politically active when the technology becomes operational.

Pearson's chi-squared tests again did not reveal any statistically significant differences in demographic characteristics between the interventions, confirming the random assignment. Consequently, demographic characteristics will be presented for the entire sample.

The sample differs significantly from the general population on several demographic characteristics, most notably education (for a detailed comparison with the 2020 Census, see Table 4.2). 58 percent of the sample are college graduates, more than double the Russian public. There are fewer widowed, divorced, or separated people in the sample than among the general population, which might be the effect of shifting the sample towards younger age groups. It appears that more people are working and fewer have kids than among the general population; however, the comparison is difficult due to different measurements.[120] The sample is also skewed towards Central and Northwestern Federal Districts, with the capital in Moscow and Saint Petersburg, respectively. Finally, the sample only includes Internet users who signed up for the online panel.

Table 2.2. Demographic Characteristics of the Sample

	N	**Sample Percentage**	**Population Percentage (2020 Census)**
Gender	**2,031**		
Men	915	45.05	46.5
Women	1,116	54.95	53.5
Age	**2,031**		
18-29	464	22.85	15.76
30-39	47	23.49	21.09
40-49	386	19.01	18.47
50-59	366	18.02	15.80
60 or more	338	16.64	28.87
Education	**2,026**		(Out of those who responded)
No high school diploma	12	0.59	9.83
High school graduate	128	6.32	18.88
Professional school graduate	569	28.08	41.05
Some college education	136	6.71	2.46
Collage graduate	1,181	58.29	26.39
Employment Status	**2,025**		(Out of those who responded)
Employed/Self-employed/Serving	1,367	67.51	62.26
Retired	305	15.06	(No detailed data on non-employed)
Unemployed	92	4.54	
Student	77	3.80	
Housewife	121	5.98	
Other	63	3.11	

120 The census uses different work status categories. The census contains childbearing data for women and the number of children in a household.

	N	Sample Percentage	Population Percentage (2020 Census)
Military Family	**2,026**		No data
Yes	238	11.75	
No	1,788	88.25	
Marital Status	**2,026**		(Out of those who responded)
Married	1,179	58.19	53.95
In a relationship	204	10.07	5.35
Widowed	79	3.90	11.57
Divorced/Separated	196	9.67	12.05
Single	368	18.16	17.09
Having Children	**2,026**		(Out of those who responded)
Yes	1,490	73.54	83.38
No	536	26.46	16.62
Federal District	**2,025**		
Central	667	32.94	27.40
Northwestern	257	12.69	9.46
Southern	173	8.54	11.38
North Caucasian	45	2.22	6.91
Volga	398	19.65	19.66
Ural	192	9.48	8.36
Siberia	241	11.90	11.41
Far Eastern	52	2.57	5.42

SOURCE: For the 2020 Census data: Rosstat Federal State Statistics Service, “All-Russian population census 2020 [Vserossiiskaya perepis’ naseleniya 2020 goda],” webpage, undated.

Using a non-representative sample limits the generalizability of the results; however, several studies show that online convenience samples can obtain similar treatment effects as nationally representative population-based samples. One study specifically compared the average treatment effects of 20 survey experiments between Mechanical Turk (MTurk) convenience samples and population-based probability samples to examine the generalizability of survey experiments. The authors found that 15 out of 20 experiments produced the same inferences with respect to the direction of the treatment effect and statistical significance, whereas the remaining five diverged on the significance of the results. None of them diverged in the direction of a significant effect.[121] A comparison of treatment effects between an MTurk convenience sample and a YouGov representative sample in an experiment measuring public support for drone strikes found that the two samples produced “qualitatively similar results.”[122]

[121] Kevin J. Mullinix, Thomas J. Leeper, James N. Druckman and Jeremy Freese, “The Generalizability of Survey Experiments,” *Journal of Experimental Political Science*, Vol. 2, No. 2, 2015, pp: 109–138.

[122] James Igoe Walsh, and Marcus Schulzke, *Drones and Support for the Use of Force*, E-book, Ann Arbor, MI: University of Michigan Press, 2018, p. 23.

Findings

Treatment Effects

The experiment yielded several interesting results. First, Pearson's chi-squared test has shown that the difference in support for the hypothetical military operation between the interventions is not statistically significant at $\alpha = 0.05$ (Table 4.3).[123] Specifically, there is 36.5 percent that the null hypothesis is true (i.e., there is no difference in support between the interventions). It is even higher (41.4 percent) when the support for the hypothetical military operation is binarized (Table 4.4).[124] Overall, 60.71 percent of participants did not support the proposed operation, and 39.29 percent supported it (Table 4.4). Only 8.86 percent strongly supported it, while 27.13 percent strongly opposed it (Table 4.3). Scenarios calling for the deployment of a robotic unit seem to have a lower proportion of those who strongly opposed it, but that effect is not statistically significant.

Table 4.3. Effects of Technology and Strategic Context.

Intervention	Strongly oppose	Somewhat oppose	Somewhat support	Strongly support	χ^2 statistic and p-value
NATO, Airborne (n=510)	28.63%	32.35%	30.39%	8.63%	χ^2 = 9.8236 p-value = 0.365
NATO, Robots (n=495)	25.66%	33.94%	32.73%	7.68%	
Non-NATO, Airborne (n=541)	30.13%	33.27%	28.10%	8.50%	
Non-NATO, Robots (n=485)	23.71%	34.85%	30.72%	10.72%	
NATO (n=1,005)	27.16%	33.13%	31.54%	8.16%	χ^2 = 2.0403 p-value = 0.564
Non-NATO (n=1,026)	27.10%	34.02%	29.34%	9.55%	
Airborne (n=1,051)	29.40%	32.83%	29.21%	8.56%	χ^2 = 5.7918 p-value = 0.122
Robots (n=980)	24.69%	34.39%	31.73%	9.18%	
Whole sample (N=2,031)	**27.13%**	**33.58%**	**30.43%**	**8.86%**	

Table 4.4. Effects of Technology and Strategic Context (Binarized Outcome).

Intervention	Oppose	Support	χ^2 statistic and p-value
NATO, Airborne (n=510)	60.98%	39.02%	χ^2 = 2.8585 p-value = 0.414
NATO, Robots (n=495)	59.60%	40.40%	
Non-NATO, Airborne (n=541)	63.40%	36.60%	

[123] All the statistical analysis was conducted in STATA17.

[124] The results are presented for Q10 (see Appendix B), under the assumption that the series of questions about factors influencing the decision led to more informed and thoughtful responses. The results for Q4 are qualitatively the same, i.e., the difference is not statistically significant at $\alpha = 0.05$, with even higher probability that the null hypothesis is true.

Intervention	Oppose	Support	χ^2 statistic and p-value
Non-NATO, Robots (n=485)	58.56%	41.44%	
NATO (n=1,005)	60.30%	38.89%	χ^2 = 0.1405 p-value = 0.708
Non-NATO (n=1,026)	61.11%	39.70%	
Airborne (n=1,051)	62.23%	37.77%	χ^2 = 2.1026 p-value = 0.147
Robots (n=980)	59.08%	40.92%	
Whole sample (N=2,031)	60.71%	39.29%	

The effect of technology seems to be slightly bigger than the effect of the strategic context, but it is not statistically significant at $\alpha = 0.05$. The difference in support appears to be the biggest on technology among non-NATO countries: the support for the hypothetical operation was 4.84 percentage points higher when the proposal called for the deployment of a robotic unit. However, the difference was still well below ten percentage points and not statistically significant (Table 4.5).

Table 4.5. Effects of Technology in Non-NATO Sample.

Intervention	Oppose	Support	χ^2 statistic and p-value
Non-NATO, Airborne (n=541)	63.40%	36.60%	χ^2 = 2.5254 p-value = 0.112
Non-NATO, Robots (n=485)	58.56%	41.44%	
Non-NATO (N=1,026)	61.11%	39.70%	

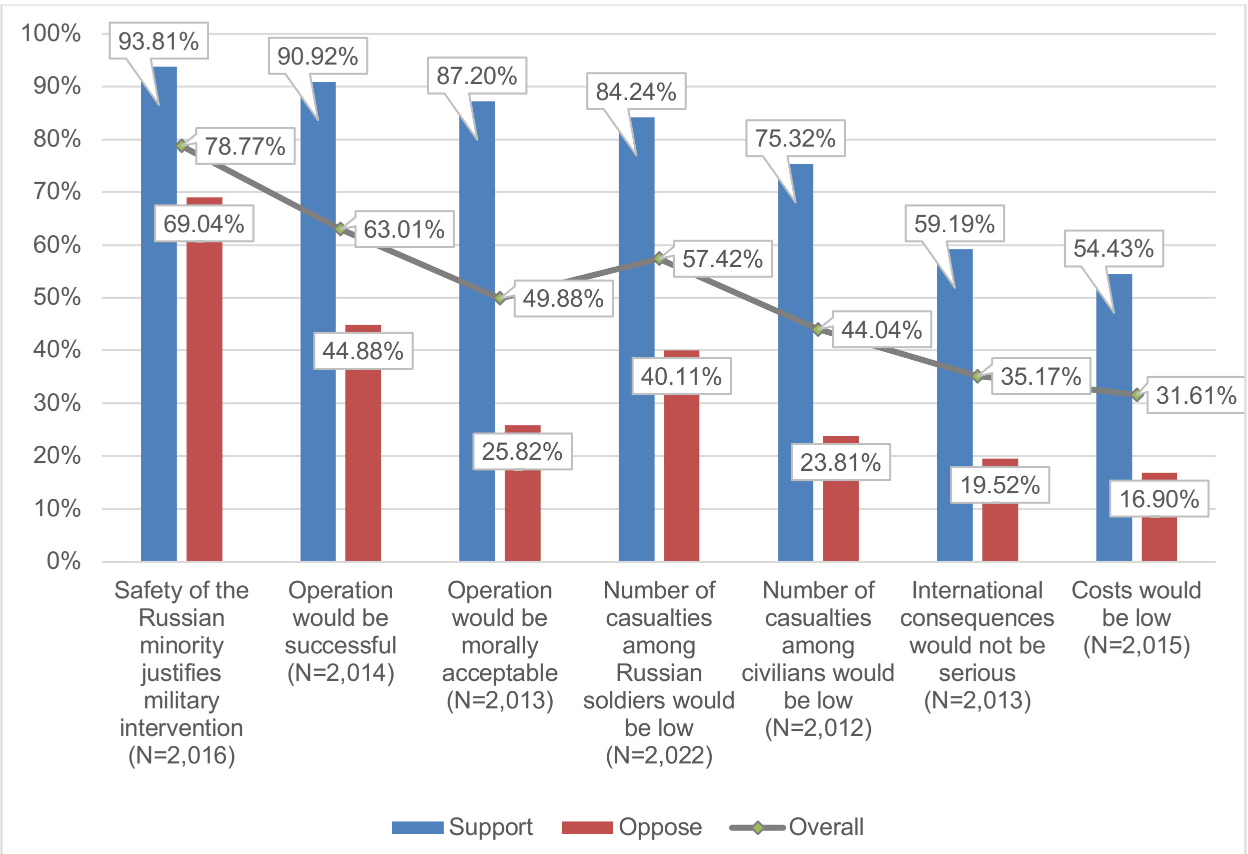
Therefore, this result differs from similar studies conducted on the American public measuring the effects of drones on the support for military operations. There might be several reasons for that, which I will discuss at the end of this chapter.

Decision Factors

While treatments did not affect the outcome, those who supported the proposed operation viewed it in a much better light than those who opposed it (Figure 4.4). For example, among those who supported the operation (represented by the red bars below), 87 percent strongly agreed or somewhat agreed with the statement that the operations would be morally acceptable. Only 26 percent of the opponents of the operation (blue bar below) shared that view. 91 percent of supporters believed that the operation would be successful compared to only 45 percent of those who opposed the operation. The comparison is presented in Figure 4.4.

Overall, the prevailing views among the sample were that the safety of the Russian minority is a very important issue that warrants a military intervention, the proposed operation would be successful, and the number of casualties among Russian soldiers would be low (regardless of the type of unit deployed). On the other hand, most participants did not agree with the statements that the number of civilian casualties would be low, that costs would be low, or that the international consequences would not be serious. The sample was almost evenly split on the issue of the morality of the proposed military operation. Yet, despite all the positive beliefs about the proposed operation, approximately 60 percent oppose the plan.

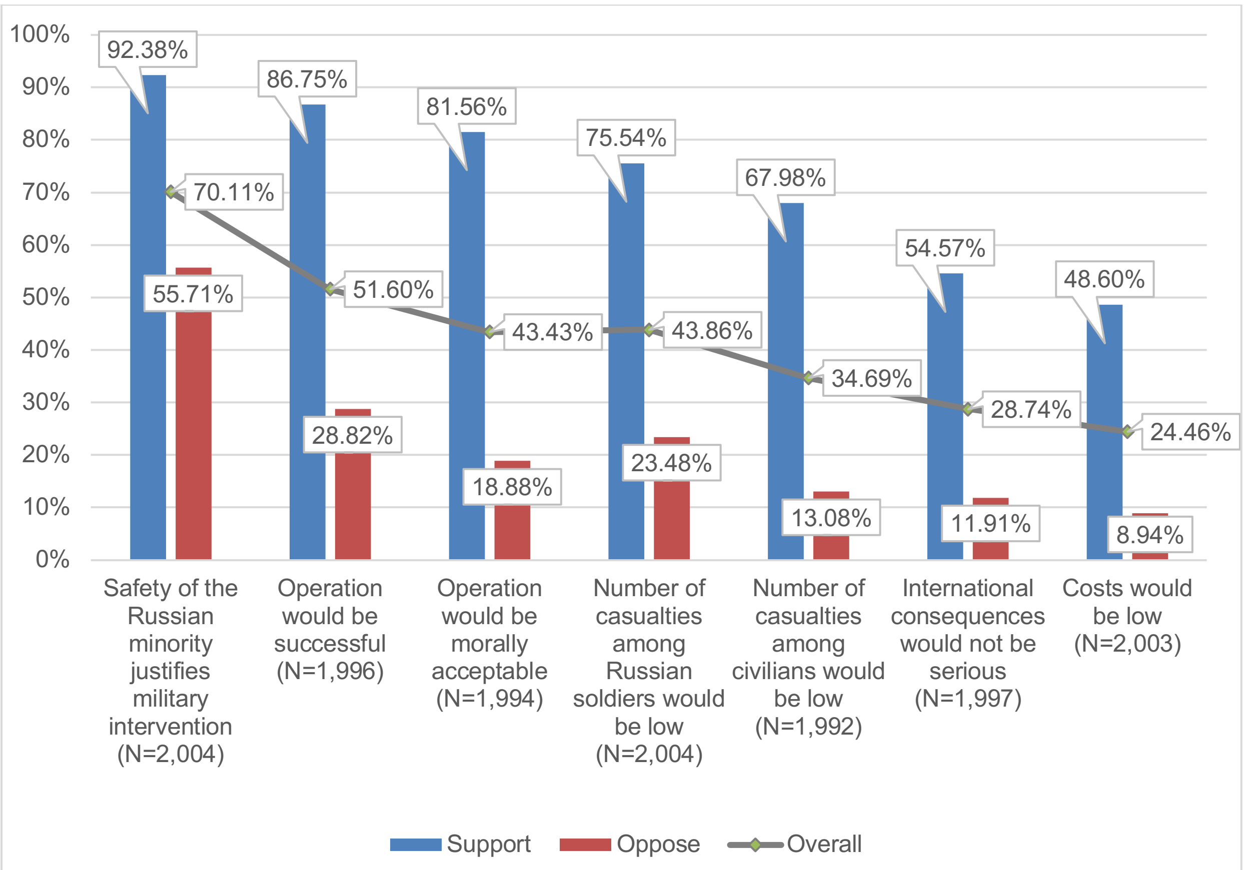
Figure 4.4. Percentage of Those Who Strongly Agreed or Somewhat Agreed with Positive Framing of the Proposed Military Operation.



NOTE: The plot presents the percentage of those who responded “Strongly agree” or “Somewhat agree” to “Q5. How much do you agree or disagree with the following statements when you think about the government's hypothetical plan?” for each of the seven statements displayed on Figure 4.2. Blue and red bars represent the share of those responses among the participants who also responded, “Strongly support,” or “Somewhat support,” and “Strongly oppose” or “Somewhat oppose,” respectively, to “Q10. How much do you now support the government’s hypothetical plan to launch this military intervention?” For details, see Appendix B.

Figure 4.4 does not account for the situation when a respondent agreed with a proposed statement about the operation (for example, “operation would be successful;” for the exact wording, see Figure 4.2) but did not believe it spoke in favor of the proposed military operation. Conceptually, weights people assign to different factors are subjective, and some factors might be discounted altogether. Furthermore, the proposed list of factors is unlikely to be exhaustive. Case in point, some respondents saw the proposed military operation in a very positive light—they agreed with the positive framing for all seven factors—and yet they did not support it. And so, when we look at the percentage of those who believed a given factor spoke in favor of the hypothetical government plan, the separation between the supporters and opponents of the operation grows. The only factor that a strong majority saw as an argument in favor of the operation was the motivation for the proposed intervention, that is, the safety of the Russian minority (Figure 4.5).

Figure 4.5. Percentage of Participants Who Agreed with Positive Framing of Factor Statements and Declared the Statements as Arguments in Favor of the Hypothetical Plan.



NOTE: The plot presents the percentage of those who responded “Speaks strongly in favor” or “Speaks somewhat in favor” to “Q6. What, in your opinion, speaks in favor of the government’s hypothetical plan?” for each of the seven statements displayed on Figure 4.2. Blue and red bars represent the share of those responses among the participants who also responded, “Strongly support,” or “Somewhat support,” and “Strongly oppose” or “Somewhat oppose,” respectively, to “Q10. How much do you now support the government’s hypothetical plan to launch this military intervention?” For details, see Appendix B.

Logistic Regression

To estimate the impact of each factor on the support for the proposed military intervention, I used logistic regression with binarized support as the dependent variable regressed on the intervention, seven independent variables signifying the decision factors (Figure 4.2), and demographic controls. Several models were tested, varying the number of demographic controls and the approach to the decision factors’ variables. The decision factors’ variables were ordinal variables combining respondents' views on the proposed operation (Figure 4.2) and how much the given factor spoke in favor or against the hypothetical deployment plan (Figure 4.3).

Two different variants were tested. The more granular variant included all twelve possible levels for each decision factor. The lowest level of one was assigned when a respondent strongly disagreed with the given factor statement and indicated that the statement—reframed as negative—was a strong argument against the proposed operation. The highest level of twelve was assigned when a respondent strongly agreed with the factor statement and indicated that it was a strong argument in favor of the proposed operation. The assignment of levels in the twelve-level independent variable (12LIV) model is presented in Table 4.6.

Table 4.6. Categorical Indicators of Factors' Importance (Twelve Levels).

	Strongly disagree	Somewhat disagree		Somewhat agree	Strongly agree
Strongly against	1	2	Not in favor	7	8
Somewhat against	3	4	Somewhat in favor	9	10
Not against	5	6	Strongly in favor	11	12

NOTE: Levels two and three, as well as ten and eleven, could be assigned as three and two, and eleven and ten, respectively, but it would not impact marginal effects. Column names refer to responses to "How much do you agree or disagree with the following statements when you think about the government's hypothetical plan?" Row names refer to responses to "What, in your opinion, speaks against the government's hypothetical plan?" and "What, in your opinion, speaks in favor of the government's hypothetical plan?," respectively. For details, see Appendix B.

The second variant comprised four levels, grouping all those who indicated that the given factor was an argument in favor of the operation and vice versa. The assignment of levels in the four-level independent variable (4LIV) model is presented in Table 4.7.

Table 4.7. Categorical Indicators of Factors' Importance (Four Levels).

<table>
<tr><th></th><th>Strongly disagree</th><th>Somewhat disagree</th><th></th><th>Somewhat agree</th><th>Strongly agree</th></tr>
<tr><td>Strongly against</td><td colspan="2" rowspan="2">1</td><td>Not in favor</td><td colspan="2">3</td></tr>
<tr><td>Somewhat against</td><td>Somewhat in favor</td><td colspan="2" rowspan="2">4</td></tr>
<tr><td>Not against</td><td colspan="2">2</td><td>Strongly in favor</td></tr>
</table>

NOTE: Column names refer to responses to "How much do you agree or disagree with the following statements when you think about the government's hypothetical plan?" Row names refer to responses to "What, in your opinion, speaks against the government's hypothetical plan?" and "What, in your opinion, speaks in favor of the government's hypothetical plan?," respectively. For details, see Appendix B.

All tested models produced qualitatively the same results. Namely, out of seven independent variables, six were significant predictors of support for the hypothetical plan. The beliefs about the number of casualties among Russian soldiers were consistently not a statistically significant predictor across all tested models. This result might be counterintuitive given the disparity in this factor between supporters and opponents of the hypothetical plan. However, the regression model indicates that this variance is, in fact, explained by other variables. On the other hand, consistently across models, views on the morality of the operation, followed by the prospects of success, were the strongest predictor of support.

Except for age, the demographic controls were also not statistically significant. Older people tend to be less likely to support the proposed operation, which contrasts with surveys conducted during the ongoing war with Ukraine.[125] One hypothesis why this could be the case is the difference between hypothetical and realistic support for war. Specifically, for older cohorts, support for war in both cases is “cheap talk”: there is no cost of supporting war because older people are not expected to participate. Meanwhile, younger people are expected to participate in the war in Ukraine. Therefore, they might be incentivized to declare relatively lower support for the ongoing war, whereas, in the hypothetical study scenario, they did not have that incentive.

Based on the AIC metric, the simplest model—with no intervention variable, no demographic controls, and only four levels for each decision factor—performed best. However, studies on drone warfare indicated that gender, age, education, and military experience could influence the support for unmanned operations.[126] Therefore, I kept these controls, along with the intervention, in the model. Even though the 4LIV model performed better and is easier to interpret, I will highlight interesting findings for both (if applicable) since the 12LIV model provides more granular insights and both models are statistically significant (p-value = 0.0000) and reasonably good fit. Tables 4.8 and 4.9 present coefficients (odds ratios) for the 4LIV and the 12LIV model, respectively. Table C.2 in Appendix C presents the comparison of all tested models.

The regression results indicate a certain level of inconsistency between participants’ views on decision factors and their support for the proposed military operations. It would seem logical that the predicted support should monotonically increase from those who disagree with the positive framing of a given decision factor and believe that it is an argument against it to those who agree with the positive framing and see it as an argument in favor of the operation. That, however, was only the case for two decision factors: Costs and Morality. As for Motivation, the lowest predicted probability of supporting the operation have those who agree with the importance of stakes but do not see them as an argument in favor of the operation (level three). In the case of the Prospects of Success, level three has the second lowest predicted probability of support. The

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[125] Russian Field, “Year of "special military operation" in Ukraine: the attitude of Russians [God «spetsial'noi voennoi operatsii» v Ukraine: otnoshenie rossiyan],” webpage, undated.

[126] Horowitz, Scharre, and FitzGerald, 2017; Walsh and Schulzke, 2018.

highest predicted probability of support among the four levels of the International Risks has level two: those who believe the risks are high but do not see this as an argument against the operation. Somewhat surprisingly, the same level two has the lowest predicted probability of support among Civilian Casualties levels. These dynamics and their implications will be further discussed in the sections below. The discussion will be based on predictive margins, presented in Tables C.3 and C.4 in Appendix C.

Table 4.8. Four-Level Decision Factors' Independent Variables Model.

	Dependent Variable: Support						
	*****Motivation**	*****Prospects of Success**	*****International Risks**	*****Civilian Casualties**	**Military Casualties**	*****Costs**	*****Morality**
	(Reference: Disagree with the positive framing + factor speaks against the proposed intervention)						
Disagree + Does not speak against	1.401 (.613)	1.392 (.465)	***2.091 (.466)	.640 (.174)	.945 (.290)	1.267 (.240)	1.405 (.398)
Agree + Does not speak in favor	*.468 (.216)	1.114 (.339)	1.074 (.305)	1.449 (.369)	1.361 (.348)	1.544 (.441)	***3.099 (.827)
Agree + Speaks in favor	***2.147 (.570)	***3.499 (.785)	***1.635 (.303)	***2.188 (.420)	1.169 (.249)	***2.470 (.488)	***7.41 (1.372)

Intervention (Reference: Airborne, NATO)		**Gender** (Reference: Woman)		***Age** (Reference: 18-29)		**Education** (Reference: No high school diploma)		**Military Family** (Reference: No)	
Robots, NATO	.999 (.197)	Man	1.148 (.164)	30-39	1.038 (.209)	High school graduate	.753 (.754)	Yes	1.155 (.246)
Airborne, non-NATO	.904 (.178)			40-49	.816 (.180)	Professional school graduate	.677 (.661)		
Robots non-NATO	1.010 (.201)			50-59	.706 (.162)	Some college education	.981 (.980)		
				60 +	**.579 (.139)	Collage graduate	.993 (.966)		

Observations	1875
AIC	0.773
BIC	-12421.062
Hosmer–Lemeshow test p-value	0.3148

NOTE: Coefficients in the table are odds ratios. Standard errors are in parentheses. * p < .1; ** p < .05; *** p < .01. All the predictors are indicator variables; therefore, I indicate the overall significance of each variable next to a variable name.

Table 4.9. Twelve-Level Decision Factors' Independent Variables Model

	Dependent Variable: Support						
	***Motivation	***Prospects of Success	**International Risks	***Civilian Casualties	Military Casualties	***Costs	***Morality
	(Reference: Strongly disagree with the positive framing + factor speaks strongly against the proposed intervention)						
Somewhat disagree + Strongly against	1.270 (1.598)	2.231 (1.989)	*2.304 (1.022)	.804 (.353)	.770 (.473)	**.393 (.170)	3.188 (2.402)
Strongly disagree + Somewhat against	1.596 (2.140)	.484 (.549)	***3.891 (1.869)	.568 (.341)	1.602 (1.149)	.819 (.356)	1.245 (1.123)
Somewhat disagree + Somewhat against	.468 (.552)	1.603 (1.379)	***3.131 (1.274)	1.213 (.518)	.707 (.408)	.815 (.301)	2.962 (2.121)
Strongly disagree + Not against	.343 (.459)	2.012 (1.984)	**3.908 (2.186)	.468 (.280)	2.884 (2.090)	.693 (.311)	1.208 (1.015)
Somewhat disagree + Not against	1.072 (1.309)	2.560 (2.265)	***5.812 (2.564)	.688 (.340)	.571 (.354)	1.062 (.398)	**4.958 (3.685)
Somewhat agree + Not in favor	.350 (.425)	1.421 (1.234)	***3.068 (1.437)	1.001 (.473)	1.236 (.716)	1.084 (.488)	***6.876 (5.010)
Strongly agree + Not in favor	.149 (.207)	2.107 (2.067)	3.254 (3.426)	**4.255 (2.702)	1.045 (.787)	.322 (.261)	***20.230 (18.429)
Somewhat agree + Somewhat in favor	1.134 (1.294)	*4.841 (4.054)	***4.495 (1.850)	*2.190 (.965)	.857 (.481)	1.767 (.686)	***17.101 (12.080)
Strongly agree + Somewhat in favor	1.542 (1.794)	**6.543 (5.705)	***5.164 (3.235)	2.475 (1.411)	.878 (.558)	.977 (.640)	***13.545 (10.493)
Somewhat agree + Strongly in favor	1.117 (1.295)	*5.151 (4.484)	**3.462 (2.030)	1.605 (.855)	1.243 (.791)	*3.394 (2.121)	***23.951 (18.123)
Strongly agree + Strongly in favor	1.763 (2.017)	*4.905 (4.261)	2.302 (1.367)	1.159 (.629)	1.182 (.779)	1.432 (.854)	***68.793 (53.018)

Intervention (Reference: Airborne, NATO)		**Gender** (Reference: Woman)		*****Age** (Reference: 18-29)		**Education** (Reference: No high school diploma)		**Military Family** (Reference: No)	
Robots, NATO	.857 (.180)	Man	1.010 (.158)	30-39	1.003 (.215)	High school graduate	.446 (.438)	Yes	1.066 (.235)
Airborne, non-NATO	.823 (.174)			40-49	*.636 (.149)	Professional school graduate	.449 (.428)		
Robots non-NATO	.951 (.200)			50-59	**.564 (.140)	Some college education	.566 (.556)		
				60 +	***.460 (.120)	Collage graduate	.687 (.652)		

Observations	1875
AIC	0.778
BIC	-12101.049
Hosmer–Lemeshow test p-value	0.0838

NOTE: Coefficients in the table are odds ratios. Standard errors are in parentheses. * p < .1; ** p < .05; *** p < .01. All the predictors are indicator variables; therefore, I indicate the overall significance of each variable next to a variable name.

Motivation: "The Safety of the Russian Minority is a Very Important Issue, and It Would Call for Military Intervention"

Motivation was a statistically significant predictor in all models. Even though the differences in odds of support on different levels were not statistically significant in the 12LIV model, the probability that none of the levels impacted the outcome was 0.0017. This probability was even lower in the 4LIV model (0.0001). On the other hand, the marginal effects of more positive views on the stakes of the operation (i.e., the safety of the Russian minority) and their importance as an argument in support of the operation are small and inconsistent.

In the 12LIV model, the shift from level one (strongly disagree + speaks strongly against) to level twelve (strongly agree + speaks strongly in favor) increases the predicted probability of supporting the proposed operation only by 6.4 percentage points. This is not much more than shifting to level three instead (strongly disagree + speaks somewhat against), leading to a 5.3 percentage points increase in the predicted probability of support. This implies that people's beliefs on the stakes of war are not very reliable, even if significant, predictors of support. The model predicts that people can oppose the importance of stakes but support the operation at a similar rate as those who see the stakes as important.

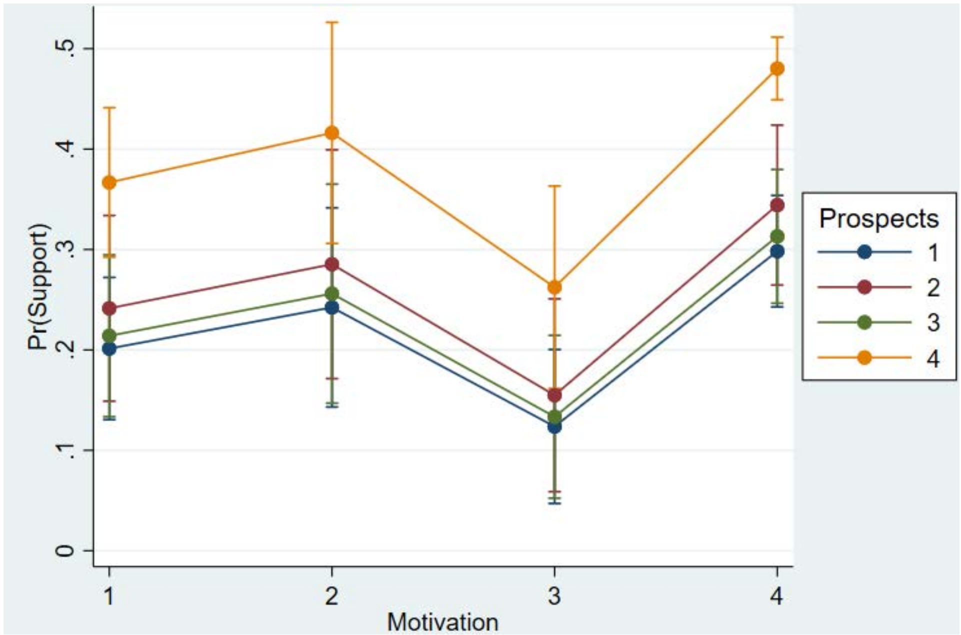
Even more surprisingly, among all twelve levels, the lowest predicted probability of supporting the operation have those who strongly agree with the statement that the safety of the Russian minority is a very important issue but indicate that it is not an argument in favor of the proposed operation (17.5 percent). This could be people who, for example, highly value the safety of their compatriots but are against war on principle. The same inconsistency is also visible in the 4LIV model where, again, the lowest predicted probability of supporting the operation have those who admit that they agree with the importance of stakes but do not see them as an argument in favor of the operation. This implies that a communication campaign aimed at weakening the Russian public's support for military operations might be more efficient if such a campaign does not target the stakes of war but points to other arguments against it.

Prospects of Success: "The Intervention Would Be Successful"

Prospects of Success were a statistically significant predictor in all models. In the 4LIV model, the odds of supporting the operation by those who believe the operations would be successful and see it as an argument in favor of it are three and a half times higher than those who have the opposite view. The predicted probability of support increases from 28.1 percent on level one to 44.7 percent on level four (i.e., the marginal effect is 16.6 percentage points).

On the other hand, we can see a similar inconsistency as in the case of motivation, albeit not of the same degree. In the 4LIV model, those who believe the operation would be unsuccessful but do not see it as an argument against the operation have a higher predicted probability of supporting it than those who think the operation would be successful yet do not see it as an argument in favor. Views on motivation seem to have a dominating effect: three groups with the lowest predicted probabilities of support are all on Motivation Level 3, i.e., those who do not believe that the motivation is an argument in favor even though they agree with their importance otherwise (Figure 4.6). Furthermore, those who share that opinion while seeing the success of the operation as an argument in favor of it (Prospects Level 4 at Motivation Level 3) still have a lower predicted probability of support than those who believe the operation would be a failure and the stakes are not sufficiently high to justify the operation but see neither as an argument against (Prospects Level 2 at Motivation Level 2).

Figure 4.6. Predictive Margins of the Prospects of Success with 95% Cis at Different Motivation Levels.

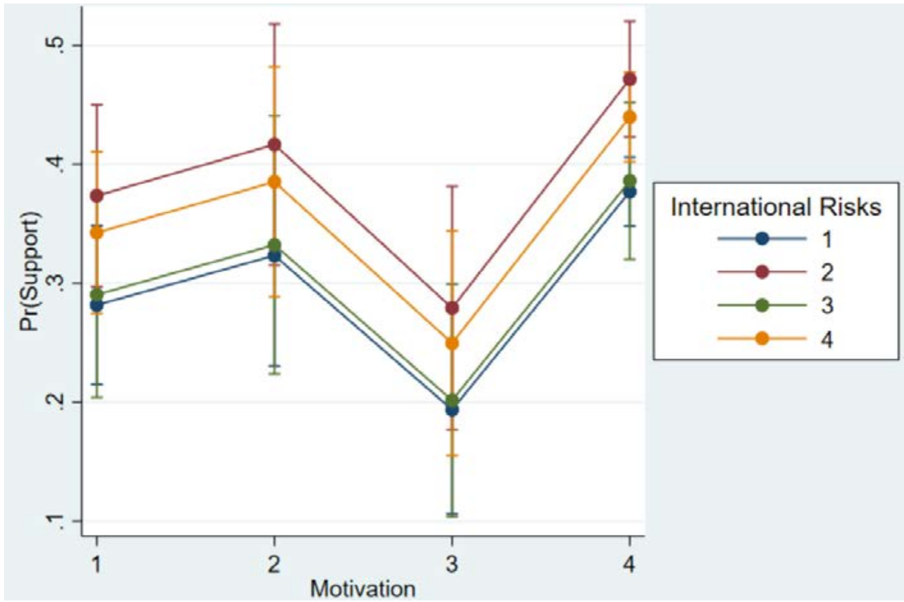


International Risks: "The International Risks Associated with This Intervention Would Be Low"

International Risks are another peculiar decision factor: among the four groups in the 4LIV model, the highest predicted probability of support for the proposed operation have those who recognize that the international risks associated with the operation could be high, but they do not believe it to be an argument against the operation (level two). More specifically, in the 12LIV model, the highest predicted probability of support had level six (somewhat disagree + not an argument against), 45.1 percent. Meanwhile, the predicted probability of support for level one was 26.1 percent, and for level twelve, only 35 percent, the lowest among those who believe that low international risks are an argument in favor of the operation. This surprising finding—the expectation that Russia will be penalized for protecting the minority increases the probability of supporting the operation—might imply that the respondents draw some additional utility from challenging the international order. This could also partially explain the lack of effect of the strategic context (NATO/non-NATO). Specifically, NATO has had no overall effect because the two effects might have canceled each other out: some respondents might have been more wary of supporting an operation against a NATO country, whereas others might have been more eager as it provides an additional opportunity to challenge the international system.

Here, again, the motivation seems to dominate the impact on the predicted probability of support: those who believe that the safety of the Russian minority is not an argument in favor of the proposed operation have the lowest predicted probability of support, regardless of the views on international risks (Figure 4.7).

Figure 4.7. Predictive Margins of the International Risks with 95% Cis at Different Motivation Levels.



Civilian Casualties: "There Would Be Few Casualties Among Civilians"

In the 4LIV model, the *Civilian Casualties* are another decision factor with a nonmonotonic behavior, but here, level two has the lowest predicted probability of support. Predictive margins in the 12LIV model indicate that the least likely to support the operation—27.9 percent—are those who strongly disagree with the assertation that there will be few civilian casualties but do not see that as an argument against the operation (level five). The most likely to support the operation—53.3 percent—are those who strongly agree with the assertion but do not see it as an argument in favor (level eight). Both effects require more research. However, the latter effect could stem from the self-image of the supporters of the operation. They might have wanted to avoid sounding cynical when using low expected numbers of civilian deaths as an argument for the operation.

Military Casualties: "There Would Be Few Casualties Among Our Soldiers"

Russian *Military Casualties* were not a statistically significant predictor in either model. In the 4LIV model, the probability that all levels had no impact on the probability of support was 57.7 percent; in the 12LIV model, that probability was 32.2 percent. This is a particularly interesting finding, both in the context of this study and in the broader context of the ongoing war in Ukraine.

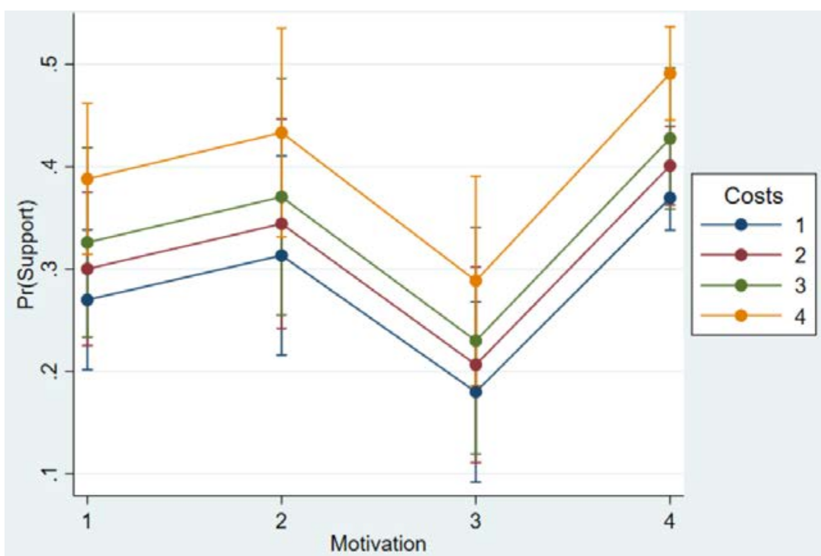
In the context of this study, the lack of a statistically significant effect of the casualties among Russian soldiers might explain why technology has no impact on the probability of support, given that saving soldiers' lives is arguably the best-advertised advantage of robotic capabilities. In the broader context of the war in Ukraine, this lack of statistically significant effect might explain why there is so little protest against the war on the grounds of mounting losses.

This is not to say that the Russian public is permanently insensitive to deaths among their own soldiers. Existing scholarship suggests that the public's sensitivity to own losses depends on the importance of the stakes.[127] So it is conceivable that both the current war and even the hypothetical scenario presented in the survey struck a nerve that allows sustaining a high casualty rate without the loss of popularity of the war effort. This finding implies that highlighting Russian losses might be an inefficient strategy for an information campaign to weaken the Russian public's support for the war.

Costs: "The Costs Associated with the Intervention Would Be Low"

In the 4LIV model, the *Costs* decision factor behaves in a predictable monotonic manner: the lowest predicted probability of support have those who believe that the costs would be low and it is an argument against the operation, while the highest predicted probability of support belongs to those have the opposite view. The marginal effect of the shift from level one to level four is 11.1 percentage points. In Figure 4.8, we can see again how motivation level three dominates the impact on the low predicted probability of supporting the operation.

Figure 4.8. Predictive Margins of the Costs with 95% Cis at Different Levels of the Motivation



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[127] Edwards, Mitchell, and Welch, 1995.

Morality: "Such an Operation Would Be Morally Acceptable"

The *Morality* decision factor also behaves in a monotonic manner. However, shifting from the lowest to the highest level of morality produces in both the 4LIV and 12LIV models the largest marginal effects of 30.1 and 58.1 percentage points, respectively. In the 12LIV model, the odds of supporting the operation by those who strongly believe the operation is moral and believe that it speaks strongly in favor of the operation are so much higher—almost 67 times—than the odds of supporting it by those who have the opposite view that one could suspect artificial inflation of the odds ratio due to multicollinearity. However, the variance inflation factor for morality is only 2.33 and, as such, confirms that this effect does not stem from multicollinearity.

The high marginal effect of increases in the morality decision factor is an important finding in the context of widespread worry that Russia could be less constrained in the development of LAWS because there is less pushback against autonomy on ethical grounds. In the face of Russian atrocities against Ukrainian civilians and the lack of a negative reaction from the Russian public, one could speculate that ethics do not matter much for Russians' support for the war. However, the results of this experiment suggest this not to be true. The link between technology and morality might be so far missing, but there is a strong foundation on which the moral assessment of different technologies of war could potentially be built.

Other Contributing Factors: Open-Ended Question

Approximately a quarter of the sample gave meaningful responses to the open-ended question about additional factors influencing their decisions. The opponents of the operation provided responses to the open-ended question twice as often as the supporters: there were 381 recorded responses from the opponents of the hypothetical government's plan and 177 responses from the supporters. The responses often reiterated factors proposed earlier, yet their analysis contributes to a deeper understanding of the difference between the supporters and the opponents of the proposed operation.

Operation Opponents

The most common reason to oppose the hypothetical government's plan was general opposition to war, as indicated in 28.6 percent of opponents' responses. Some of those statements were very brief ("No to war!"[128]); others were more elaborate ("Our country participated in infamous, terrible wars. We won, but we remember the huge losses. We do not want war" or "The use of violence is in principle unacceptable, regardless of the nationality and race of a person."). Some respondents particularly did not like the idea of Russia being the one who starts a war ("I categorically do not like the very assumption that Russia can be the first to

[128] Translations own throughout this chapter.

start any military operations on any territory.”). Either way, this principled statement might be a qualitative interpretation of the morality decision factor. Specifically, the opponents of the proposed operation might have seen the operation as morally unacceptable not because of the specifics of the operation but because they would be against any combat operations. More research is needed to explore these dynamics.

Almost a quarter of opponents’ responses pointed to peaceful solutions to the presented problem. Some responses generally called for peaceful or diplomatic resolution (“Why is it necessary to resolve conflicts through war? Diplomats of hypothetical countries eat their bread free????”), others even proposed specific solutions (“There are other non-violent ways to protect ethnic minorities. For example, a relocation program or dialogue with the government of another country and negotiations on the residence for an ethnic minority.”).

The third most common reason for the lack of support for the proposed operation was general distrust in the government (e.g., “I strongly disagree with any actions of the government. I don’t think the current government is legitimate.”), which was mentioned in 6.2 percent of opponents’ responses. Some respondents also called for solving problems at home instead of spending resources on people who chose to live abroad (“Put things in order in your country; there is no need to spend taxes and resources for citizens who have left Russia. Moreover, they will impose sanctions, and the people will have to pay for the whole adventure.”).

Figure 4.9. Additional Factors Influencing the Opposition Against the Proposed Military Operation.

Operation Supporters

On the side of the supporters of the operation, more than a third of the respondents reiterated the importance of stakes, that is, protecting the Russian minority (34.9 percent of coded supporters’ responses). Some of the responses were very brief (“The Russians don’t abandon their own people.”), others, again, more elaborate (“Russia’s military doctrine allows only the defense of the state. ‘Who lives by the sword, dies by the sword.’ Throughout the history of Rus’, enemies attacked us, and as a result, Rus’ was freed from them and rose from the ashes. I hope this time the Lord will not abandon the Russian people.”). Some responses evoked more general threats against Russia that warrant a military response (e.g., “It is high time to respond very harshly to all attacks against Russia. Lies have swept the Western media. If the West wants war, it will get it. I don’t want to fight anyone; there is no hostility toward ordinary people. But you are asking for trouble.”). They also mentioned NATO expansion as a justification (“NATO’s approach to Russia’s borders is a threat in and of itself”).

Not surprisingly, contrary to those who opposed the operation, supporters sometimes mentioned that their position stemmed from their trust in the government (“I am only for the decision of our government”). On the other hand, the supporters—especially those whose support was weak—also pointed to reasons for *not* supporting the operation. They worried about the risk of escalation (“I fear human casualties on both sides. Negative economic consequences, new sanctions, the country will stop developing, and the poverty will increase. I’m afraid of an increase in the scale of hostilities and the onset of a real war,” or “Such an operation could become a pretext for unleashing hostilities on the part of NATO. I think that we should try to remove citizens from the territory of country X within the framework of peace agreements.”). Some presented more nuanced considerations of pros and cons (e.g., “Availability of modern weapons, timely rearmament of the standing army. Opportunity to avoid strong negative reactions of other countries due to their economic dependence.”).

Figure 4.10. Additional Factors Influencing the Support for the Proposed Military Operation.

Robots

A few responses mentioned the robotic capabilities. Some of them expressed hope that they could save lives (“Robotic warfare will certainly reduce the death of our soldiers but will inflict a large number of civilian casualties, on the other hand,” or “We (ordinary people) do not have a complete understanding of robots and how they will affect all this, but if the number of victims is minimized, and the losses after using them are not great, then why not ?!”).

On the other hand, confidence in Russian robotic capabilities seems low (“The control device of autonomous robotic vehicles is not fully understood, from this point, there are some concerns about the accuracy and loyalty of performing a hypothetical military operation,” “Robotic technology is less efficient than humans,” “Robotics is not stable,” “I do not trust domestic robotics.”). Some opposed the robotization of the armed forces in general (“Robotization of the army is a bad option,” “No robots needed”). Finally, some opposed their participation on ethical grounds (“Robots should not participate in wars against humans. Such operations are not ethical and very dangerous.”)

Ukrainian Factor

Finally, both sides mentioned Ukraine, even though the country was not named in the scenario. The supporters advocated for the population of the Donbas region (“I believe that the Russian-speaking population of Donbas should sleep peacefully, and not live under shelling.”) or pointed to the narrative of Ukraine as a Western client state (“Putin is smarter and will find a more pragmatic option because the enemies surround our country and seek its collapse so that the hegemon from across the ocean rules the world. I hope [derogatory term for Ukrainians] will be smarter and will not follow the path of Saakashvili.”).

The opponents of the operation pointed out the cynicism of war against Ukraine (“This operation is propaganda, and I don’t see the point in carrying it out, and I don’t see a war between two states. Russia and Ukraine” or “A large number of the Russian-speaking population lives in Ukraine, not only in the LPR and DPR, but our government is trying to save not the entire population, but specifically from the LPR and DPR, most likely this is done as a lever of pressure on Ukraine. In general, I think that there will be no actual attack.”).

Interestingly, Ukraine was mentioned in open-ended responses regardless of the NATO variable in the hypothetical scenario. It has at least two implications. First, it may explain why NATO as a treatment did not impact the outcome—respondents might have believed that NATO would get involved regardless of the country’s membership status. Given the level of Western support for Ukraine in the ongoing war, this expectation has proven to be quite realistic.

Second, it speaks to a broader challenge of using hypothetical scenarios in social science. Namely, people might be conceptualizing a given hypothetical scenario in a specific fashion grounded in real-life events, which may influence the responses.[129] However, study participants might have various real-life benchmarks, and their influence usually remains unknown. It is less of a problem for the treatment effects in experimental studies because random assignment should guarantee that even if there is an effect of real-life reference points, those reference points are statistically evenly distributed between study groups. Non-experimental designs are more prone to biased responses. The solution could be giving up some part of generality and setting up study scenarios in more realistic settings; however, that creates another set of challenges, especially in authoritarian countries. This study, in particular, opted for a hypothetical scenario to minimize the political charge of the survey and maximize the number of responses. And yet, online panel vendors based in Russia refused to be involved in the project.

Limitations

The experiment has serious limitations. First, due to the ongoing war in Ukraine, it is impossible to replicate the study in the foreseeable future. If the tension with Ukraine impacted the responses even before Russia launched the full-scale invasion, then deploying hypotheticals during the ongoing high-intensity war would have little credibility.

Second, the survey scenario did not define the adversary country beyond the NATO/non-NATO classifications. However, as the open-ended responses suggest, some participants responded with Ukraine in mind, regardless of their scenario's NATO variable. It is impossible to discern how many respondents completed the survey as if it was specifically about the conflict with Ukraine and how many were able to remain on the hypothetical level. Thus, it is impossible to assess how biased the responses were. One way or another, this limitation should be addressed in future studies.

Third, the surveys conducted after Russia launched its war on Ukraine suggest that there might be an important decision factor not accounted for by this survey instrument. Namely, the trust in the government. Surveys indicate that a considerable number of people are ready to support any decision of President Putin, whether the decision leads to escalation or termination of the war.[130] The trust factor was mentioned by several participants in the open-ended responses, but it might not be a complete picture. Traditionally, literature studies the impact of wars on leaders' approval ratings. This study speaks to the reversal of this relationship.

Conclusions

This survey experiment aimed to measure the impact of robotic technologies and strategic context (approximated by adversaries' NATO membership status) on public support for military

[129] As discussed earlier, that also might have been the case for the India public sample in Horowitz, Scharre, and FitzGerald, 2017.

[130] Russian Field, 2023.

operations. As such, it followed the footsteps of several experimental studies on the U.S. public and the use of drones. However, this study has reached different conclusions from its predecessors. Namely, while the support for U.S. drone deployments is usually higher than the support for manned deployments, no statistically significant effect has been found in this study.

There are several potential explanations, best discussed through the lenses of motivation and costs. The study scenario posited that the Russian minority was at risk of ethnic cleansing. The participants might have perceived these stakes as so high that the marginal effect of technology became negligible. That might even more so be the case if respondents thought, in fact, about Ukraine. Given that the Russian propaganda had portrayed Ukraine as a "brotherly nation," if not a part of Russia itself,[131] the technology of war could have been completely tramped by higher-level ramifications of the proposed operation. In the language of the model, the survey scenario might have been set in a parameter regime where the motivation is so high that only a very large shift in model parameters could lead to different outcomes. For example, there might be a technology effect if this technology was a swarm of autonomous nuclear devices, albeit that would be against the conceptualization of LAWS as conventional weapons.

That leads to the other factor that might have impacted no statistically significant effect of technology: costs. Given the context of targeted strikes in which the drone studies were conducted, the difference in costs between control (manned capabilities) and treatment (drones) might have been perceived as much higher than in this study. Specifically, this study scenario posited that some human soldiers would still be involved in combat operations even in the robotic condition. Furthermore, the scenario assumed a larger scale operation—even if still limited in scope—that surgical target killings. I opted for such a scenario because it seemed more realistic, given the applications of robotic capabilities discussed in the Russian military literature.[132] However, the results could be different if the scenario assumed no humans were involved in the robotic condition. But even that is far from guaranteed, given that the deaths of Russian soldiers were not a significant predictor of support when the motivation seems so high.

Nevertheless, the survey leads to interesting broader conclusions about Russians' support for war in general and possibly the war in Ukraine in particular. On the one hand, the open-ended responses of war supporters indicate that protecting compatriots is a compelling motivation. On the other, many Russians seem to believe that war is simply not an acceptable way of solving conflicts with neighbors. This group is barely visible during the ongoing war.[133] Yet, regression analysis suggests that moral arguments are the most promising avenue to influence public support for war. More research is needed to understand how this avenue could be exploited.

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[131] Denis Timoshenko, "'Brotherly people' - the concept of Kyiv scribes, which has become a social doctrine of Russia [«Bratskii narod» – kontsept kievskikh knizhnikov, kotoryi prevratilsya v sotsial'nuyu doktrinu Rossii]," Radio Freedom [Radio Svoboda], September 7, 2020.

[132] Marcinek and Han, 2023, pp: 23–26.

[133] Marco Hernandez, "Decoding the Antiwar Messages of Miniature Protesters in Russia," *The New York Times*, June 16, 2023.

On the other end of the spectrum are casualties among Russian soldiers, which seems to be the least promising avenue to weaken Russians' support for war. Russians' seeming insensitivity to casualties among Russian soldiers in the ongoing war remains a puzzle for many Western commentators.[134] The regression analysis suggests that military casualties indeed have no significant impact on the support for military operations in the proposed hypothetical scenario. Unfortunately, the study gives no indications of the threshold of motivation where soldiers' death could become a significant factor. More research is also needed to understand the non-monotonicity of support in certain decision factors and its implications for operations in the information environment.

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[134] Julian E. Barnes, "Russian Public Appears to Be Souring on War Casualties, Analysis Shows," *The New York Times*, May 26, 2023; Timothy Frye, "Casualties Won't Topple Putin," Foreign Policy, April 10, 2023; Alexander Smith, "The 'stunning' scale of Russian deaths in Ukraine signals trouble ahead for Putin," NBC News, May 2, 2023.

Chapter 5. Conclusion, Policy Implications and Recommendations, and Areas for Future Research

It is often the case that new technologies designed to solve existing problems perturb the system in which those problems exist and create new challenges. The same technologies can be viewed as a peril by some and a new hope by others. Autonomous weapons systems will likely be no exception. The promise of unmanned warfare is the minimization of harm inflicted on human soldiers. The hope is that emotionless algorithms will also minimize the harm inflicted on enemy combatants and non-combatants. A futuristic fantasy could be that wars delegated to AI will never reach the kinetic phase because algorithms will simulate their actions and determine the winner without a single missile fired.

But risks abound, too. From ethnic cleansings stemming from biases inscribed in AI algorithms by human programmers to indiscriminate violence unleashed by rogue algorithms to uncontrolled machine-speed escalation to shifting wars onto new areas in search of novel ways to influence adversary's political will, the negative consequences of robotic and AI revolution can be countless. Many of these potential dangers have to do with machine decision-making. This study, however, focused on the impact of the robotic revolution on human decision-making. Namely, it explored how the characteristics of LAWS can impact leaders' propensity for war.

Maybe in some distant future, humans will embrace the idea of missile-less conflicts adjudicated by impartial algorithms. Perhaps we will even realize that, as a species, we have more shared than diverging interests and will give up wars altogether. But until then, the study of incentives and disincentives for wars remains relevant. The mix of human passions, false optimism, and the chaotic nature of war leads to a situation where armed conflicts rarely play out the way leaders imagine when they first send soldiers to the battlefield. Thus, it is critically important to ensure that the development of capabilities designed to put fewer lives in harm's way does not create incentives to do it more frequently.

Consequently, I sought to answer three questions: Under what conditions could lethal autonomous weapons systems increase the likelihood of war? Could they increase Russia's propensity for war? And what should the United States do to mitigate such risk?

To that end, I developed a framework to provide a formalized and systematic way to reason, anticipate, and analyze the implications of different directions the policy and technological advancements could take. Through formal modeling, I studied the impact of emerging technologies on extended deterrence and explored mitigation options for the United States. Finally, I conducted a survey experiment to better understand the potential impact of robotic capabilities on Russian public support for military deployments.

The framework conceptualizes the decision to go to war as a function of four factors: motivation, prospects of success, economic costs, and net political benefits. The propensity for war marginally increases when emerging technology increases the motivation, prospects of success, and/or net political benefits and/or decreases the economic costs of war. The model confirms this intuition in the context of major power competition over a third county while providing several nuanced insights about the interactions of decision factors in such a decision scenario.

I found that LAWS are unlikely to influence the primary motivation for war. Even if they were perceived as a force shifting the balance of power, war seems an unlikely response. That is because LAWS' advantage lies in algorithms that will likely be developed in a distributed environment and, as such, can hardly be destroyed in combat operations.

On the other hand, there is a common expectation that LAWS will decrease the costs of wars: they are expected to save soldiers' lives, they might also lower labor and sustainment costs, and lessen the requirements on defensive systems today deployed to protect the crew of manned capabilities. However, the economic benefit of introducing LAWS will differ across countries because it depends on their current cost functions: how much they value human life and what is the price differential between capital and labor. It will also depend on whether LAWS will augment or substitute the existing capabilities. Finally, the cost will be influenced by the new dependencies the demand for LAWS' components will produce.

Importantly, the model suggests that a decrease in economic costs of fighting has an almost universally negative impact on extended deterrence, regardless of whose costs of fighting decreased. The mechanisms through which deterrence fails are slightly different depending on which player's costs decline, but the result is the same: the lower the costs, the less likely deterrence is to survive. Lowering mobilization costs, on the other hand, seems to have an unambiguously positive impact on deterrence.

LAWS are often seen as force multipliers that can enable currently impossible missions. If that was the case, the perceived prospects of success of some missions would increase, and so would the leaders' propensity to use them. But this might be overly simplistic, and the LAWS' potential impact on perceived prospects of success might be ambiguous. Firstly, the prospects of success are predominantly affected when the rate of adoption is unequal, and early adopters have the upper hand. However, this implies that the effect will be temporal and localized, and in the case of LAWS, such windows of opportunity might be particularly small: only very few countries can acquire aircraft carriers or nuclear weapons, but almost all can afford programmers.

Secondly, such windows of opportunity have a particularly destabilizing effect if the technology favors offense. But if LAWS were proven useful in defense, they might have a stabilizing impact. Similarly, technology can be destabilizing if prospects of success are higher for the first mover. But despite AI's decision-making speed, moving first might not be perceived as beneficial since it would provide a training set for the adversary's AI. Leaders might also be wary of deploying LAWS unless absolutely necessary if they are uncertain about their performance.

In the model, the prospects of success were represented as contest success functions combining the strategic choice of the number of troops committed to war with their quality. The model confirms the intuition that when the challenger's troops' quality increases, deterrence—in most cases—weakens. But it also suggests that this effect could be nullified if the defender improves the quality of own troops, as long as the quality ratio remains constant.

The model also indicates that deterrence is also in peril when the challenger's political benefits increase, or the defender's political benefits decrease. The latter has an adverse effect on the defender's ability to uphold commitments to allies and partners. However, the impact of LAWS on the political benefits of fighting remains unclear and will depend on expectations regarding the systems' performance and the public's views on their legality and ethics. Political leaders might see them as a liability: the difficulty of assigning responsibility for their mistakes might shift said responsibility up the command chain to the political leaders who authorize the deployment.

The survey experiment drilled down on the potential impact of the robotic revolution on the Russian public support for military operations. I found that robotic technology does not seem to have an impact on the support for military operations. This finding, coupled with the fact that Russian robotization efforts seem to be aimed at improving the prospects of success, suggests that if Russia's robotization program succeeds, Moscow's propensity to use force might further increase.

However, the survey also suggests that there are several significant predictors of support, which could potentially be used to influence the public's views, even if, at the time of survey deployment, there was no difference in how these factors were viewed by the manned and unmanned study groups. Specifically, I found that belief about the morality of the proposed operation is by far the strongest predictor of support for it, whereas the views on casualties among Russian soldiers have no statistically significant effect. Interestingly, most of the factors had a non-monotonic impact on the outcomes. Only two predictors—costs and morality—largely followed the expected pattern where the probability of support was an increasing function of the beliefs about low costs and high morality of the war. More research is needed to understand better the non-monotonic effects in motivation, prospects of success, international risks, and civilian casualties.

Policy Implications and Recommendations

This research abstracted away from the technological feasibility of different levels of autonomy and AI decision-making. It also did not seek to understand why or under what conditions political leaders could be willing to delegate the decision-making to a possibly untransparent algorithm while the political responsibility for its actions would still rest on them. Both of those questions are worth years of research. Instead, this study sought to understand how the propensity to deploy military force could change if LAWS become operational and leaders are overall willing to use them.

The key overarching implication of this study is that the future of warfare in the presence of LAWS is not predetermined. There are possible futures where AI and robotic revolution could incentivize conflict and those where technological advances strengthen stability. Pathways leading to these futures will be shaped in part by the decisions and actions of the U.S. Government and U.S. allies and partners. The agency and ability to influence the future are not unlimited. More research is needed to understand what the most robust pathways might be to achieve a more stable and peaceful future. Robust decision-making and technology foresight are two quantitative and qualitative methods that come to mind for such analytical exercises. However, in this concluding section, I will present some preliminary recommendations that stem from the research conducted for this study.

Motivation

As indicated earlier, primary political reasons why countries might go to war are unlikely to change due to the emergence of LAWS; however, one recommendation that stems from this research is to avoid centralized research and development centers, which could be destroyed in a kinetic strike.

Prospects of Success

To prevent destabilizing shifts in perceived prospects of success, the United States, its allies, and partners can leverage the software and hardware components of LAWS. The characteristics of robotics hardware can be used to demonstrate the defensive intent of LAWS, whereas properties of algorithmic control systems can be exploited to ensure that potential adversaries do not overestimate their relative prospects of success.

And so, one of the key ways the United States and its allies and partners could aid the emergence of a more peaceful future is to heavily invest in defensive autonomous capabilities. This could prevent the emergence of windows of opportunity, where the local shifts in offense-defense balance would favor adversaries' offensive capabilities. Sharing such defensive capabilities with allies and partners would improve their capacity and resiliency while demonstrating the defensive intent behind technological innovation to potential adversaries.

Admittedly, distinguishing between offensive and defensive capabilities is famously difficult, but on the LAWS' hardware side, there might be avenues to demonstrate intent that were not available to legacy capabilities. For example, the size of robotic capabilities constraints the operational range. Thus, investing in small autonomous drones might make it clear that their purpose is not to attack targets deep inside the enemy's territory but to protect critical infrastructure, communication nodes, and command posts. The impact of such capabilities on the decision calculus could be twofold: they could ease potential adversaries' threat perception without leading them to overestimate the prospects of success in attack.

The U.S. Government has a limited ability to influence which types of LAWS applications the potential adversaries pursue, but diplomatic efforts could be undertaken to promote the defensive LAWS. If such efforts fail, the U.S. national security community could leverage the software side of LAWS to work against adversaries' trust in their offensive autonomous capabilities. This might be attempted through cyber, information, and psychological operations. The inherently complex and unpredictable algorithms underlying LAWS might create avenues for influence operations unavailable with legacy systems. While it might be very difficult to change foreign leaders' perceptions about the performance of their own tanks or missiles, it might be easier to sow doubt in capabilities that are hard to comprehend to begin with. Meanwhile, if adversaries have little trust in the performance of their robotic capabilities, they might be hesitant to deploy them, thus limiting LAWS impact on their decision-making.

At the same time, it is critically important for peace and stability that the United States strengthens or at least maintains its technological edge in algorithmic warfare. As the model suggests, as long as the troops' quality ratio remains constant, the advances in adversaries' quality would have no impact on extended deterrence. Hence the importance of Verification and Validation (V&V) and Testing and Evaluation (T&E) of autonomous weapons. Its primary purpose is "to ensure autonomous and semi-autonomous weapon systems function as anticipated in realistic operational environments against adaptive adversaries and are sufficiently robust to minimize failures."[135] Therefore, its impact on deterrence can be twofold: the U.S. leaders will build confidence to deploy them when necessary, whereas the potential aggressive behavior of the U.S. adversaries might be restrained by the hesitance to confront LAWS proven effective.

V&V and T&E could potentially signal the ability of U.S. algorithms to learn and adapt quickly. This could have a stabilizing effect as the potential adversaries could be wary of providing opportunities for U.S. algorithms to train and improve. Thus, instead of providing a seemingly cheap avenue to force new realities on weaker countries, the emergence of LAWS could make war a mean of the very last resort.

Economic Costs

Affecting potential adversaries' cost function could be difficult, as evidenced by the limited impact of sanctions and export controls imposed on Russia in the aftermath of the full-scale invasion of Ukraine in February 2022. Even though the sanctions were unprecedented in scope, they have not choked Russia's ability to continue high-intensity operations. The particular challenge of affecting the cost component of LAWS is the components' dual use. Russia, for one, has a long history of evading sanctions and export controls, dating back to the Cold War, when the Soviet intelligence agencies developed elaborate schemes of importing critical

[135] Department of Defense Directive 3000.09, 2023.

components seemingly for civilian use.[136] Such schemes increase the price due to the risk premium, but it might be insufficient to affect the outcome.

Consequently, a more promising avenue could be aiding brain drain from countries of interest: the United States and its allies and partners could adopt immigration policies that would lower the supply of high-skilled workers aiding adversaries' scientific and technological innovation.

The key takeaway about the cost functions of LAWS developed by the United States, its allies, and partners is the emphasis on lowering the mobilization costs instead of the costs of fighting. It might be difficult to disaggregate the two in real life, but one way to conceptualize this difference is through two paths of technology development and deployment. The United States could invest in large quantities of capabilities that are cheap, expendable, and easy to use and pre-deploy them in allied and partner countries. The second path would instead focus on quality: these would be sophisticated but expensive capabilities the U.S. military brings onto the theater when a crisis develops. This way, the United States could credibly signal that the allies and partners are protected under the umbrella of U.S. technological superiority without fostering a dangerous belief that war can be cheap.

This is an important point because many experts perceive the low costs of unmanned capabilities as their biggest advantage, and indeed, many countries produce drones whose low cost justifies their high attrition rate in a contested environment. Consequently, there might be pressure on the U.S. military to follow suit. However, my research suggests that this could have an adverse effect on stability, and therefore, the dual path where one strain of research and development focuses on improving quality instead of price might be more beneficial.

Net Political Benefits

To lower the likelihood of war in the presence of LAWS, the U.S. Government could also influence the shifts in political benefits both domestically and internationally.

First, the government should ensure that there is no a priori backlash against autonomous weapons systems as this could impede the U.S. commitments to allies and partners and, consequently, undermine extended deterrence. To that end, the U.S. Government should lead dialog on LAWS with the American people and the expert community. The same applies to the dialog with allies and partners, who might have even stronger concerns about the legality and ethics of LAWS than the U.S. public. Those concerns should not be disregarded. The Euromissile crisis is a good example of a situation when capabilities developed to respond to the security demand of the European allies created additional political challenges and thus put

[136] Central Intelligence Agency, *Soviet Acquisition of Militarily Significant Western Technology: An Update*, Washington, D.C.: U.S. Department of Defense, September 1985.

NATO in peril.[137] The U.S. Government should spare no diplomatic efforts to avoid such a scenario with LAWS.

To that end, the U.S. Government could publicize the efforts to ensure LAWS safety and reliability, i.e., to the extent possible, use the V&V and T&E as a confidence-building measure, both domestically and internationally. Another avenue could be to highlight LAWS' applications in defense and their advantage in saving the lives of American and allied soldiers. The research present in this study can provide additional arguments in favor of LAWS. Namely, if the potential adversaries continue to pursue them, not maintaining the technological edge can have an adverse effect on extended deterrence. Finally, the government should continue to engage in the debate on ethics as there might be strong ethical arguments in favor of military robots, even if their performance is not error-free. Specifically, some war crimes and crimes against humanity are unlikely to be committed by LAWS, including rape and torture. And so, there might be an ethical tradeoff we cannot easily disregard. Namely, the tradeoff between killing the wrong people and saving others from abuses.

When it comes to influencing adversaries' calculation of political benefits of fighting, the impact of LAWS remains highly uncertain as it will likely depend on the specific context in which LAWS might be used. For one, the Russian public confronted in the experimental setting with the risk of ethnic violence against the Russian minority in a neighboring country showed the same support for military intervention regardless of war technology. On the other hand, the fact that the morality of war was such a strong predictor of support for it suggests that this could be an avenue of influence to be exploited through operations in the information environment.

Areas for Future Research

The future of LAWS remains uncertain, and it is up to the U.S. Government, its allies, and partners to shape it. But what are the most robust pathways to the future where LAWS support, not undermine stability? This question requires more research on the technology itself and the guiding policies and incentive structures. As indicated earlier, technology foresight and robust decision-making under deep uncertainty are some of the candidate methods, but there might be countless others.

Some of the questions that stem from this study are: what could be the best ways to affect adversaries' perception of the effectiveness of LAWS, how to communicate defensive intent, and how to avoid the perception of first mover advantage? As suggested, the nature of LAWS hardware and software might be more conducive to such signaling than the manned systems, but so far, it remains an open empirical question. Another set of questions pertains to prospects of

[137] The Euromissile were two theater nuclear missiles developed in the 1970s and 1980s by the United States for respond to Europeans concerns about the modernization of the Soviet nuclear force. The 1979 decision to deploy them caused historically unprecedented mass protests in the host countries. See: Susan Colbourn, *Euromissiles. The Nuclear Weapons That Nearly Destroyed NATO*, Ithaca, NY: Cornell University Press.

success in an alliance context: How do we ensure that LAWS are interoperable within alliances, e.g., NATO, so that they improve and not degrade the Blue's prospects of success? Finally, some of the key technical questions pertain to the United States' ability to maintain the technological edge not only in a controlled environment but also once the new capabilities are deployed. There will be a continued demand for research on LAWS vulnerability to enemy countermeasures and deception, including network attacks, spoofing, and decoys, aimed at neutralizing or degrading the U.S. and allied LAWS advantages. In the context of my research, those issues raise an important question of the impact of those countermeasures on our decision-making.

Another set of questions pertains to cost functions associated with LAWS: How does the development of LAWS change the demand for—and therefore the price of—inputs, and how can they change the costs of warfighting under different development and deployment assumptions? How could the costs be impacted by the demand of a sustained conflict, i.e., can the U.S. and allied defense industries provide adequate amounts of LAWS, and how production at scale will impact the price? What are the LAWS supply chain bottlenecks, and how to mitigate the risk of supply chain disruptions?

The net political benefits might be the most elusive and challenging research area. Some of the questions raised by this study are: How significant could their impact be in different countries? How do the inputs differ, and how could they be affected? What would be the impact of LAWS used within alliances, e.g., NATO, where deployment decisions are negotiated between countries with different moral and ethical rules and sensitivities. The importance of this issue cannot be overstated: the European Allies, including Germany, hosting the biggest number of U.S. troops in Europe, tend to have stronger concerns about the use of LAWS.[138] Meanwhile, any operations in the European theater (and beyond, as evidenced by the wars in Iraq and Afghanistan) will crucially rely on the Allies' support.

The analysis presented in this study remained on an abstract level of LAWS as a broad category of potential future capability. However, the discussed effect could differ depending on the type of LAWS. I discussed how the impact can greatly vary if the offense–defense distinction is possible. But there are also other important characteristics, including LAWS complexity (and resulting training time) and domain of operations. Arguably, LAWS in the land domain might be technologically most challenging while politically most controversial since such LAWS would have the highest potential for interactions with the civilian population. On the other end of the spectrum might be underwater drones, where the likelihood of encountering civilian submarines is negligible. Such discrepancies and their impact on decision-making should be carefully studied, given the growing concerns about the stability of the Indo-Pacific region.

The methods used in this study also invite further exploration. In particular, more research could shed light on the non-monotonic effects of costs and troops' quality: how common those

[138] IPSOS, "Six in Ten (61%) Respondents Across 26 Countries Oppose the Use of Lethal Autonomous Weapons Systems," webpage, January 22, 2019.

might be and what are their policy implications. Those questions could potentially be tackled with simulation methods. One extension of the model could be a formal introduction of uncertainty: How full-scale war deterrence could be affected when players are uncertain about the values of parameters. This would be particularly relevant for LAWS as uncertainty is inherent to their nature. Simulations can be especially useful to study the impact of emerging technology, including LAWS, on decision-making in an alliance context where multiple players guided by their own set of rules seek coordinated behavior. Such simulations could help study the heterogeneity of moral and ethical considerations. In this context, agent-based modeling comes to mind as a particularly useful approach. Yet another extension of the model could be an analysis of the impact of LAWS on low-intensity conflicts.

The survey also opens new avenues of future research. The experiment found no statistically significant effect of technology or NATO when the Russian minority was at risk of ethnic cleansing, but the result might be different when the stakes are lower. Consequently, in future experiments, other motivations for military operations might be tested. For example, technology could have a bigger impact if the proposed operation would support Russia's ally (as opposed to the Russian people living in a neighboring country). In such a scenario, expected casualties among Russian soldiers might be a significant predictor of support. Relatedly, future experiments could test whether technology has a larger effect when human soldiers are completely removed from the experimental scenario. More research is also needed to better understand the non-monotonic effects of some of the decision factors or how those factors can be exploited to shift the Russian public support for military operations. Another next step could also be a longitudinal study where the same cohort is surveyed over a longer period of time, as the war in Ukraine—and advances in the robotic revolution—progress.

From the methodological perspective, the experiment—the first of this kind on a Russian sample of convenience—demonstrated the feasibility of such studies. It took only one week to collect 2,031 complete responses, and the honesty of open-ended responses indicates that scenarios in future studies might be more specific (e.g., ask about operations against specific countries) than the scenario used in this study. Admittedly, the survey was deployed before the outbreak of Russia's full-scale war against Ukraine, which might impact the feasibility of future studies. However, the Russian public continues to respond to surveys about the ongoing war, which indicates that as long as the panel provider is based outside of Russia, online surveys might still be a viable methodology to study the sentiments and beliefs of the Russian public. Furthermore, as much as the war poses additional research challenges, it might also lead to more realistic responses as the questions about the support for military operations ceased to be hypothetical.

Above all, the war in Ukraine, like so many wars before it, reminds us of the true stakes of this research. No matter how low-cost or low-effort wars might seem in the minds of those who launch them, the reality of war remains barbaric, cruel, and chaotic. Once unleashed, wars seem to last far beyond any reasonable expectation of strategic victory. And so, while conventional military technologies, old and new, might seem like a decision input of little significance. Similarly, this study is a tiny contribution to a vast research field of deterrence and war decision-making. Still, the price of getting it wrong could be somebody's house, kindergarten, or hospital getting bombed, lives upended and prematurely lost. But the prize of getting it right might be somebody's life saved. Therefore, I hope this research will continue so that the dramatic pace of technological progress does not lead us to the regress of humanity.

Appendix A. Tripartite Model of Extended Deterrence: Proofs

Let $\overline{t_D}$ be the value of t_D where Challenger is indifferent between backing down and fighting. Let t_D^0 be the value of t_D where Challenger earns 0 from fighting. Let t_D^L be the lowest value of t_D where Challenger fights with t_L. Let t_D^* be Defender's optimal value of t_D conditional on Challenger threatening but not backing down (even if it was optimal for him to do so). Let t_D' be Defender's optimal fighting force if there were no constraints on her fighting payoff maximization problem. Let $\widehat{K}$ be the minimum $K(t_D)$ for Protégé not to concede. Let $\widehat{t_D}$ be the minimum number of Defender's troops necessary for Protégé not to concede if $K(0) < \widehat{K}$. Let t_C^* be Challenger's optimal fighting force. Let $g(t_D) = U_D(\overline{t_D}) - U_D(t_D^*)$. Then there is full deterrence when $g(t_D) > 0$.

Proposition 1.

1. If $t_D^* < t_D^L$, then there is full deterrence if $U_D(t_D^*) \leq U_D(\overline{t_D})$ but if $U_D(t_D^*) > U_D(\overline{t_D})$, there is no deterrence.
2. If $t_D^L \leq t_D^*$, then there is partial deterrence if $t_D^* < t_D^0$ and $U_D(t_D^*) > U_D(\overline{t_D})$. Otherwise, there is full deterrence.

Proof:

Challenger's payoff of not threatening in 0. Therefore, he does not threaten when the alternative outcome leads to a negative payoff. Since Challenger's payoff when Protégé concedes is always positive, Challenger does not threaten when Defender and Protégé commit such a number of troops that at the last node, either (1) Challenger would be better off backing down than fighting, or (2) Challenger would be better off fighting than backing down but his payoff off fighting is negative. I exclude from analysis trivial parameter regimes where Protégé alone has enough troops to guarantee deterrence. Thus, deterrence depends on Defender's commitment, where Defender chooses optimal support based on the comparison of payoffs of all possible outcomes.

Conditions under which these two deterrence cases materialize were found through backwards induction. At the last node Challenger decides whether to attack or back down. Challenger backs down when his maximized payoff of fighting is lower than the payoff of backing down and vice versa.

If Defender supports Protégé then Challenger attacks when

$$-b_C < m\frac{q_C t_C}{q_C t_C + q_D(t_D + 1)} - c_C t_C$$

where t_C is Challenger's payoff maximizing force size such that the marginal benefit of t_C (increased probability of winning times the value of motivation) equals marginal cost (unit costs of force employment).[139]

$$Max\ U_C(t_C) = \frac{q_C t_C}{q_C t_C + q_D(t_D + 1)} m - c_C t_C$$

$$\frac{dU_C}{dt_C} = \frac{m q_C q_D(t_D + 1)}{(q_C t_C + q_D(t_D + 1))^2} - c_C = 0 \Leftrightarrow t_C^* = \sqrt{\frac{m q_D(t_D + 1)}{c_C q_C}} - \frac{q_D(t_D + 1)}{q_C}$$

$t_C^*(t_D)$ is Challenger's best response to forces committed by Defender and Protégé and his maximized payoff of fighting is[140]

$$U_C(t_C^*) = \frac{q_C t_C^*}{q_C t_C^* + q_D(t_D + 1)} m - c_C t_C^*$$

$$= \frac{q_C\left(\sqrt{\frac{m q_D(t_D + 1)}{c_C q_C}} - \frac{q_D(t_D + 1)}{q_C}\right)}{q_C\left(\sqrt{\frac{m q_D(t_D + 1)}{c_C q_C}} - \frac{q_D(t_D + 1)}{q_C}\right) + q_D(t_D + 1)} m - c_C\left(\sqrt{\frac{m q_D(t_D + 1)}{c_C q_C}} - \frac{q_D(t_D + 1)}{q_C}\right)$$

$$= \left(\sqrt{m} - \sqrt{\frac{c_C q_D(t_D + 1)}{q_C}}\right)^2$$

However, there is a minimum $t_C = t_L$ that Challenger has to commit when he decides to fight. When $t_C^* < t_L$, Challenger fights with suboptimal force t_L and his payoff of fighting is

$$U_C(t_L) = \frac{q_C t_L}{q_C t_L + q_D(t_D + 1)} m - c_C t_L$$

We can therefore define the thresholds Challenger fights with t_L:

[139] If Defender does not support the Protégé $U_C(t_C) = \frac{q_C t_C}{q_C t_C + K} m - c_C t_C + \beta_C$ but $\frac{dU_C}{dt_C}$ remains the same.

[140] When Defender does not support the Protégé, but Protégé does not concede, Challenger maximizes his payoff by attacking with $t_C^* = \sqrt{\frac{m q_P}{c_C q_C}} - \frac{q_P}{q_C}$ and his payoff of fighting is $\left(\sqrt{m} - \sqrt{\frac{c_C q_P}{q_C}}\right)^2 + \beta_C$.

$$\sqrt{\frac{mq_D(t_D+1)}{c_Cq_C}}-\frac{q_D(t_D+1)}{q_C}\le t_L \Leftrightarrow \frac{mq_D(t_D+1)}{c_Cq_C}\le\left(t_L+\frac{q_D(t_D+1)}{q_C}\right)^2$$
$$\Leftrightarrow 0\le\left(\frac{1}{q_C}\right)^2(q_D(t_D+1))^2+\left(\frac{2t_L}{q_C}-\frac{m}{c_Cq_C}\right)q_D(t_D+1)+t_L{}^2$$
$$\Leftrightarrow m<4c_Ct_L$$
$$\vee\begin{cases}4c_Ct_L\le m\\ t_D\le\dfrac{q_C}{2q_Dc_C}\left(m-2c_Ct_L-\sqrt{m(m-4c_Ct_L)}\right)-1\end{cases}$$
$$\vee\begin{cases}4c_Ct_L\le m\\ \dfrac{q_C}{2q_Dc_C}\left(m-2c_Ct_L+\sqrt{m(m-4c_Ct_L)}\right)-1\le t_D\end{cases}$$

There is also a minimum $t_D=t_L$ that Defender has to commit when she decides to support Protégé. I also assumed that $q_D(t_L+1)>q_Ct_L$.

Lemma A.1.

$$q_D(t_L+1)>q_Ct_L\Rightarrow\nexists t_D\ge t_L: t_D<\tfrac{q_C}{2q_Dc_C}\left(m-2c_Ct_L-\sqrt{m(m-4c_Ct_L)}\right)-1$$

Lemma A.1 is true when it can be shown that

$$q_D(t_L+1)>q_Ct_L\Rightarrow\nexists t_L: t_L<\frac{q_C}{2q_Dc_C}\left(m-2c_Ct_L-\sqrt{m(m-4c_Ct_L)}\right)-1$$

Proof:

$$t_L<\frac{q_C}{2q_Dc_C}\left(m-2c_Ct_L-\sqrt{m(m-4c_Ct_L)}\right)-1$$
$$\Leftrightarrow\sqrt{m(m-4c_Ct_L)}<m-2c_Ct_L-\frac{2c_Cq_D(t_L+1)}{q_C}$$
$$\Leftrightarrow\begin{cases}0<m-2c_Ct_L-\dfrac{2c_Cq_D(t_L+1)}{q_C}\\ m(m-4c_Ct_L)<\left(m-2c_Ct_L-\dfrac{2c_Cq_D(t_L+1)}{q_C}\right)^2\end{cases}$$
$$\Leftrightarrow\begin{cases}2c_Ct_L\dfrac{q_Ct_L+q_D(t_L+1)}{q_Ct_L}<m\\ m<c_Ct_L\dfrac{\left(q_Ct_L+q_D(t_L+1)\right)^2}{q_Ct_Lq_D(t_L+1)}\end{cases}$$

$$\Leftrightarrow \begin{cases} 2c_C t_L \dfrac{q_C t_L + q_D(t_L+1)}{q_C t_L} < m < c_C t_L \dfrac{\left(q_C t_L + q_D(t_L+1)\right)^2}{q_C t_L q_D(t_L+1)} \\ 2c_C t_L \dfrac{q_C t_L + q_D(t_L+1)}{q_C t_L} < c_C t_L \dfrac{\left(q_C t_L + q_D(t_L+1)\right)^2}{q_C t_L q_D(t_L+1)} \end{cases}$$

$$\Leftrightarrow \begin{cases} 2c_C t_L \dfrac{q_C t_L + q_D(t_L+1)}{q_C t_L} < m < c_C t_L \dfrac{\left(q_C t_L + q_D(t_L+1)\right)^2}{q_C t_L q_D(t_L+1)} \\ q_D(t_L+1) < q_C t_L \end{cases}$$

Consequently,

$$q_C t_L < q_D(t_L+1) \Rightarrow \nexists t_L : t_L < \frac{q_C}{2q_D c_C}\left(m - 2c_C t_L - \sqrt{m(m-4c_C t_L)}\right) - 1$$

Q.E.D.

And so,

$$\sqrt{\frac{mq_D(t_D+1)}{c_C q_C}} - \frac{q_D(t_D+1)}{q_C} < t_L \Leftrightarrow$$

$$\Leftrightarrow m < 4c_C t_L$$

$$\vee \begin{cases} 4c_C t_L \le m \\ \dfrac{q_C}{2q_D c_C}\left(m - 2c_C t_L + \sqrt{m(m-4c_C t_L)}\right) - 1 \le t_D \end{cases}$$

These two cases are cases of partial deterrence, where Challenger only fights with minimal force t_L. The first one, when $m < 4c_C t_L$, means that the stakes are so low for Challenger that he would never commit more than t_L, regardless of troops committed by other players. The second one occurs when t_D^* happens to be so high that Challenger only responds with t_L. And so, t_D^L as the lowest value of t_D where Challenger fights with t_L is:

$$t_D^L = \frac{q_C}{2q_D c_C}\left(m - 2c_C t_L + \sqrt{m(m-4c_C t_L)}\right) - 1$$

It can be shown that when Challenger fights with his optimal force t_C^*, i.e., when

$$\begin{cases} 4c_C t_L \le m \\ \dfrac{q_C}{2q_D c_C}\left(m - 2c_C t_L - \sqrt{m(m-4c_C t_L)}\right) - 1 < t_D < \dfrac{q_C}{2q_D c_C}\left(m - 2c_C t_L + \sqrt{m(m-4c_C t_L)}\right) - 1 \end{cases}$$

then his payoff if always positive.

Lemma A.2.

$$\forall t_C^* > t_L : U_C(t_C^*) > 0$$

Proof:

$$U_C(t_C^*) > 0 \Leftrightarrow \left(\sqrt{m} - \sqrt{\frac{c_C q_D (t_D + 1)}{q_C}}\right)^2 > 0 \Leftrightarrow m \neq \frac{c_C q_D (t_D + 1)}{q_C} \Leftrightarrow t_D \neq \frac{q_C m}{q_D c_C} - 1$$

This is always true if $t_D = \frac{q_C m}{q_D c_C} - 1$ is always in the region where Challenger fights with t_L:

$$\frac{q_C}{2 q_D c_C}\left(m - 2 c_C t_L + \sqrt{m(m - 4 c_C t_L)}\right) - 1 < \frac{q_C m}{q_D c_C} - 1$$
$$\Leftrightarrow \sqrt{m(m - 4 c_C t_L)} < m + 2 c_C t_L$$
$$\Leftrightarrow 0 < 8 m c_C t_L + (2 c_C t_L)^2$$

$$\forall m, c_C, t_L > 0 : 0 < 8 m c_C t_L + (2 c_C t_L)^2$$
$$\Rightarrow \forall t_D < \frac{q_C}{2 q_D c_C}\left(m - 2 c_C t_L + \sqrt{m(m - 4 c_C t_L)}\right) - 1 : t_D \neq \frac{q_C m}{q_D c_C} - 1$$
$$\Rightarrow \forall t_C^* > t_L : U_C(t_C^*) > 0$$

Q.E.D.

However, in the regions of partial deterrence Challenger's payoff of fighting can be negative:

$$\frac{q_C t_L}{q_C t_L + q_D (t_D + 1)} m - c_C t_L < 0 \Leftrightarrow \frac{q_C t_L}{q_D} \frac{m - c_C t_L}{c_C t_L} - 1 < t_D$$

And so, t_D^0 as the value of t_D where Challenger earns 0 from fighting is

$$t_D^0 = \frac{q_C t_L}{q_D} \frac{m - c_C t_L}{c_C t_L} - 1$$

Finally, Challenger is better off backing down when

$$\frac{q_C t_L}{q_C t_L + q_D (t_D + 1)} m - c_C t_L < -b_C \Leftrightarrow \frac{q_C t_L}{q_C t_L + q_D (t_D + 1)} m < c_C t_L - b_C$$
$$\Leftrightarrow \begin{cases} 0 < c_C t_L - b_C \\ \frac{q_C t_L}{q_D} \frac{m - (c_C t_L - b_C)}{c_C t_L - b_C} - 1 < t_D \end{cases}$$

And so, $\overline{t_D}$ as the value of t_D where Challenger is indifferent between backing down and fighting is

$$\overline{t_D} = \frac{q_C t_L}{q_D} \frac{m - (c_C t_L - b_C)}{c_C t_L - b_C} - 1$$

When Defender chooses whether to support Protégé and if so, with how much force, she compares three alternative payoffs and associated force sizes:

- payoff of not supporting Protégé (i.e., payoff of Protégé conceding or fighting alone against Challenger, depending on Protégé own force size), i.e., $t_D = t_D^* = 0$
- maximized payoff of fighting, i.e., $t_D = t_D^* = \max\{t_D', \widehat{t_D}, t_L\}$
- payoff of Challenger backing down, i.e., $t_D = \overline{t_D}$.

Lemma A.2 implies that out of the two deterrence scenarios—Challenger backs down or Challenger attacks but his payoff of fighting is negative—only the former is feasible when $t_D^* < t_D^L$. That is, when Defender's optimal support conditional of Challenger not backing down t_D^* would be such that Challenger would attack with t_C^*, then Challenger can only be deterred when Defender is better off committing to deterrence instead. In other words, if $t_D^* < t_D^L$, then there is full deterrence if $U_D(t_D^*) \leq U_D(\overline{t_D})$ but if $U_D(t_D^*) > U_D(\overline{t_D})$, there is no deterrence.

Conversely, in the cases of partial deterrence when $t_D^L \leq t_D^*$, there are two possible full deterrence scenarios: either $t_D^0 \leq t_D^*$ (Defender is ready to fight with her optimal force which gives Challenger a negative payoff) or $t_D^* < t_D^0$ but $U_D(t_D^*) < U_D(\overline{t_D})$ (Defender's optimal fighting force size is such that Challenger would attack with t_L and earn positive payoff but Defender is better off committing to deterrence instead). In other words, if $t_D^L \leq t_D^*$, then there is partial deterrence if $t_D^* < t_D^0$ and $U_D(t_D^*) > U_D(\overline{t_D})$. Otherwise, there is full deterrence.

Q.E.D.

I limit further analysis to cases, where Defender chooses between full deterrence and no deterrence, i.e., $t_D^* < t_D^L$. I call this parameter regime *Potential Full-Scale War* (PFSW) regime since $t_D^* < t_D^L$ implies $t_C^* > t_L$. And so, this regime is in the region where

$$\begin{cases} 4c_C t_L \leq m \\ \frac{q_C}{2c_C}\left(m - 2c_C t_L - \sqrt{m(m - 4c_C t_L)}\right) < K < \frac{q_C}{2c_C}\left(m - 2c_C t_L + \sqrt{m(m - 4c_C t_L)}\right) \end{cases}$$

where

$$K(t_D) = \begin{cases} q_D(t_D + 1), & \text{if } t_D > 0 \\ q_P(t_D + 1) = q_P, & \text{if } t_D = 0 \end{cases}$$

In this regime, Protégé own force size, as well as $t_L, \widehat{t_D}, t'_D$ are such that should Challenger attack Protégé, $t^*_C > t_L$.

Lemma A.3. $t^*_C > t_L$ when

$$\begin{cases} b_C < c_C t_L < c_P - b_P \\ c_C t_L \dfrac{(q_C t_L + q_P)^2}{q_C t_L q_P} < m \\ c_C t_L \dfrac{\left(q_C t_L + q_D(t_L + 1)\right)^2}{q_C t_L q_D (t_L + 1)} < m \\ \dfrac{(c_P - b_P)^2}{c_P - b_P - c_C t_L} < m \\ \dfrac{4 c_C t_L}{2\alpha r_D - (\alpha r_D)^2} < m \\ \alpha r_D < 2 \end{cases}$$

To prove Lemma A.3, first I need to define $\widehat{t_D}$ and t'_D. $\widehat{t_D}$ stems from Protégé's decision to concede or fight, whereas t'_D is a result of Defender's payoff maximization problem.

Protégé prefers to fight than to concede when the net cost of fighting $c_P - b_P$ is less than expected value of the motivation $\frac{K}{q_C t_C + K} m$:

$$-b_P \le \frac{K}{q_C t_C + K} m - c_P \Leftrightarrow c_P - b_P \le \frac{K}{q_C t_C + K} m$$

Hence, one can calculate the threshold K, when Protégé does not concede.

$$c_P - b_P \le \frac{K}{q_C t_C + K} m \Leftrightarrow (c_P - b_P) q_C t_C \le \left(m - (c_P - b_P)\right) K$$

First, note that if Protégé's net cost of fighting $c_P - b_P$ is higher than the value of stakes m, then Protégé will back down, regardless of forces committed by Challenger and Defender.

$$m < c_P - b_P \Rightarrow \forall K: \left(\left(m - (c_P - b_P)\right) K \le (c_P - b_P) q_C t_C\right)$$

I assume that scenario away.

Conversely,

$$c_P - b_P < m \Rightarrow \left((c_P - b_P) q_C t_C \le \left(m - (c_P - b_P)\right) K \Leftrightarrow \frac{c_P - b_P}{m - (c_P - b_P)} q_C t_C \le K\right)$$

It also implies that if Protégé's economic cost is lower than their political benefit then Protégé would never concede.

$$c_C < b_C \Leftrightarrow c_C - b_C < 0 \Rightarrow \forall K\colon \left(\frac{c_P - b_P}{m - (c_P - b_P)} q_C t_C \leq K\right)$$

I assume that scenario away.

Otherwise, if total benefit ($m + b_P$) is greater than economic cost of fighting ($b_P < c_P < m + b_P \Leftrightarrow 0 < c_P - b_P < m$) then Protégé does not concede when

$$\frac{c_P - b_P}{m - (c_P - b_P)} q_C t_C \leq K$$

Where

$$t_C = \sqrt{\frac{mK}{c_C q_C}} - \frac{K}{q_C}$$

$$\frac{c_P - b_P}{m - (c_P - b_P)} q_C t_C \leq K \Leftrightarrow \frac{c_P - b_P}{m - (c_P - b_P)} q_C \left(\sqrt{\frac{mK}{c_C q_C}} - \frac{K}{q_C}\right) \leq K \Leftrightarrow \frac{(c_P - b_P)^2 q_C}{c_C m} \leq K$$

That is, $\widehat{K}$ as the minimum $K(t_D)$ for Protégé not to concede is

$$\widehat{K} = \frac{(c_P - b_P)^2 q_C}{c_C m}$$

Protégé does not concede even without Defender's support when

$$\frac{(c_P - b_P)^2 q_C}{c_C m} < q_P \Leftrightarrow \frac{(c_P - b_P)^2}{c_C} \frac{q_C}{q_P} < m$$

Conversely, Protégé concedes without Defender's support when

$$m < \frac{(c_P - b_P)^2}{c_C} \frac{q_C}{q_P}$$

In such a case, Protégé does not concede when Defender provides such support that

$$\frac{(c_P - b_P)^2 q_C}{c_C m} < q_D(t_D + 1) \Leftrightarrow \frac{(c_P - b_P)^2}{c_C} \frac{q_C}{q_D} \frac{1}{m} - 1 < t_D$$

That is, $\widehat{t_D}$ as the minimum number of Defender's troops necessary for Protégé not to concede if $K(0) < \widehat{K}$ is

$$\widehat{t_D} = \frac{(c_P - b_P)^2}{c_C}\frac{q_C}{q_D}\frac{1}{m} - 1$$

When Defender and Protégé fight together and $t_C = t_C^*$, Defender's payoff is:

$$U_D(t_D) = \frac{q_D(t_D+1)}{q_C\left(\sqrt{\frac{mq_D(t_D+1)}{c_C q_C}} - \frac{q_D(t_D+1)}{q_C}\right) + q_D(t_D+1)}\alpha m - c_D t_D$$
$$= \sqrt{\frac{c_C q_D(t_D+1)}{mq_C}}\alpha m - c_D t_D = \alpha\sqrt{\frac{c_C q_D(t_D+1)m}{q_C}} - c_D t_D$$

Thus, Defender is maximizing her payoff by committing t_D' such that

$$\frac{dU_D}{dt_D} = \frac{\alpha}{2}\sqrt{\frac{mc_C q_D}{q_C(t_D+1)}} - c_D = 0 \Leftrightarrow t_D' = \frac{\alpha^2 mc_C q_D}{4c_D^2 q_C} - 1 = \frac{\alpha^2 r_D}{4c_D}m - 1$$

Then Defender's payoff of fighting is

$$U_D(t_D') = \sqrt{\frac{c_C q_D(t_D'+1)}{mq_C}}\alpha m - c_D t_D' = \frac{\alpha c_C q_D}{2c_D q_C}\alpha m - c_D\left(\frac{\alpha^2 mc_C q_D}{4c_D^2 q_C} - 1\right)$$
$$= \frac{\alpha r_D}{2}\alpha m - c_D\left(\frac{\alpha^2 m r_D}{4c_D} - 1\right) = \frac{\alpha^2 m r_D}{4} + c_D = \frac{\alpha^2 mc_C q_D}{4c_D q_C} + c_D$$

However, $t_D^* = t_D' \Leftrightarrow \max\{t_D', \widehat{t_D}, t_L\} = t_D'$.

Lemma A.4. $\max\{t_D', \widehat{t_D}, t_L\} = t_D' \Rightarrow U_D(t_D') > U_D(0)$

Proof:

When Defender does not support Protégé then Protégé either fights alone or concedes depending on their own force size. Since

$$\forall b_D > 0: -b_D < \alpha\sqrt{\frac{c_C q_P m}{q_C}} - b_D < \alpha\sqrt{\frac{c_C q_P m}{q_C}}$$

then $U_D(t'_D) > U_D(0)$ is always true when

$$\alpha\sqrt{\frac{c_C q_P m}{q_C}} < \frac{\alpha^2 m c_C q_D}{4 c_D q_C} + c_D$$

Let

$$\frac{\alpha^2 m c_C}{q_C} = x$$

Then

$$\sqrt{x q_P} < x\frac{q_D}{4c_D} + c_D \Leftrightarrow 0 < \left(\frac{q_D}{4c_D}\right)^2 x^2 + \left(\frac{q_D}{2} - q_P\right)x + {c_D}^2$$

Roots of $\left(\frac{q_D}{4c_D}\right)^2 x^2 + \left(\frac{q_D}{2} - q_P\right)x + {c_D}^2$:

$$x = \frac{-\left(\frac{q_D}{2} - q_P\right) \pm \sqrt{\left(\frac{q_D}{2} - q_P\right)^2 - 4\left(\frac{q_D}{4c_D}\right)^2 {c_D}^2}}{2\left(\frac{q_D}{4c_D}\right)^2} = \frac{-\left(\frac{q_D}{2} - q_P\right) \pm \sqrt{-(q_D - q_P)(q_P)}}{2\left(\frac{q_D}{4c_D}\right)^2}$$

$$q_D > q_P \Rightarrow -(q_D - q_P)(q_P) < 0 \Rightarrow \forall x\colon \left(0 < \left(\frac{q_D}{4c_D}\right)^2 x^2 + \left(\frac{q_D}{2} - q_P\right)x + {c_D}^2\right)$$
$$\Rightarrow \forall t'_D\colon U_D(t'_D) > U_D(0)$$

Q.E.D.

Conversely,

$$\left(\max\{t'_D, \widehat{t_D}, t_L\} = \widehat{t_D} \wedge U_D(\widehat{t_D}) > U_D(0)\right) \Rightarrow t_D^* = \widehat{t_D}$$

$$\left(\max\{t'_D, \widehat{t_D}, t_L\} = t_L \wedge U_D(t_L) > U_D(0)\right) \Rightarrow t_D^* = t_L$$

$$\left(\left(\max\{t'_D, \widehat{t_D}, t_L\} = \widehat{t_D} \wedge U_D(\widehat{t_D}) < U_D(0)\right) \vee \left(\max\{t'_D, \widehat{t_D}, t_L\} = t_L \wedge U_D(t_L) < U_D(0)\right)\right)$$
$$\Rightarrow t_D^* = 0$$

Lemma A.3 proof:

Protégé own force size is such that Challenger responds with $t_C^* > t_L$ when

$$\frac{q_C}{2c_C}\left(m - 2c_C t_L - \sqrt{m(m - 4c_C t_L)}\right) < q_P < \frac{q_C}{2c_C}\left(m - 2c_C t_L + \sqrt{m(m - 4c_C t_L)}\right)$$

$$\Leftrightarrow \left(2c_C t_L \frac{q_C t_L + q_P}{q_C t_L} - m\right)^2 < m(m - 4c_A f_L) \Leftrightarrow c_C t_L \frac{(q_C t_L + q_P)^2}{q_C t_L q_P} < m$$

t_L is such that Challenger responds with $t_C^* > t_L$ when

$$\frac{q_C}{2c_C}\left(m - 2c_C t_L - \sqrt{m(m - 4c_C t_L)}\right) < q_D(t_L + 1) < \frac{q_C}{2c_C}\left(m - 2c_C t_L + \sqrt{m(m - 4c_C t_L)}\right)$$

$$\Leftrightarrow c_C t_L \frac{(q_C t_L + q_D(t_L + 1))^2}{q_C t_L q_D(t_L + 1)} < m$$

$\widehat{t_D}$ is such that Challenger responds with $t_C^* > t_L$ when

$$\frac{q_C}{2c_C q_D}\left(m - 2c_C t_L - \sqrt{m(m - 4c_C t_L)}\right) - 1 < \frac{(c_P - b_P)^2}{c_C}\frac{q_C}{q_D}\frac{1}{m} - 1$$

$$< \frac{q_C}{2c_C q_D}\left(m - 2c_C t_L + \sqrt{m(m - 4c_C t_L)}\right) - 1$$

$$\Leftrightarrow \left(\frac{2(c_C - b_C)^2}{m} - (m - 2c_C t_L)\right)^2 < m(m - 4c_C t_L)$$

$$\Leftrightarrow ((c_P - b_P)^2)^2 + (c_P - b_P)^2 2c_C t_L m + ((c_C t_L)^2 - (c_P - b_P)^2)m^2 < 0$$

$$\Leftrightarrow \begin{cases} \dfrac{(c_P - b_P)^2}{c_P - b_P - c_C t_L} < m \\ c_C t_L < c_P - b_P \end{cases}$$

t_D' is such that Challenger responds with $t_C^* > t_L$ when

$$\frac{q_C}{2c_C q_D}\left(m - 2c_C t_L - \sqrt{m(m - 4c_C t_L)}\right) - 1 < \frac{\alpha^2 m r_D}{4c_D} - 1$$

$$< \frac{q_C}{2c_C q_D}\left(m - 2c_C t_L + \sqrt{m(m - 4c_C t_L)}\right) - 1$$

$$\Leftrightarrow \begin{cases} \dfrac{4c_C t_L}{2\alpha r_D - (\alpha r_D)^2} < m \\ \alpha r_D < 2 \end{cases}$$

Thus, when

$$\begin{cases} b_C < c_C t_L < c_P - b_P \\ c_C t_L \dfrac{(q_C t_L + q_P)^2}{q_C t_L q_P} < m \\ c_C t_L \dfrac{\left(q_C t_L + q_D (t_L + 1)\right)^2}{q_C t_L q_D (t_L + 1)} < m \\ \dfrac{(c_P - b_P)^2}{c_P - b_P - c_C t_L} < m \\ \dfrac{4 c_C t_L}{2\alpha r_D - (\alpha r_D)^2} < m \\ \alpha r_D < 2 \end{cases}$$

then $t_C^* > t_L$.

Q.E.D.

Lemma 1. Challenger would never back down regardless of troops committed by other players when $c_C t_L < b_C$. Thus, full-scale war deterrence can exist only when $\frac{b_C}{t_L} < c_C$.

Proof:

Recall that Challenger is better off backing down than fighting when

$$\frac{q_C t_L}{q_C t_L + q_D (t_D + 1)} m - c_C t_L < -b_C \Leftrightarrow \frac{q_C t_L}{q_C t_L + q_D (t_D + 1)} m < c_C t_L - b_C$$

Then

$$c_C t_L - b_C < 0 \Rightarrow \nexists t_D > 0 \colon \frac{q_C t_L}{q_C t_L + q_D (t_D + 1)} m < c_C t_L - b_C$$

Conversely,

$$\exists t_D > 0 \colon \frac{q_C t_L}{q_C t_L + q_D (t_D + 1)} m < c_C t_L - b_C \Leftrightarrow c_C t_L - b_C > 0 \Leftrightarrow \frac{b_C}{t_L} < c_C$$

Q.E.D.

Lemma 2.1. In the PFSW regime, t_C^* decreases in c_C.

Proof:

In the PFSW regime, t_C^* can take four forms:

$$t_D^* = 0 \text{ and } K(0) \geq \widehat{K} \Rightarrow t_C^* = \sqrt{\frac{mK}{c_C q_C}} - \frac{K}{q_C} = \sqrt{\frac{mq_P}{c_C q_C}} - \frac{q_P}{q_C}$$

$$\frac{\partial t_C^*}{\partial c_C} = -\frac{1}{2}\sqrt{\frac{mq_P}{c_C^3 q_C}} < 0$$

$$t_D^* = t_L \Rightarrow t_C^* = \sqrt{\frac{mK}{c_C q_C}} - \frac{K}{q_C} = \sqrt{\frac{mq_D(t_L+1)}{c_C q_C}} - \frac{q_D(t_L+1)}{q_C}$$

$$\frac{\partial t_C^*}{\partial c_C} = -\frac{1}{2}\sqrt{\frac{mq_D(t_L+1)}{c_C^3 q_C}} < 0$$

$$t_D^* = \widehat{t_D} \Rightarrow t_C^* = \sqrt{\frac{mK}{c_C q_C}} - \frac{K}{q_C} = \frac{c_P - b_P}{c_C}\frac{m - (c_P - b_P)}{m}$$

$$\frac{\partial t_C^*}{\partial c_C} = -\frac{c_P - b_P}{c_C^2}\frac{m - (c_P - b_P)}{m} < 0$$

$$t_D^* = t_D' \Rightarrow t_C^* = \sqrt{\frac{mK}{c_C q_C}} - \frac{K}{q_C} = \frac{\alpha m q_D}{2q_C c_D} - \left(\frac{\alpha q_D}{2q_C c_D}\right)^2 mc_C$$

$$\frac{\partial t_C^*}{\partial c_C} = -\left(\frac{\alpha q_D}{2q_C c_D}\right)^2 m < 0$$

Thus, $\forall t_D^*: \frac{\partial t_C^*}{\partial c_C} < 0$

Q.E.D.

Lemma 2.2. In the PFSW regime, t_D' increases in c_C and q_D, and decreases in c_D and q_C.

Proof:

$$t_D' = \frac{\alpha^2 m c_C q_D}{4c_D^2 q_C} - 1 \Rightarrow \begin{cases} \dfrac{\partial t_D'}{\partial c_C} = \dfrac{\alpha^2 m q_D}{4c_D^2 q_C} > 0 \\ \dfrac{\partial t_D'}{\partial q_D} = \dfrac{\alpha^2 m c_C}{4c_D^2 q_C} > 0 \\ \dfrac{\partial t_D'}{\partial q_C} = -\dfrac{\alpha^2 m c_C q_D}{4c_D^2 q_C^2} < 0 \\ \dfrac{\partial t_D'}{\partial c_D} = -\dfrac{\alpha^2 m c_C q_D}{8c_D^3 q_C^2} < 0 \end{cases}$$

Q.E.D.

Lemma 2.3. $\widehat{K}$ decreases in c_C, and increases in q_C.

Proof:

$$\widehat{K} = \frac{(c_P - b_P)^2 q_C}{c_C m} \Rightarrow \begin{cases} \dfrac{\partial \widehat{K}}{\partial c_C} = \dfrac{\alpha^2 m q_D}{4c_D^2 q_C} > 0 \\ \dfrac{\partial \widehat{K}}{\partial q_C} = -\dfrac{\alpha^2 m c_C q_D}{4c_D^2 q_C^2} < 0 \end{cases}$$

Q.E.D.

Lemma 2.4. $\widehat{t_D}$ decreases in c_C and q_D, and increases in q_C.

Proof:

$$\widehat{t_D} = \frac{(c_P - b_P)^2}{c_C} \frac{q_C}{q_D} \frac{1}{m} - 1 \Rightarrow \begin{cases} \dfrac{\partial \widehat{t_D}}{\partial c_C} = -\dfrac{(c_P - b_P)^2}{c_C^2} \dfrac{q_C}{q_D} \dfrac{1}{m} < 0 \\ \dfrac{\partial \widehat{t_D}}{\partial q_D} = -\dfrac{(c_P - b_P)^2}{c_C} \dfrac{q_C}{q_D^2} \dfrac{1}{m} < 0 \\ \dfrac{\partial \widehat{t_D}}{\partial q_C} = \dfrac{(c_P - b_P)^2}{c_C} \dfrac{1}{q_D m} > 0 \end{cases}$$

Q.E.D.

Lemma 2.5. t_D^L decreases in c_C and q_D, and increases in q_C.

Proof:

$$t_D^L = \frac{q_C}{2c_C q_D}\left(m - 2c_C t_L + \sqrt{m(m - 4c_C t_L)}\right) - 1$$

$$\Rightarrow \begin{cases} \dfrac{\partial t_D^L}{\partial c_C} = -\dfrac{q_C m}{2c_C^2 q_D} - \dfrac{q_C m(m - 2c_C t_L)}{2c_C^2 q_D \sqrt{m(m - 4c_C t_L)}} < 0 \\ \dfrac{\partial t_D^L}{\partial q_D} = -\dfrac{q_C}{2c_C q_D^2}\left(m - 2c_C t_L + \sqrt{m(m - 4c_C t_L)}\right) < 0 \\ \dfrac{\partial t_D^L}{\partial q_C} = \dfrac{m - 2c_C t_L + \sqrt{m(m - 4c_C t_L)}}{2c_C q_D} > 0 \end{cases}$$

Q.E.D.

Lemma 2.6. $\overline{t_D}$ decreases in c_C and q_D, and increases in q_C and b_C.

Proof:

$$\overline{t_D} = \frac{q_C t_L}{q_D}\frac{m - (c_C t_L - b_C)}{c_C t_L - b_C} - 1 \Rightarrow \begin{cases} \dfrac{\partial \overline{t_D}}{\partial c_C} = -\dfrac{q_C t_L^2}{q_D}\dfrac{m}{(c_C t_L - b_C)^2} < 0 \\ \dfrac{\partial \overline{t_D}}{\partial q_D} = -\dfrac{q_C t_L}{q_D^2}\dfrac{m - (c_C t_L - b_C)}{c_C t_L - b_C} < 0 \\ \dfrac{\partial \overline{t_D}}{\partial q_C} = \dfrac{t_L}{q_D}\dfrac{m - (c_C t_L - b_C)}{c_C t_L - b_C} > 0 \\ \dfrac{\partial \overline{t_D}}{\partial b_C} = \dfrac{q_C t_L}{q_D}\dfrac{m}{(c_C t_L - b_C)^2} > 0 \end{cases}$$

Q.E.D.

Lemma 3.1. In the PFSW regime, $U_D(\overline{t_D})$ increases in c_C and q_D, and decreases in q_C, b_C, and d_D.

Proof:

$$U_D(\overline{t_D}) = \alpha m - d_D\overline{t_D} = \alpha m - d_D\left(\frac{m-(c_Ct_L-b_C)}{c_Ct_L-b_C}\frac{q_Ct_L}{q_D}-1\right)$$

$$\Rightarrow\begin{cases}\dfrac{\partial U_D(\overline{t_D})}{\partial c_C} = d_D\dfrac{q_Ct_L^2}{q_D}\dfrac{m}{(c_Ct_L-b_C)^2} > 0\\ \dfrac{\partial U_D(\overline{t_D})}{\partial q_D} = d_D\dfrac{q_Ct_L}{q_D^2}\dfrac{m-(c_Ct_L-b_C)}{c_Ct_L-b_C} > 0\\ \dfrac{\partial U_D(\overline{t_D})}{\partial q_C} = -d_D\dfrac{t_L}{q_D}\dfrac{m-(c_Ct_L-b_C)}{c_Ct_L-b_C} < 0\\ \dfrac{\partial U_D(\overline{t_D})}{\partial b_C} = -d_D\dfrac{q_Ct_L}{q_D}\dfrac{m}{(c_Ct_L-b_C)^2} < 0\\ \dfrac{\partial U_D(\overline{t_D})}{\partial d_D} = -\left(\dfrac{m-(c_Ct_L-b_C)}{c_Ct_L-b_C}\dfrac{q_Ct_L}{q_D}-1\right) < 0\end{cases}$$

Q.E.D.

Lemma 3.2. In the PFSW regime, if $t_D^* > 0$ then $U_D(t_D^*)$ increases in c_C and q_D, and decreases in q_C and c_D.

Proof:

If $t_D^* > 0$ then $U_D(t_D^*) = \sqrt{\frac{c_Cq_D(t_D^*+1)}{mq_C}}\alpha m - c_Dt_D^*$ can take three forms:

$$\max\{t_D', \widehat{t_D}, t_L\} = t_L \Rightarrow U_D(t_D^*) = \sqrt{\frac{c_Cq_D(t_D^*+1)}{mq_C}}\alpha m - c_Dt_D^* = \sqrt{\frac{c_Cq_D(t_L+1)}{mq_C}}\alpha m - c_Dt_L$$

$$\Rightarrow\begin{cases}\dfrac{\partial U_D(t_D^*)}{\partial c_C} = \dfrac{1}{2}\sqrt{\dfrac{q_D(t_L+1)}{mc_Cq_C}}\alpha m > 0\\ \dfrac{\partial U_D(t_D^*)}{\partial q_D} = \dfrac{1}{2}\sqrt{\dfrac{c_C(t_L+1)}{mq_Dq_C}}\alpha m > 0\\ \dfrac{\partial U_D(t_D^*)}{\partial q_C} = -\dfrac{1}{2}\sqrt{\dfrac{c_Cq_D(t_L+1)}{mq_C^3}}\alpha m < 0\\ \dfrac{\partial U_D(t_D^*)}{\partial c_D} = -t_L < 0\end{cases}$$

$$\max\{t_D', \widehat{t_D}, t_L\} = \widehat{t_D} \Rightarrow U_D(t_D^*) = \sqrt{\frac{c_C q_D (t_D^* + 1)}{m q_C}} \alpha m - c_D t_D^*$$

$$= \alpha(c_P - b_P) - (c_P - b_P)^2 \frac{c_D q_C}{c_C q_D} \frac{1}{m} + c_D \Rightarrow \begin{cases} \dfrac{\partial U_D(t_D^*)}{\partial c_C} = (c_P - b_P)^2 \dfrac{c_D q_C}{c_C^2 q_D} \dfrac{1}{m} > 0 \\ \dfrac{\partial U_D(t_D^*)}{\partial q_D} = (c_P - b_P)^2 \dfrac{c_D q_C}{c_C q_D^2} \dfrac{1}{m} > 0 \\ \dfrac{\partial U_D(t_D^*)}{\partial q_C} = -(c_P - b_P)^2 \dfrac{c_D}{c_C q_D} \dfrac{1}{m} < 0 \\ \dfrac{\partial U_D(t_D^*)}{\partial c_D} = 1 - (c_P - b_P)^2 \dfrac{q_C}{c_C q_D} \dfrac{1}{m} < 0 \end{cases}$$

$$1 - (c_P - b_P)^2 \frac{q_C}{c_C q_D} \frac{1}{m} < 0 \Leftrightarrow m < \frac{(c_P - b_P)^2}{c_C} \frac{q_C}{q_P}$$

Recall that $m < \frac{(c_P - b_P)^2}{c_C} \frac{q_C}{q_P}$ is the condition when Protégé requires $\widehat{t_D}$ not to concede.

$$\max\{t_D', \widehat{t_D}, t_L\} = t_D' \Rightarrow U_D(t_D^*) = \sqrt{\frac{c_C q_D (t_D^* + 1)}{m q_C}} \alpha m - c_D t_D^* = \frac{\alpha^2 m}{4} \frac{c_C q_D}{c_D q_C} + c_D$$

$$\Rightarrow \begin{cases} \dfrac{\partial U_D(t_D^*)}{\partial c_C} = \dfrac{\alpha^2 m}{4} \dfrac{q_D}{c_D q_C} > 0 \\ \dfrac{\partial U_D(t_D^*)}{\partial q_D} = \dfrac{\alpha^2 m}{4} \dfrac{c_C}{c_D q_C} > 0 \\ \dfrac{\partial U_D(t_D^*)}{\partial q_C} = -\dfrac{\alpha^2 m}{4} \dfrac{c_C q_D}{c_D q_C^2} < 0 \\ \dfrac{\partial U_D(t_D^*)}{\partial c_D} = 1 - \dfrac{\alpha^2 m}{4} \dfrac{c_C q_D}{c_D^2 q_C} < 0 \end{cases}$$

$$1 - \frac{\alpha^2 m}{4} \frac{c_C q_D}{c_D^2 q_C} < 0 \Leftrightarrow \frac{\alpha^2 m}{4} \frac{c_C q_D}{c_D^2 q_C} - 1 > 0$$

Recall that $t_D' = \frac{\alpha^2 m}{4} \frac{c_C q_D}{c_D^2 q_C} - 1$, so $\forall t_D^* = t_D'$: $\frac{\alpha^2 m}{4} \frac{c_C q_D}{c_D^2 q_C} - 1 > 0$

Thus,

$$\forall t_D^* > 0: \begin{cases} \dfrac{\partial U_D(t_D^*)}{\partial c_C} > 0 \\ \dfrac{\partial U_D(t_D^*)}{\partial q_D} > 0 \\ \dfrac{\partial U_D(t_D^*)}{\partial q_C} < 0 \\ \dfrac{\partial U_D(t_D^*)}{\partial c_D} < 0 \end{cases}$$

Q.E.D.

Lemma 3.3. In the PFSW regime, if $t_D^* = 0$ and $K(0) \geq \widehat{K}$ then $U_D(t_D^*)$ increases in c_C, and decreases in q_C and b_D.

Proof:

If $t_D^* = 0$ and $K(0) \geq \widehat{K}$ then $U_D(t_D^*) = \sqrt{\frac{c_C q_P}{m q_C}} \alpha m - b_D$ and

$$\begin{cases} \dfrac{\partial U_D(t_D^*)}{\partial c_C} = \dfrac{1}{2}\sqrt{\dfrac{q_P}{m c_C q_C}} \alpha m > 0 \\ \dfrac{\partial U_D(t_D^*)}{\partial q_C} = -\dfrac{1}{2}\sqrt{\dfrac{c_C q_P}{m q_C^3}} \alpha m < 0 \\ \dfrac{\partial U_D(t_D^*)}{\partial b_D} = -1 < 0 \end{cases}$$

Q.E.D.

Lemma 3.4. In the PFSW regime, if $t_D^* = 0$ and $K(0) < \widehat{K}$ then $U_D(t_D^*)$ decreases in b_D.

Proof:

If $t_D^* = 0$ and $K(0) \geq \widehat{K}$ then $U_D(t_D^*) = -b_D$ and $\frac{\partial U_D(t_D^*)}{\partial b_D} = -1 < 0$

Q.E.D.

Proposition 2. Let $g(t_D) = U_D(\overline{t_D}) - U_D(t_D^*)$. Then

$$\lim_{c_C \to \frac{b_C}{t_L}^+} g(t_D) = -\infty$$

Proof:

$$U_D(\overline{t_D}) = \alpha m - d_D \left(\frac{m - (c_C t_L - b_C)}{c_C t_L - b_C} \frac{q_C t_L}{q_D} - 1 \right) = \alpha m - d_D \frac{m}{c_C t_L - b_C} \frac{q_C t_L}{q_D} + d_D \frac{q_D + q_C t_L}{q_D}$$
$$\Rightarrow \lim_{c_C \to \frac{b_C}{t_L}^+} U_D(\overline{t_D}) = -\infty$$

$U_D(t_D^*)$ can take five different forms (see, Lemma 3.2 through 3.4), but in all five cases

$$\lim_{c_C \to \frac{b_C}{t_L}^+} U_D(t_D^*) \geq -b_D$$

This is because $\min U_D(0) = -b_D$ and $t_D^* > 0$ when $U_D(\max\{t_D', \widehat{t_D}, t_L\}) > U_D(0)$. Thus, $\min U_D(t_D^*) = -b_D$. Hence,

$$\lim_{c_C \to \frac{b_C}{t_L}^+} g(t_D) = -\infty + b_D = -\infty$$

Q.E.D.

Proposition 3. $\frac{\partial g}{\partial c_C}$ is ambiguous.

Corollary 1. There exist such parameter regimes where decreasing c_C leads to the emergence of full deterrence.

Proof:

Proposition 3 is true if there exists such $c_{C_{low}} < c_{C_{medium}} < c_{C_{high}}$ that $g\big(t_D(c_{C_{medium}})\big) > g\big(t_D(c_{C_{low}})\big)$ and $g\big(t_D(c_{C_{medium}})\big) > g\left(t_D(c_{C_{high}})\right)$. Corollary 1 is true when there is no deterrence when $c_C = c_{C_{high}}$ but there is deterrence when $c_C = c_{C_{medium}}$. I will demonstrate existence of such parameter regimes numerically with one of parameter regimes depicted in Figure 3.7.

Suppose $m = 50, \alpha = .65, t_L = .1, b_C = .2, q_C = 1, b_P = .05, c_P = 5.5, q_P = .3, b_D = .05, c_D = 15, q_D = 13, d_D = 1.15, c_{C_{low}} = 2.5, c_{C_{medium}} = 3, c_{C_{high}} = 3.5$. For such parameter values $\max\{t_D', \widehat{t_D}, t_L\} = t_L, U_D(t_L) > U_D(0)$, and

$$g(t_D) = \alpha m - d_D\left(\frac{m-(c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\sqrt{\frac{c_C q_D (t_L+1)}{m q_C}}\alpha m - c_D t_L\right) =$$

$$= \begin{cases} -1.17, & \text{if } c_C = 2.5 \\ 0.63, & \text{if } c_C = 3 \\ -0.31, & \text{if } c_C = 3.5 \end{cases}$$

Thus, $g\left(t_D\left(c_{C_{medium}}\right)\right) > g\left(t_D\left(c_{C_{low}}\right)\right)$ and $g\left(t_D\left(c_{C_{medium}}\right)\right) > g\left(t_D\left(c_{C_{high}}\right)\right)$. Furthermore, there is full deterrence when $c_C = c_{C_{medium}}$ and there is no deterrence when $c_C = c_{C_{low}}$ or $c_C = c_{C_{high}}$.

Q.E.D.

Proposition 4.1. There exist parameter regimes where decreasing c_C leads to deterrence failure where Defender withdraws all the support for Protégé.

Proof:

I will demonstrate the existence of such parameter regimes numerically with one of parameter regimes depicted in Figure 3.4. Suppose $m = 20, \alpha = .7, t_L = .1, b_C = .05, q_C = 1, b_P = .05, c_P = 4.45, q_P = .75, b_D = .05, c_D = 8, q_D = 1.1, d_D = .55$, $c_{C\_low} = .55, c_{C\_high} = 1.5$.

When $c_C = c_{C_{low}}$, $t_D^* = 0$ and $K(0) < \widehat{K}$ (that is, Protégé concedes without Defender's support), whereas when $c_C = c_{C_{high}}$, $t_D^* = t_L$ (that is, without Defender's support, Protégé does not concede). Hence,

$$g(t_D)$$

$$= \begin{cases} \alpha m - d_D\left(\frac{m-(c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - (-b_D), & \text{if } c_C = .55 \\ \alpha m - d_D\left(\frac{m-(c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\sqrt{\frac{c_C q_D (t_L+1)}{m q_C}}\alpha m - c_D t_L\right), & \text{if } c_C = 1.5 \end{cases}$$

$$= \begin{cases} -185.35, & \text{if } c_C = .55 \\ 1.18, & \text{if } c_C = 1.5 \end{cases}$$

Therefore, when $c_C = c_{C_{high}}$ there is full deterrence, i.e., Defender supports Protégé with $\overline{t_D}$, whereas at $c_C = c_{C_{low}}$ there is no deterrence, i.e., Defender does not support Protégé and Protégé concedes.

Q.E.D.

Proposition 4.2. In a specific parameter regime, where Defender's cost of fighting is sufficiently high, there exist two values of Challenger's cost, $c_{C_{low}} < c_{C_{high}}$, such that Defender does not support Protégé and there is no deterrence at $c_{C_{high}}$ but at $c_{C_{low}}$ Defender supports Protégé with $\overline{t_D}$ and there is full deterrence.

Proof:

I will demonstrate the existence of such parameter regimes numerically with one of parameter regimes depicted in Figure 3.4. Suppose $m = 20, \alpha = .7, t_L = .1, b_C = .05, q_C = 1, b_P = .05, c_P = 4.45, q_P = .75, b_D = .05, c_D = 25, q_D = 1.1, d_D = .55, c_{C_{very\ low}} = .55, c_{C_{low}} = 1.25, c_{C_{high}} = 1.35, c_{C_{very\ high}} = 1.5$.

$$g(t_D)$$

$$= \begin{cases} \alpha m - d_D\left(\frac{m - (c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - (-b_D), & \text{if } c_C = .55 \\ \alpha m - d_D\left(\frac{m - (c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\sqrt{\frac{c_C q_D (t_L + 1)}{m q_C}}\alpha m - c_D t_L\right), & \text{if } c_C = 1.25 \\ \alpha m - d_D\left(\frac{m - (c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\sqrt{\frac{c_C q_P}{m q_C}}\alpha m - b_D\right), & \text{if } c_C \in \{1.35, 1.5\} \end{cases}$$

$$= \begin{cases} -185.35, & \text{if } c_C = .55 \\ .92, & \text{if } c_C = 1.35 \\ -.26, & \text{if } c_C = 1.35 \\ 1.33, & \text{if } c_C = 1.5 \end{cases}$$

$c_{C_{very\ low}}$ and $c_{C_{very\ high}}$ represent the situation describer by the Proposition 4.1, where the decrease in c_C leads to deterrence failure where Defender withholds the support. However, $c_{C_{low}}$ and $c_{C_{high}}$ represent a narrow parameter region where the opposite is true: there is no deterrence at $c_{C_{high}}$ and Protégé fights alone, and there is full deterrence at $c_{C_{low}}$, where otherwise Protégé would concede without Defender's support.

Q.E.D.

Proposition 5.1. $\lim\limits_{q_C \to +\infty} g(t_D) = -\infty$

Proof:

As q_C increases, Protégé requires more support not to concede (Lemma 2.3, which implies that $\lim\limits_{q_C \to +\infty} K(0) < \widehat{K}$ and Lemma 2.4, which implies that $\lim\limits_{q_C \to +\infty} \widehat{t_D}(q_C) = \infty$), whereas Defender is willing to commit less support (Lemma 2.2; note that $\lim\limits_{q_C \to +\infty} t'_D < 0$). Thus, $\lim\limits_{q_C \to +\infty} \max\{t'_D, \widehat{t_D}, t_L\} = \widehat{t_D} = \infty$. And so, as q_C approaches infinity, Protégé concedes unless Defender provides infinite support. That infinite support would maximize Defender's payoff of fighting, but the payoff $U_D(\widehat{t_D})$ would still be infinitely negative. Thus, when choosing between supporting with $\widehat{t_D}$ and not supporting at all, Defender is better off not supporting at the costs of $-b_D$. Hence, when q_C approaches infinity, $g(t_D) = \alpha m - d_D\left(\frac{m-(c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - (-b_D)$ and $\lim\limits_{q_C \to +\infty} g(t_D) = -\infty$.

Q.E.D.

Proposition 5.2. $\frac{\partial g}{\partial q_C}$ is ambiguous.

Proof:

Proposition 5.1 is true if there exists such $q_{C_{low}} < q_{C_{medium}} < q_{C_{high}}$ that $g\big(t_D(q_{C_{medium}})\big) > g\big(t_D(q_{C_{low}})\big)$ and $g\big(t_D(q_{C_{medium}})\big) > g\left(t_D(q_{C_{high}})\right)$. I will demonstrate existence of such parameter regimes numerically.

Suppose $m = 50, \alpha = .65, t_L = .1, b_C = .2, c_C = 2.75, b_P = .05, c_P = 5.5, q_P = .3, b_D = .05, c_D = 15, q_D = 15, d_D = 1.15, q_{C_{low}} = 1, q_{C_{medium}} = 2, q_{C_{high}} = 4$. For such parameter values

$$g(t_D) = \alpha m - d_D\left(\frac{m - (c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\sqrt{\frac{c_C q_D (t_L + 1)}{m q_C}}\alpha m - c_D t_L\right)$$
$$= \begin{cases} -0.91, & \text{if } q_C = 1 \\ 3.05, & \text{if } q_C = 2 \\ -0.74, & \text{if } q_C = 4 \end{cases}$$

Thus, $g\left(t_D\big(q_{C_{medium}}\big)\right) > g\left(t_D\big(q_{C_{low}}\big)\right)$ and $g\left(t_D\big(q_{C_{medium}}\big)\right) > g\left(t_D\left(q_{C_{high}}\right)\right)$.

Q.E.D.

Proposition 5.3. $\frac{\partial g}{\partial b_C} < 0$

Proof:

b_C only affects $\overline{t_D}$ and $U_D(\overline{t_D})$. Thus, regardless of $U_D(t_D^*)$,

$$\forall t_D^* \in \{0, t_D', \widehat{t_D}, t_L\}: \frac{\partial g(t_D)}{\partial b_C} = -d_D \frac{q_C t_L}{q_D} \frac{m}{(c_C t_L - b_C)^2} < 0$$

Q.E.D.

Let $\overline{c_C}$ be the lowest value of c_C where full deterrence exists and, therefore, at that point, $\frac{\partial g}{\partial c_C} > 0$. Let $\overline{q_C}$ be the highest value of q_C where full deterrence exists and, therefore, at that point, $\frac{\partial g}{\partial q_C} < 0$. Let $\overline{b_C}$ be the highest value of b_C where full deterrence exists.

Proposition 6. Suppose in the status quo $c_C = \overline{c_C}$, $q_C = \overline{q_C}$, and $b_C = \overline{b_C}$. For sufficiently small Δ_{j_i}, if $\Delta_{c_C} < 0$ then there is deterrence when

$$\left(\Delta_{q_C} < 0 \vee \Delta_{b_C} < 0\right) \wedge \frac{\frac{\partial g}{\partial q_C}}{\frac{\partial g}{\partial c_C}} \Delta_{q_C} + \frac{\frac{\partial g}{\partial b_C}}{\frac{\partial g}{\partial c_C}} \Delta_{b_C} \geq -\Delta_{c_C}$$

Proof:

For sufficiently small Δ_{j_i}, the total change in the value of g caused by shifts in Challenger's economic costs and political benefits of fighting and the quality of his troops can be approximated by $\partial g = \frac{\partial g}{\partial c_C} \Delta_{c_C} + \frac{\partial g}{\partial q_C} \Delta_{q_C} + \frac{\partial g}{\partial b_C} \Delta_{b_C}$. In the case of precarious deterrence, where, in the status quo, $c_C = \overline{c_C}$, $q_C = \overline{q_C}$, and $b_C = \overline{b_C}$, deterrence survives changes in Challenger's economic costs and political benefits of fighting and the quality of his troops when the changes are such that

$$\frac{\partial g}{\partial c_C} \Delta_{c_C} + \frac{\partial g}{\partial q_C} \Delta_{q_C} + \frac{\partial g}{\partial b_C} \Delta_{b_C} \geq 0 \overset{\frac{\partial g}{\partial c_C} > 0}{\Longleftrightarrow} \frac{\frac{\partial g}{\partial q_C}}{\frac{\partial g}{\partial c_C}} \Delta_{q_C} + \frac{\frac{\partial g}{\partial b_C}}{\frac{\partial g}{\partial c_C}} \Delta_{b_C} \geq -\Delta_{c_C}$$

$$\Delta_{c_C} < 0 \Rightarrow \left(\exists \Delta_{q_C}, \Delta_{b_C}: \frac{\frac{\partial g}{\partial q_C}}{\frac{\partial g}{\partial c_C}} \Delta_{q_C} + \frac{\frac{\partial g}{\partial b_C}}{\frac{\partial g}{\partial c_C}} \Delta_{b_C} \geq -\Delta_{c_C} \Leftrightarrow: \frac{\frac{\partial g}{\partial q_C}}{\frac{\partial g}{\partial c_C}} \Delta_{q_C} + \frac{\frac{\partial g}{\partial b_C}}{\frac{\partial g}{\partial c_C}} \Delta_{b_C} > 0 \right)$$

$$\begin{cases} \dfrac{\partial g}{\partial c_C} > 0 \\ \dfrac{\partial g}{\partial q_C} < 0 \\ \dfrac{\partial g}{\partial b_C} < 0 \end{cases} \Rightarrow \begin{cases} \dfrac{\frac{\partial g}{\partial q_C}}{\frac{\partial g}{\partial c_C}} < 0 \\ \dfrac{\frac{\partial g}{\partial b_C}}{\frac{\partial g}{\partial c_C}} < 0 \end{cases} \Rightarrow \nexists(\Delta_{q_C} > 0 \wedge \Delta_{b_C} > 0): \dfrac{\frac{\partial g}{\partial q_C}}{\frac{\partial g}{\partial c_C}}\Delta_{q_C} + \dfrac{\frac{\partial g}{\partial b_C}}{\frac{\partial g}{\partial c_C}}\Delta_{b_C} > 0$$

$$\Rightarrow \left(\frac{\partial g}{\partial c_C}\Delta_{c_C} + \frac{\partial g}{\partial q_C}\Delta_{q_C} + \frac{\partial g}{\partial b_C}\Delta_{b_C} \geq 0 \Leftrightarrow \left(\Delta_{q_C} < 0 \vee \Delta_{b_C} < 0\right) \wedge \frac{\frac{\partial g}{\partial q_C}}{\frac{\partial g}{\partial c_C}}\Delta_{q_C} + \frac{\frac{\partial g}{\partial b_C}}{\frac{\partial g}{\partial c_C}}\Delta_{b_C} \geq -\Delta_{c_C} \right)$$

Q.E.D.

Proposition 7.1. Suppose in the status quo, $t_D^* > 0$. Then $\frac{\partial g}{\partial c_D} > 0$ and $\frac{\partial g}{\partial b_D} = 0$.

Proof:

When $t_D^* > 0$, $g(t_D)$ can take three forms:

$$\max\{t_D', \widehat{t_D}, t_L\} = t_D' = \frac{\alpha^2 m}{4}\frac{c_C q_D}{c_D^2 q_C} - 1$$

$$\Rightarrow g(t_D) = \alpha m - d_D\left(\frac{m - (c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\frac{\alpha^2 m}{4}\frac{c_C q_D}{c_D q_C} + c_D\right)$$

$$\Rightarrow \begin{cases} \dfrac{\partial g(t_D)}{\partial c_D} = \dfrac{\alpha^2 m}{4}\dfrac{c_C q_D}{c_D^2 q_C} - 1 > 0 \\ \dfrac{\partial g(t_D)}{\partial b_D} = 0 \end{cases}$$

$$\max\{t_D', \widehat{t_D}, t_L\} = \widehat{t_D} = \frac{(c_P - b_P)^2}{c_C}\frac{q_C}{q_D}\frac{1}{m} - 1$$

$$\Rightarrow g(t_D) = \alpha m - d_D\left(\frac{m - (c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\alpha(c_P - b_P) - (c_P - b_P)^2\frac{c_D q_C}{c_C q_D}\frac{1}{m} + c_D\right)$$

$$\Rightarrow \begin{cases} \dfrac{\partial g(t_D)}{\partial c_D} = \dfrac{(c_P - b_P)^2}{c_C}\dfrac{q_C}{q_D}\dfrac{1}{m} - 1 > 0 \\ \dfrac{\partial g(t_D)}{\partial b_D} = 0 \end{cases}$$

$$\max\{t_D', \widehat{t_D}, t_L\} = t_L$$

$$\Rightarrow\ g(t_D) = \alpha m - d_D\left(\frac{m - (c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\sqrt{\frac{c_C q_D (t_L + 1)}{m q_C}}\alpha m - c_D t_L\right)$$

$$\Rightarrow \begin{cases} \dfrac{\partial g(t_D)}{\partial c_D} = t_L > 0 \\ \dfrac{\partial g(t_D)}{\partial b_D} = 0 \end{cases}$$

Thus,

$$\forall t_D^* > 0\colon \begin{cases} \dfrac{\partial g(t_D)}{\partial c_D} > 0 \\ \dfrac{\partial g(t_D)}{\partial b_D} = 0 \end{cases}$$

Q.E.D.

Proposition 7.2. Suppose in the status quo, $t_D^* = 0$. Then $\frac{\partial g}{\partial c_D} = 0$ and $\frac{\partial g}{\partial b_D} > 0$.

Proof:

When $t_D^* = 0$, $g(t_D)$ can take two forms:

$$K(0) \geq \widehat{K}$$

$$\Rightarrow\ g(t_D) = \alpha m - d_D\left(\frac{m - (c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\sqrt{\frac{c_C q_P}{m q_C}}\alpha m - b_D\right) \Rightarrow \begin{cases} \dfrac{\partial g(t_D)}{\partial c_D} = 0 \\ \dfrac{\partial g(t_D)}{\partial b_D} = 1 > 0 \end{cases}$$

$$K(0) < \widehat{K} \Rightarrow\ g(t_D) = \alpha m - d_D\left(\frac{m - (c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - (-b_D) \Rightarrow \begin{cases} \dfrac{\partial g(t_D)}{\partial c_D} = 0 \\ \dfrac{\partial g(t_D)}{\partial b_D} = 1 > 0 \end{cases}$$

Thus,

$$\forall t_D^* = 0\colon \begin{cases} \dfrac{\partial g(t_D)}{\partial c_D} > 0 \\ \dfrac{\partial g(t_D)}{\partial b_D} = 0 \end{cases}$$

Q.E.D.

Proposition 8.1. $\lim_{q_D \to 0} g(t_D) = -\infty$

Proof:

As q_D decreases, Protégé requires more support not to concede (Lemma 2.4, which implies that $\lim_{q_D \to 0} \widehat{t_D}(q_C) = \infty$), whereas Defender is willing to commit less support (Lemma 2.2; note that $\lim_{q_D \to 0} t_D' < 0$). Thus, $\lim_{q_D \to 0} \max\{t_D', \widehat{t_D}, t_L\} = \widehat{t_D} = \infty$. And so, as q_D approaches 0, Protégé concedes unless Defender provides infinite support. That infinite support would maximize Defender's payoff of fighting, but the payoff $U_D(\widehat{t_D})$ would still be infinitely negative. Thus, when choosing between supporting with $\widehat{t_D}$ and not supporting at all, Defender is better off not supporting at the costs of $-b_D$. Hence, when q_D approaches 0, $g(t_D) = \alpha m - d_D\left(\frac{m-(c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - (-b_D)$ and $\lim_{q_D \to 0} g(t_D) = -\infty$.

Q.E.D.

Proposition 8.2. $\frac{\partial g}{\partial q_D}$ is ambiguous

Proof:

Proposition 8.2 is true if there exists such $q_{D_{low}} < q_{D_{medium}} < q_{D_{high}}$ that $g\left(t_D(q_{D_{medium}})\right) > g\left(t_D(q_{D_{low}})\right)$ and $g\left(t_D(q_{D_{medium}})\right) > g\left(t_D(q_{D_{high}})\right)$. I will demonstrate existence of such parameter regimes numerically.

Suppose $m = 50, \alpha = .65, t_L = .1, b_C = .2, c_C = 2.75, q_C = 1, b_P = .05, c_P = 5.5, q_P = .3, b_D = .05, c_D = 15, d_D = 1.15, q_{D_{low}} = 3, q_{D_{medium}} = 8, q_{D_{high}} = 15$. For such parameter values

$$g(t_D) = \alpha m - d_D\left(\frac{m-(c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\sqrt{\frac{c_C q_D (t_L + 1)}{m q_C}}\alpha m - c_D t_L\right)$$
$$= \begin{cases} -4.21, & \text{if } q_D = 3 \\ 2.97, & \text{if } q_D = 8 \\ -0.91, & \text{if } q_D = 15 \end{cases}$$

Thus, $g\left(t_D(q_{D_{medium}})\right) > g\left(t_D(q_{D_{low}})\right)$ and $g\left(t_D(q_{D_{medium}})\right) > g\left(t_D\left(q_{D_{high}}\right)\right)$.

Q.E.D.

Proposition 8.3. $-\frac{q_D}{q_C}\frac{\partial g}{\partial q_D} = \frac{\partial g}{\partial q_C}$, unless $t_D^* = 0$ and $K(0) \geq \widehat{K}$. For any change in q_C and q_D, changes in the status quo depend on the ratio $\frac{q_D}{q_C}$.

Proof:
$g(t_D)$ can take five forms:

$$\max\{t_D', \widehat{t_D}, t_L\} = t_D' \Rightarrow g(t_D) = \alpha m - d_D\left(\frac{m-(c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\frac{\alpha^2 m}{4}\frac{c_C q_D}{c_D q_C} + c_D\right)$$

$$\Rightarrow \begin{cases} \dfrac{\partial g(t_D)}{\partial q_D} = d_D \dfrac{q_C t_L}{q_D^2}\dfrac{m-(c_C t_L - b_C)}{c_C t_L - b_C} - \dfrac{\alpha^2 m}{4}\dfrac{c_C}{c_D q_C} \\ \dfrac{\partial g(t_D)}{\partial q_C} = -d_D \dfrac{t_L}{q_D}\dfrac{m-(c_C t_L - b_C)}{c_C t_L - b_C} + \dfrac{\alpha^2 m}{4}\dfrac{c_C q_D}{c_D q_C^2} \end{cases} \Rightarrow -\frac{q_D}{q_C}\frac{\partial g}{\partial q_D} = \frac{\partial g}{\partial q_C}$$

$$\left(\max\{t_D', \widehat{t_D}, t_L\} = \widehat{t_D} \wedge U_D(\widehat{t_D}) > U_D(0)\right)$$

$$\Rightarrow g(t_D) = \alpha m - d_D\left(\frac{m-(c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\alpha(c_P - b_P) - (c_P - b_P)^2\frac{c_D q_C}{c_C q_D}\frac{1}{m} + c_D\right)$$

$$\Rightarrow \begin{cases} \dfrac{\partial g(t_D)}{\partial q_D} = d_D \dfrac{q_C t_L}{q_D^2}\dfrac{m-(c_C t_L - b_C)}{c_C t_L - b_C} - (c_P - b_P)^2\dfrac{c_D q_C}{c_C q_D^2}\dfrac{1}{m} \\ \dfrac{\partial g(t_D)}{\partial q_C} = -d_D \dfrac{t_L}{q_D}\dfrac{m-(c_C t_L - b_C)}{c_C t_L - b_C} + (c_P - b_P)^2\dfrac{c_D}{c_C q_D}\dfrac{1}{m} \end{cases} \Rightarrow -\frac{q_D}{q_C}\frac{\partial g}{\partial q_D} = \frac{\partial g}{\partial q_C}$$

$$\left(\max\{t_D', \widehat{t_D}, t_L\} = t_L \wedge U_D(t_L) > U_D(0)\right)$$

$$\Rightarrow g(t_D) = \alpha m - d_D\left(\frac{m-(c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\sqrt{\frac{c_C q_D(t_L+1)}{m q_C}}\alpha m - c_D t_L\right)$$

$$\Rightarrow \begin{cases} \dfrac{\partial g(t_D)}{\partial q_D} = d_D \dfrac{q_C t_L}{q_D^2}\dfrac{m-(c_C t_L - b_C)}{c_C t_L - b_C} - \dfrac{1}{2}\sqrt{\dfrac{c_C(t_L+1)}{m q_D q_C}}\alpha m \\ \dfrac{\partial g(t_D)}{\partial q_C} = -d_D \dfrac{t_L}{q_D}\dfrac{m-(c_C t_L - b_C)}{c_C t_L - b_C} + \dfrac{1}{2}\sqrt{\dfrac{c_C q_D(t_L+1)}{m q_C^3}}\alpha m \end{cases} \Rightarrow -\frac{q_D}{q_C}\frac{\partial g}{\partial q_D} = \frac{\partial g}{\partial q_C}$$

$$\left(U_D(\max\{t_D', \widehat{t_D}, t_L\}) < U_D(0) \wedge K(0) \geq \widehat{K}\right)$$

$$\Rightarrow\ g(t_D) = \alpha m - d_D\left(\frac{m-(c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - \left(\sqrt{\frac{c_C q_P}{m q_C}}\alpha m - b_D\right)$$

$$\Rightarrow \begin{cases} \dfrac{\partial g(t_D)}{\partial q_D} = d_D \dfrac{q_C t_L}{q_D^2}\dfrac{m-(c_C t_L - b_C)}{c_C t_L - b_C} \\ \dfrac{\partial g(t_D)}{\partial q_C} = -d_D \dfrac{t_L}{q_D}\dfrac{m-(c_C t_L - b_C)}{c_C t_L - b_C} + \dfrac{1}{2}\sqrt{\dfrac{c_C q_P}{m q_C^3}}\alpha m \end{cases} \Rightarrow -\frac{q_D}{q_C}\frac{\partial g}{\partial q_D} + \frac{1}{2}\sqrt{\frac{c_C q_P}{m q_C^3}}\alpha m = \frac{\partial g}{\partial q_C}$$

$$\left(U_D(\max\{t_D', \widehat{t_D}, t_L\}) < U_D(0) \wedge K(0) < \widehat{K}\right)$$

$$\Rightarrow\ g(t_D) = \alpha m - d_D\left(\frac{m-(c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1\right) - (-b_D)$$

$$\Rightarrow \begin{cases} \dfrac{\partial g(t_D)}{\partial q_D} = d_D \dfrac{q_C t_L}{q_D^2}\dfrac{m-(c_C t_L - b_C)}{c_C t_L - b_C} \\ \dfrac{\partial g(t_D)}{\partial q_C} = -d_D \dfrac{t_L}{q_D}\dfrac{m-(c_C t_L - b_C)}{c_C t_L - b_C} \end{cases} \Rightarrow -\frac{q_D}{q_C}\frac{\partial g}{\partial q_D} = \frac{\partial g}{\partial q_C}$$

Thus,

$$\forall\left(t_D^* > 0 \vee \left(t_D^* = 0 \wedge K(0) < \widehat{K}\right)\right): -\frac{q_D}{q_C}\frac{\partial g}{\partial q_D} = \frac{\partial g}{\partial q_C}$$

This implies that

$$\frac{\partial g}{\partial q_C}\Delta_{q_C} + \frac{\partial g}{\partial q_D}\Delta_{q_D} = 0 \Leftrightarrow -\frac{q_D}{q_C}\frac{\partial g}{\partial q_D}\Delta_{q_C} + \frac{\partial g}{\partial q_D}\Delta_{q_D} = 0 \Leftrightarrow -\frac{q_D}{q_C}\Delta_{q_C} + \Delta_{q_D} = 0 \Leftrightarrow \frac{\Delta_{q_D}}{\Delta_{q_C}} = \frac{q_D}{q_C}$$

Q.E.D.

Proposition 9.

$$\frac{\partial g}{\partial d_D} < 0$$

Proof:

d_D only affects $U_D(\overline{t_D})$. Thus, regardless of $U_D(t_D^*)$,

$$\forall t_D^* \in \{0, t_D', \widehat{t_D}, t_L\}: \frac{\partial g(t_D)}{\partial d_D} = -\frac{m-(c_C t_L - b_C)}{c_C t_L - b_C}\frac{q_C t_L}{q_D} - 1 < 0$$

Q.E.D.

Appendix B. Survey Instrument

Consent Form

My name is Krystyna Marcinek. I am a Ph.D. student at the Pardee RAND Graduate School, based in the United States. This survey is conducted purely for academic purposes as part of my doctoral research. Thank you for your interest in my survey!

This 16 item survey will take about 6 minutes. You will first be asked a few screening questions to determine whether you are eligible to participate and whether the quotas for your demographic groups have not been met yet. Only adult residents of the Russian Federation are eligible to participate. If the screening questions indicate you are not eligible or the quotas have been met, you will be redirected back to TGM Research. Otherwise, you will be presented with a short hypothetical scenario and asked about your support for the plan laid out in the scenario. The study aims to better understand the factors contributing to public support for military interventions. At the end of the survey, you will also be asked a few demographic questions that will contribute to my data analysis. I will summarize the findings in my doctoral dissertation.

This survey will be completely anonymous. I am not collecting any information that could connect you or your computer's IP address with your responses. The company paying for your participation, TGM Research, will only receive your completion status but will not have access to your responses. There are no direct risks or benefits to participation, except for the payment you will receive for submitting the survey.

This study is voluntary. You may stop participation at any time. However, if you choose to stop your participation, you will not be paid. To get paid you will need to submit your responses, as instructed on the last page of this questionnaire. Thank you for your time.

If you have questions about this study, you can contact the research coordinator for this study at marcinek@prgs.edu. If you have questions about your rights as a research participant or need to report a research-related injury or concern, contact RAND's Human Subject Protection Committee by emailing hspcinfo@rand.org. If possible, when you contact the committee, please reference study #2021-N0514.

Q0. Do you agree to participate?

- Yes (1)
- No (0)

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Please answer the following screening questions. If you are not eligible to participate in the survey or your quotas have been met, you will be redirected back to TGM Research.

Q1. Are you an adult resident of the Russian Federation?

- Yes (1)
- No (0)

Q2. How old are you?

- 18-29 (1)
- 30-39 (2)
- 40-49 (3)
- 50-59 (4)
- 60 or more (5)

Q3. What is your gender?

- Male (1)
- Female (0)

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You will now be presented with a brief hypothetical scenario. You will be asked whether you support the government's plan laid out in the scenario or not and why.

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The participants will be randomly assigned to one of four scenarios:

Scenario 1: "Russian minority living in country X is in danger of ethnic cleansing. Country X is Russia's neighbor, which is a member of NATO. The Russian government is considering a limited armed intervention to protect the minority. The plan calls for a deployment of airborne forces."

Scenario 2: "Russian minority living in country X is in danger of ethnic cleansing. Country X is Russia's neighbor, which is a member of NATO. The Russian government is considering a limited armed intervention to protect the minority. The plan calls for a deployment of a robotic unit, comprising of ground and aerial autonomous robots, with minimal support from military personnel."

Scenario 3: "Russian minority living in country X is in danger of ethnic cleansing. Country X is Russia's neighbor, which is not a member of NATO. The Russian government is considering a limited armed intervention to protect the minority. The plan calls for a deployment of airborne forces."

Scenario 4: "Russian minority living in country X is in danger of ethnic cleansing. Country X is Russia's neighbor, which is a not member of NATO. The Russian government is considering a limited armed intervention to protect the minority. The plan calls for a deployment of a robotic unit, comprising of ground and aerial autonomous robots, with minimal support from military personnel."

[Display scenario]

Q4. What is your first reaction to the government's hypothetical plan to launch a limited military intervention? Would you be for or against such an operation?

- Strongly oppose (1)
- Somewhat oppose (2)
- Somewhat support (3)
- Strongly support (4)

Q5. How much do you agree or disagree with the following statements when you think about the government's hypothetical plan?

	Strongly disagree (1)	Somewhat disagree (2)	Somewhat agree (3)	Strongly agree (4)
I believe that the safety of the Russian minority is a very important issue, and it would call for military intervention.	○	○	○	○
I believe the intervention would be successful.	○	○	○	○
I believe that the international risks associated with this intervention would be low. I would not expect any major negative consequences.	○	○	○	○
I believe there would be few casualties among civilians.	○	○	○	○
I believe there would be few casualties among our soldiers.	○	○	○	○
I believe that the costs associated with the intervention would be low.	○	○	○	○
I believe that such an operation would be morally acceptable.	○	○	○	○

---PAGE BREAK---

[Display scenario]

Q6. What, in your opinion, speaks in favor of the government's hypothetical plan?

[Display only those statements with which the participant has agreed in Q5]

	Does not speak in favor (1)	**Speaks somewhat in favor (2)**	**Speaks strongly in favor (3)**
I believe that the safety of the Russian minority is a very important issue, and it would call for military intervention.	○	○	○
I believe the intervention would be successful.	○	○	○
I believe that the international risks associated with this intervention would be low. I would not expect any major negative consequences.	○	○	○
I believe there would be few casualties among civilians.	○	○	○
I believe there would be few casualties among our soldiers.	○	○	○
I believe that the costs associated with the intervention would be low.	○	○	○
I believe that such an operation would be morally acceptable.	○	○	○

Q7. What, in your opinion, speaks against the government's hypothetical plan?

[Recode and display only those statements with which the participant has not agreed in Q5]

	Does not speak against (1)	**Speaks somewhat against (2)**	**Speaks strongly against (3)**
I believe that the safety of the Russian minority is not an important enough issue for military intervention.	○	○	○
I do not believe the intervention would be successful.	○	○	○
I worry that international risks associated with the intervention would be high. I expect major negative consequences.	○	○	○
I worry there would be many casualties among civilians.	○	○	○
I worry there would be many casualties among our soldiers.	○	○	○
I worry that the costs associated with the intervention would be high.	○	○	○
I believe that such an operation would be morally unacceptable.	○	○	○

Q8. Are there other factors that influenced your level of support of the government's hypothetical plan? If so, please elaborate.

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Q9. After this consideration, has your level of approval for the hypothetical plan changed?

- Yes, I approve of it less. (1)
- Yes, I approve of it more. (2)
- No (3)

Q10. How much do you now support the government's hypothetical plan to launch this military intervention?

- Strongly oppose (1)
- Somewhat oppose (2)
- Somewhat support (3)
- Strongly support (4)

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Thank you for your responses. To conclude the survey, please answer these basic demographic questions to support my data analysis.

Q11. How would you describe your education level?

- No high school diploma (1)
- High school graduate (2)
- Professional school graduate (3)
- Some college education (4)
- Collage graduate (5)

Q12. How would you describe your employment status?

- Employed/Self-employed/Serving (1)
- Pensioner (2)
- Unemployed (3)
- Student (4)
- Housewife (5)
- Other (6)

Q13. How would you describe your marital status?

- Married (1)

- In a relationship (2)
- Widowed (3)
- Divorced/Separated (4)
- Single (5)

Q14. Do you have children?

- Yes (1)
- No (0)

Q15. Have you or has anyone in your household ever served in the military?

- Yes (1)
- No (0)

Q16. In which federal district do you live?

- Central (1)
- Northwestern (2)
- Southern (3)
- North Caucasian (4)
- Volga (5)
- Ural (6)
- Siberia (7)
- Far Eastern (8)

Thank you for your time. Please submit your responses below. After that, you will be redirected to TGM Research for payment.

Appendix C. Logistic Regression Models Comparison

Table C.3. Levels Coding and Interpretation in LIV12 and LIV4 Models.

Level	Twelve-Level Model	Four-Level Model
1	Strongly negative view is a strong argument against (Q5: Strongly disagree and Q7: Speaks strongly against)	Negative view is an argument against (Q5: (Strongly disagree or Somewhat disagree) and Q7: (Speaks strongly against or Speaks somewhat against))
2	Somewhat negative view is a strong argument against (Q5: Somewhat disagree and Q7: Speaks strongly against)	Negative view is not an argument against (Q5: (Strongly disagree or Somewhat disagree) and Q7: Does not speak against)
3	Strongly negative view is a weak argument against (Q5: Strongly disagree and Q7: Speaks somewhat against)	Positive view is not an argument for (Q5: (Strongly agree or Somewhat agree) and Q6: Does not speak in favor)
4	Somewhat negative view is a weak argument against (Q5: Somewhat disagree and Q7: Speaks somewhat against)	Positive view is an argument for (Q5: (Strongly agree or Somewhat agree) and Q6: (Speaks strongly in favor or Speaks somewhat in favor))
5	Strongly negative view is not an argument against (Q5: Strongly disagree and Q7: Does not speak against)	
6	Somewhat negative view is not an argument against (Q5: Somewhat disagree and Q7: Does not speak against)	
7	Somewhat positive view is not an argument for (Q5: Somewhat agree and Q6: Does not speak in favor)	
8	Strongly positive view is not an argument for (Q5: Strongly agree and Q6: Does not speak in favor)	
9	Somewhat positive view is a weak argument for (Q5: Somewhat agree and Q6: Speaks somewhat in favor)	
10	Strongly positive view is a weak argument for (Q5: Strongly agree and Q6: Speaks somewhat in favor)	
11	Somewhat positive view is a strong argument for (Q5: Somewhat agree and Q6: Speaks strongly in favor)	
12	Strongly positive view is a strong argument for (Q5: Strongly agree and Q6: Speaks strongly in favor)	

Table C.2. Logistic Regression Models Comparison.

	Dependent Variable: Support					
	Twelve-Level Models			Four-Level Models		
	(1)	(2)	(3)	(4)	(5)	(6)
Intervention (Reference: Airborne, NATO)						
Robots, NATO	.855 (.183)	.857 (.180)		1.003 (.200)	.999 (.197)	
Airborne, non-NATO	.831 (178)	.823 (.174)		.914 (.182)	.904 (.178)	
Robots non-NATO	.941 (.202)	.951 (.200)		1.026 (.207)	1.010 (.201)	
The safety of the Russian minority is a very important issue, and it would call for military intervention. (Reference: Level 1)						
2	1.647 (2.070)	1.270 (1.598)	1.243 (1.536)	1.438 (.623)	1.401 (.613)	1.345 (.577)
3	1.728 (2.303)	1.596 (2.140)	1.482 (1.9450	*.455 (.213)	*.468 (.216)	*.465 (.210)
4	.594 (.697)	.468 (.552)	.530 (.614)	***2.036 (.540)	***2.147 (.570)	***1.965 (.502)
5	.485 (.648)	.343 (.459)	.385 (.507)			
6	1.269 (1.541)	1.072 (1.309)	1.029 (1.242)			
7	.450 (.545)	.350 (.425)	.387 (.465)			
8	.139 (.196)	.149 (.207)	.148 (.203)			
9	1.329 (1.513)	1.134 (1.294)	1.127 (1.270)			
10	1.728 (2.009)	1.542 (1.794)	1.449 (1.668)			
11	1.318 (1.524)	1.117 (1.295)	1.073 (1.227)			
12	2.064 (2.353)	1.763 (2.017)	1.597 (1.805)			
The intervention would be successful. (Reference: Level 1)						
2	2.139 (1.915)	2.231 (1.989)	2.269 (1.998)	1.283 (.433)	1.392 (.465)	1.387 (.454)
3	.551 (.626)	.484 (.549)	.608 (.682)	1.091 (.336)	1.114 (.339)	1.073 (.320)
4	1.608 (1.396)	1.603 (1.379)	1.871 (1.584)	***3.602 (.821)	***3.499 (.785)	***3.336 (.731)
5	1.726 (1.735)	2.012 (1.984)	2.150 (2.100)			
6	2.392 (2.133)	2.560 (2.265)	2.776 (2.429)			

	Dependent Variable: Support					
7	1.362 (1.192)	1.421 (1.234)	1.496 (1.285)			
8	2.377 (2.339)	2.107 (2.067)	2.199 (2.133)			
9	*4.888 (4.129)	*4.841 (4.054)	*5.074 (4.209)			
10	**6.945 (6.133)	**6.543 (5.705)	**6.863 (5.933)			
11	*5.547 (4.901)	*5.151 (4.484)	**5.717 (4.927)			
12	*4.962 (4.359)	*4.905 (4.261)	*5.101 (4.391)			
The international risks associated with this intervention would be low. I would not expect any major negative consequences. (Reference: Level 1)						
2	*2.371 (1.055)	*2.304 (1.022)	*2.173 (.939)	***2.065 (.468)	***2.091 (.466)	***2.090 (.463)
3	***4.016 (1.929)	***3.891 (1.869)	**3.388 (1.570)	1.031 (.302)	1.074 (.305)	1.045 (.297)
4	***3.183 (1.300)	***3.131 (1.274)	***2.915 (1.152)	**1.629 (.309)	***1.635 (.303)	**1.576 (.288)
5	**3.632 (2.048)	**3.908 (2.186)	**3.629 (2.007)			
6	***5.836 (2.587)	***5.812 (2.564)	***5.381 (2.308)			
7	**2.842 (1.354)	***3.068 (1.437)	**2.808 (1.286)			
8	3.565 (3.762)	3.254 (3.426)	3.018 (3.186)			
9	***4.576 (1.897)	***4.495 (1.850)	***4.055 (1.625)			
10	**4.628 (2.911)	***5.164 (3.235)	**4.546 (2.773)			
11	**3.675 (2.183)	**3.462 (2.030)	**3.261 (1.863)			
12	2.111 (1.261)	2.302 (1.367)	1.956 (1.135)			
There would be few casualties among civilians. (Reference: Level 1)						
2	.776 (.346)	.804 (.353)	.834 (.360)	.667 (.184)	.640 (.174)	.661 (.178)
3	.484 (.297)	.568 (.341)	.605 (.355)	1.394 (.362)	1.449 (.369)	1.429 (.359)
4	1.222 (.532)	1.213 (.518)	1.128 (.473)	***2.141 (.419)	***2.188 (.420)	***2.199 (.417)
5	.475 (.290)	.468 (.280)	.474 (.276)			
6	.732 (.366)	.688 (.340)	.678 (.329)			

	Dependent Variable: Support					
7	.937 (.450)	1.001 (.473)	.972 (.451)			
8	**4.268 (2.743)	**4.255 (2.702)	**3.668 (2.283)			
9	*2.155 (.965)	*2.190 (.965)	*2.126 (.922)			
10	2.488 (1.451)	2.475 (1.411)	2.257 (1.252)			
11	1.584 (.855)	1.605 (.855)	1.695 (.893)			
12	1.052 (.580)	1.159 (.629)	1.158 (.618)			
There would be few casualties among Russian soldiers. (Reference: Level 1)						
2	.791 (.486)	.770 (.473)	.880 (.532)	.986 (.306)	.945 (.290)	.958 (.289)
3	1.452 (1.056)	1.602 (1.149)	1.476 (1.033)	1.393 (.361)	1.361 (.348)	1.372 (.346)
4	.725 (.419)	.707 (.408)	.835 (.472)	1.199 (.260)	1.169 (.249)	1.202 (.251)
5	3.021 (2.201)	2.884 (2.090)	3.040 (2.168)			
6	.599 (.373)	.571 (.354)	.679 (.411)			
7	1.281 (.743)	1.236 (.716)	1.343 (.766)			
8	.984 (.740)	1.045 (.787)	1.305 (.968)			
9	.904 (.510)	.857 (.481)	.971 (.534)			
10	.982 (.632)	.878 (.558)	1.007 (.632)			
11	1.148 (.736)	1.243 (.791)	1.533 (.962)			
12	1.274 (.844)	1.182 (.779)	1.289 (.833)			
The costs associated with the intervention would be low. (Reference: Level 1)						
2	**.375 (.166)	**.393 (.170)	**.393 (.165)	1.257 (.241)	1.267 (.240)	1.208 (.226)
3	.900 (.396)	.819 (.356)	.823 (.346)	1.525 (.444)	1.544 (.441)	1.399 (.395)
4	.831 (.311)	.815 (.301)	.734 (.263)	***2.521 (.507)	***2.470 (.488)	***2.273 (.441)
5	.717 (.327)	.693 (.311)	.666 (.294)			
6	1.077 (.408)	1.062 (.398)	.909 (.333)			

	Dependent Variable: Support					
7	1.108 (.506)	1.084 (.488)	.900 (.392)			
8	.316 (.263)	.322 (.261)	.337 (.266)			
9	1.814 (.715)	1.767 (.686)	1.480 (.560)			
10	.999 (.668)	.977 (.640)	.807 (.521)			
11	**3.465 (2.181)	*3.394 (2.121)	*3.106 (1.940)			
12	1.585 (.954)	1.432 (.854)	1.322 (.764)			
The operation would be ethically acceptable. (Reference: Level 1)						
2	3.111 (2.324)	3.188 (2.402)	3.526 (2.607)	1.398 (.403)	1.405 (.398)	1.319 (.370)
3	1.218 (1.091)	1.245 (1.123)	1.332 (1.175)	***3.030 (.819)	***3.099 (.827)	***3.077 (.811)
4	2.933 (2.077)	2.962 (2.121)	*3.304 (2.321)	***7.639 (1.444)	***7.416 (1.372)	***7.146 (1.295)
5	1.136 (.960)	1.208 (1.015)	1.272 (1.061)			
6	**4.973 (3.663)	**4.958 (3.685)	**4.966 (3.628)			
7	***6.524 (4.696)	***6.876 (5.010)	***7.608 (5.452)			
8	***21.063 (19.114)	***20.230 (18.429)	***20.017 (17.815)			
9	***17.674 (12.363)	***17.101 (12.080)	***17.499 (12.150)			
10	***12.988 (9.996)	***13.545 (10.493)	***15.196 (11.628)			
11	***22.749 (17.167)	***23.951 (18.123)	***24.002 (17.901)			
12	***72.186 (55.443)	***68.793 (53.018)	***67.372 (51.074)			
Gender (Reference: Woman)						
Man	.999 (.166)	1.010 (.158)		1.103 (.168)	1.148 (.164)	
Age (Reference: 18-29)						
30-39	1.031 (.242)	1.003 (.215)		1.093 (.242)	1.038 (.209)	
40-49	*.626 (.161)	*.636 (.149)		.8163 (.197)	.816 (.180)	
50-59	*.610 (.169)	**.564 (.140)		.742 (.191)	.706 (.162)	
60 or more	**.476	***.460		.604	**.579	

	Dependent Variable: Support				
	(.160)	(.120)		(.191)	(.139)
Education (Reference: No high school diploma)					
High school graduate	.640 (.639)	.446 (.438)		.988 (1.022)	.753 (.754)
Professional school graduate	.696 (.679)	.449 (.428)		.919 (.927)	.677 (.661)
Some college education	.786 (.787)	.566 (.556)		1.215 (1.258)	.981 (.980)
Collage graduate	1.112 (1.084)	.687 (.652)		1.414 (1.431)	.993 (.966)
Employment Status (Reference: Employed/Self-employed/Serving)					
Retired	1.114 (.328)			1.117 (.305)	
Unemployed	*1.926 (.678)			1.657 (.530)	
Student	1.424 (.571)			1.467 (.564)	
Housewife	.930 (.286)			.838 (.250)	
Other	1.201 (.553)			.979 (.416)	
Military Family (Reference: No)					
Yes	1.119 (.252)	1.066 (.235)		1.210 (.263)	1.155 (.246)
Marital Status (Reference: Married)					
In a relationship	1.493 (.404)			1.397 (.353)	
Widowed	.575 (.264)			.604 (.255)	
Divorced/Separated	1.222 (.365)			1.178 (.322)	
Single	1.404 (.380)			1.390 (.354)	
Having Children (Reference: No)					
Yes	1.419 (.334)			1.386 (.309)	
Federal District (Reference: Central)					
Northwestern	**.561 (.140)			.710 (.165)	
Southern	*.597 (.163)			**.602 (.155)	
North Caucasian	.471 (.251)			.437 (.222)	
Volga	**.587 (.128)			**.664) (.136)	

	Dependent Variable: Support					
Ural	.971 (.273)			1.065 (.284)		
Siberia	1.051 (.276)			1.116 (.269)		
Far Eastern	.845 (.411)			.930 (.431)		
Observations	1874	1875	1880	1874	1875	1880
AIC	0.789	0.778	0.774	0.785	0.773	0.765
BIC	-11957.056	-12101.049	-12247.459	-12274.870	-12421.062	-12573.666
Hosmer–Lemeshow test p-value	0.2792	0.0838	0.3657	0.7767	0.3148	0.4810

NOTE: Standard errors in parentheses; * p < .1; ** p < .05; *** p < .01. For levels interpretation and coding see Table C.1.

Table C.3. Predictive Margins for Four-Level Decision Factors' Independent Variables Model.

	Dependent Variable: Support						
Level	**Motivation**	**Prospects of Success**	**International Risks**	**Civilian Casualties**	**Military Casualties**	**Costs**	**Morality**
1	.321	.281	.359	.352	.379	.351	.218
2	.361	.324	.446	.296	.373	.380	.264
3	.233	.295	.368	.400	.414	.405	.381
4	.412	.447	.417	.454	.397	.462	.519

Intervention		**Gender**		**Age**		**Education**		**Military Family**	
Airborne, NATO	.396	Woman	.386	18-29	.411	No high school diploma	.408	No	.391
Robots, NATO	.396	Man	.401	30-39	.415	High school graduate	.377	Yes	.407
Airborne, non-NATO	.384			40-49	.389	Professional school graduate	.365		
Robots non-NATO	.397			50-59	.373	Some college education	.406		
				60 +	.351	Collage graduate	.408		

NOTE: For levels interpretation and coding see Table C.1.

Table C.4. Predictive Margins for Twelve-Level Decision Factors' Independent Variables Model

	Dependent Variable: Support						
Level	Motivation	Prospects of Success	International Risks	Civilian Casualties	Military Casualties	Costs	Morality
1	.378	.248	.261	.364	.401	.390	.113
2	.405	.340	.350	.339	.374	.284	.238
3	.431	.174	.407	.300	.450	.367	.132
4	.293	.301	.383	.387	.365	.367	.228
5	.259	.328	.407	.279	.512	.348	.129
6	.386	.357	.451	.322	.342	.397	.299
7	.261	.287	.381	.365	.423	.399	.347
8	.175	.333	.387	.533	.406	.262	.513
9	.392	.435	.423	.455	.385	.455	.487
10	.427	.472	.438	.470	.387	.387	.451
11	.390	.442	.394	.419	.424	.528	.539
12	.442	.436	.350	.381	.419	.431	.694

Intervention		Gender		Age		Education		Military Family	
Airborne, NATO	.403	Woman	.393	18-29	.422	No high school diploma	.450	No	.392
Robots, NATO	.387	Man	.394	30-39	.422	High school graduate	.366	Yes	.399
Airborne, non-NATO	.383			40-49	.376	Professional school graduate	.366		
Robots non-NATO	.398			50-59	.364	Some college education	.390		
				60 +	.343	Collage graduate	.411		

NOTE: For levels interpretation and coding see Table C.1.

Abbreviations

4LIV Model	Four-Level Independent Variable Model
12LIV Model	Twelve-Level Independent Variable Model
AI	Artificial Intelligence
CCW	Convention on Certain Conventional Weapons
GGE	Group of Governmental Experts
HSPC	Human Subjects Protection Committee
LAWS	Lethal Autonomous Weapon Systems
MTurk	Mechanical Turk
NATO	North Atlantic Treaty Organization
T&E	Testing and Evaluation
U.S.	The United States
V&V	Verification and Validation

References

"How Ukrainians modify civilian drones for military use," *The Economist*, May 8, 2023.

Allan Dafoe, Jonathan Renshon, Paul Huth, "Reputation and Status as Motives for War," *Annual Review of Political Science*, Vol. 17, No. 1, 2014, pp: 371–393.

Allen, Gregory C., "Russia Probably Has Not Used AI-Enabled Weapons in Ukraine, but That Could Change," Center for Strategic and International Studies, May 26, 2022. As of June 2, 2023: https://www.csis.org/analysis/russia-probably-has-not-used-ai-enabled-weapons-ukraine-could-change.

Altman, Daniel, "The Strategist's Curse: A Theory of False Optimism as a Cause of War," *Security Studies*, Vol. 24, No. 2, 2015, pp: 284–315.

Altmann, Jürgen, and Frank Sauer, "Autonomous Weapon Systems and Strategic Stability," *Survival*, Vol. 59, No. 5, 2017, pp: 117–142.

Arai, Koki, and Masakazu Matsumoto, "Public perceptions of autonomous lethal weapons systems," *AI Ethics*, 2023.

Atlantic Council, "Ending Keynote: Art, Narrative, and the Third Offset," YouTube video, May 2, 2016. As of April 23, 2023: https://www.youtube.com/watch?v=R9PswCdTi2E.

Barnes, Julian E., "Russian Public Appears to Be Souring on War Casualties, Analysis Shows," *The New York Times*, May 26, 2023.

Baum, Matthew A., "How public opinion constrains the use of force: The case of Operation Restore Hope," *Presidential Studies Quarterly*, Vol. 34, No. 2, 2004, pp: 187–226.

Bell, Mark S., and Kai Quek, "Authoritarian public opinion and the democratic peace," *International Organization*, Vol. 72, No. 1, 2018, pp: 227–242.

Bendett, Samuel, "The Ukraine war and its impact on Russian development of autonomous weapons," Atlantic Council, August 30, 2022. As of June 2, 2023: https://www.atlanticcouncil.org/content-series/airpower-after-ukraine/the-ukraine-war-and-its-impact-on-russian-development-of-autonomous-weapons.

Blainey, Geoffrey, *The Causes of War*, New York: Free Press, 1988.

Blandford, Andrew C., "Reputational Costs beyond Treaty Exclusion: International Law Violations as Security Threat Focal Points," *Washington University Global Studies Law Review*, Vol. 10, No. 4, 2011, pp: 669–726.

Bode, Ingvild, Hendrik Huelss, Anna Nadibaidze, Guangyu Qiao-Franco, and Tom F.A. Watts, "Prospects for the global governance of autonomous weapons: comparing Chinese, Russian, and US practices," *Ethics and Information Technology*, Vol. 25, No. 5, 2023.

Boulanin, Vincent, Neil Davison, Netta Goussac, and Moa Peldán Carlsson, *Limits on Autonomy in Weapon Systems. Identifying Practical Elements of Human Control*, Stockholm International Peace Research Institute and the International Committee of the Red Cross, June 2020.

Brewster, Rachel, "Unpacking the State's Reputation," *Harvard International Law Journal*, Vol. 50, No. 2, 2009, pp: 231–270.

Central Intelligence Agency, *Soviet Acquisition of Militarily Significant Western Technology: An Update*, Washington, D.C.: U.S. Department of Defense, September 1985.

Clare, Joe, and Vesna Danilovic, "Reputation for Resolve, Interests, and Conflict," *Conflict Management and Peace Science*, Vol. 29, No. 1, 2012, pp: 3–27.

Colbourn, Susan, *Euromissiles. The Nuclear Weapons That Nearly Destroyed NATO*, Ithaca, NY: Cornell University Press.

Corbetta, Renato, "Determinants of Third Parties' Intervention and Alignment Choices in Ongoing Conflicts, 1946–2001," *Foreign Policy Analysis*, Vol. 6, No. 1, 2010, pp: 61–85.

Davis, Paul K., Angela O'Mahony, Christian Curriden, and Jonathan Lamb, *Influencing Adversary States: Quelling Perfect Storms*, RAND Corporation, RR-A161-1, 2021.

Davis, Paul K., Jonathan Kulick, and Michael Egner, *Implications of Modern Decision Science for Military Decision-Support Systems*, RAND Corporation, MG-360-AF, 2005.

Davis, Zachary, "Artificial Intelligence on the Battlefield: Implications for Deterrence and Surprise," *Prism: A Journal of the Center for Complex Operations*, Vol. 8, No. 2, 2019, pp: 114–131.

De Mesquita, Bruce Bueno, and Randolph M. Siverson, "War and the survival of political leaders: A comparative study of regime types and political accountability," *American Political Science Review*, Vol. 89, No. 4, 1995, pp: 841–855.

Edmonds, Jeffrey, Samuel Bendett, Anya Fink, Mary Chesnut, Dmitry Gorenburg, Michael Kofman, Kasey Stricklin, and Julian Waller, *Artificial Intelligence and Autonomy in Russia*, Arlington, Va.: CNA, DRM-2021-U-029303-Final, May 2021.

Edwards, George C., William Mitchell, and Reed Welch, "Explaining Presidential Approval: The Significance of Issue Salience." *American Journal of Political Science*, Vol. 39, No. 1, 1995, pp: 108–34.

Fearon, James D., “Domestic Political Audiences and the Escalation of International Disputes,” American *Political Science Review*, Vol. 88, No. 3, 1994, pp: 577–92.

Fearon, James D., “Rationalist Explanations for War,” *International Organization*, Vol. 49, No. 3, 1995, pp: 379–414.

Fedasiuk, Ryan, “Chinese Perspectives on AI and Future Military Capabilities,” *CSET Policy Brief*, Center for Security and Emerging Technology, August 2020.

Feickert, Andrew, Jennifer K. Elsea, Lawrence Kapp, and Laurie A. Harris, *U.S. Ground Forces Robotics and Autonomous Systems (RAS) and Artificial Intelligence (AI): Considerations for Congress*, Congressional Research Service, R45392, Updated November 20, 2018.

Feinstein, Yuval, *Rally'round the Flag: The Search for National Honor and Respect in Times of Crisis*, Oxford University Press, 2022.

Frye, Timothy, “Casualties Won’t Topple Putin,” *Foreign Policy*, April 10, 2023.

Future of Life Institute, “An Open Letter to the United Nations Convention on Certain Conventional Weapons,” webpage, August 20, 2017. As of April 26, 2023: https://futureoflife.org/autonomous-weapons-open-letter-2017.

Gaines, Michael J., *Whither the Liberal International Order? Does an Attenuated Liberal International Order Affect the Propensity, Incidence, and Intensity of Conflict?*, RAND Corporation, RGSD-A2685-1, 2023.

Gartner, Scott Sigmund, “The multiple effects of casualties on public support for war: An experimental approach,” *American political science review*, Vol. 102, No. 1, 2008, pp: 95–106.

Gartner, Scott Sigmund, and Gary M. Segura, “War, casualties, and public opinion,” *Journal of conflict resolution*, Vol. 42, No. 3, 1998, pp: 278–300.

Gartzke, Erik, “Blood and Robots: How Remotely Piloted Vehicles and Related Technologies Affect the Politics of Violence,” Journal of Strategic Studies, Vol.. 44, No. 7, 2021, pp: 983–1013.

Gartzke, Erik, and Yonatan Lupu, “Still looking for audience costs,” *Security Studies*, Vol. 21, No. 3, 2012, pp: 391–397.

Gerasimov, Valerii, “The value of’ science is in foresight [Tsennost’ nauki v predvidenii],” *Military-Industrial Courier [Voenno-promyshlennyi kur’er]*, No. 8, 2013, p. 1.

Geys, Benny, “Wars, Presidents, and Popularity the Political Cost(S) of War Re-Examined,” *The Public Opinion Quarterly*, Vol. 74, No. 2, 2010, pp: 357–374.

Gilani, Iftikhar, “India's border dispute with neighbors,” *Anadolu Agency*, May 31, 2020.

GIP Digital Watch, “GGE on lethal autonomous weapons systems,” webpage, undated. As of April 26, 2023: https://dig.watch/processes/gge-laws.

Gordeev, Vladislav, “The Ministry of Defense promised to bring the share of robots in the armament structure to 30% [Minoborony poobeshchalo dovesti dolyu robotov v strukture vooruzhenii do 30%],” *RBK*, November 6, 2014.

Guzman, Andrew T., *How international law works: a rational choice theory*, Oxford University Press, 2008.

Hernandez, Marco, “Decoding the Antiwar Messages of Miniature Protesters in Russia,” *The New York Times*, June 16, 2023.

Hoehn, John R., Kelley M. Sayler, Michael E. DeVine, *Unmanned Aircraft Systems: Roles, Missions, and Future Concepts*, Congressional Research Service, R47188, July 18, 2022.

Horowitz, Michael C., “Public opinion and the politics of the killer robots debate,” *Research and Politics*, Vol. 3, No. 1, 2016, pp: 1–8.

Horowitz, Michael C., “When Speed Kills: Lethal Autonomous Weapon Systems, Deterrence and Stability,” *Journal of Strategic Studies*, Vol. 42, No. 6, 2019, pp: 764–88.

Horowitz, Michael C., “Do Emerging Military Technologies Matter for International Politics?,” *Annual Review of Political Science*, Vol. 23, No. 1, 2020, pp: 386–7.

Horowitz, Michael C., and Matthew S. Levendusky, “Drafting support for war: Conscription and mass support for warfare,” *The Journal of Politics*, Vol. 73, No. 2, 2011, pp: 524–534.

Horowitz, Michael C., Shira Pindyck, and Casey Mahoney, “AI, the International Balance of Power, and National Security Strategy,” in *The Oxford Handbook of AI Governance*, Justin B. Bullock and others (eds), Online edition, Oxford Academic, 2022.

Horowitz, Michael C., Paul Scharre, and Ben FitzGerald, “Drone Proliferation and the Use of Force: An Experimental Approach,” Center for a New American Security, 2017.

IPSOS, “Six in Ten (61%) Respondents Across 26 Countries Oppose the Use of Lethal Autonomous Weapons Systems,” webpage, January 22, 2019. As of August 7, 2023: https://www.ipsos.com/en-us/news-polls/human-rights-watch-six-in-ten-oppose-autonomous-weapons.

Jervis, Robert, *The Logic of Images in International Relations, Princeton*, N.J.: Princeton University Press, 1970.

Jervis, Robert, *Perception and Misperception in International Politics*, Princeton, NJ: Princeton University Press, 1976.

Jervis, Robert, “Cooperation under the Security Dilemma,” *World Politics*, Vol. 30, no. 2, 1978, pp: 186–214

Jervis, Robert, “War and Misperception,” *Journal of Interdisciplinary History*, Vol. 18, No. 4, 1988, pp: 675–700.

Jervis, Robert, Keren Yarhi-Milo, and Don Casler, “Redefining the debate over reputation and credibility in international security: Promises and Limits of New Scholarship,” *World Politics*, Vol. 73, No. 1, 2021, pp: 167–203.

Johnson, James S., “Artificial Intelligence: A Threat to Strategic Stability,” *Strategic Studies Quarterly*, Vol. 14, No. 1, 2020, pp: 16–39.

Kallenborn, Zachary, “Russia may have used a killer robot in Ukraine. Now what?,” *Bulletin of the Atomic Scientists*, March 15, 2022.

Kania, Elsa B., “’AI weapons’ in China’s military innovation,” The Brookings Institution, April 2020.

Keller, Elena, Jade E. Newman, Andreas Ortmann, Louisa R. Jorm, Georgina M. Chambers, “How Much Is a Human Life Worth? A Systematic Review,” *Value in Health*, Vol. 24, No. 10, 2021, pp: 1531–1541.

Keohane, Robert O., *After Hegemony: Cooperation and Discord in the World Political Economy*, Princeton University Press, 1984.

Kertzer, Joshua D., and Ryan Brutger, “Decomposing audience costs: Bringing the audience back into audience cost theory,” *American Journal of Political Science*, Vol, 60, No. 1, 2016, pp: 234–249.

Kuijpers, Dieuwertje, “Rally around all the flags: the effect of military casualties on incumbent popularity in ten countries 1990–2014,” *Foreign Policy Analysis*, Vol. 15, No. 3, 2019, pp: 392–412.

Langlois, Catherine C., “Power and Deterrence in Alliance Relationships: The Ally’s Decision to Renege” *Conflict Management and Peace Science*, Vol. 29, No. 2, 2012, pp: 148–169.

Larson, Eric V., *Casualties and Consensus: The Historical Role of Casualties in Domestic Support for U.S. Military Operations*, RAND Corporation, MR-726-RC, 1996.

Larson, Eric V., and Bogdan Savych, *Misfortunes of War: Press and Public Reactions to Civilian Deaths in Wartime*, RAND Corporation, MG-441-AF, 2006.

Levada-Center, “Escalation in Donbas [Obostrenie v Donbasse],” webpage, December 14, 2021. As of May 19, 2023: https://www.levada.ru/2021/12/14/obostrenie-v-donbasse.

Levada-Center, “Ukraine and Donbas,” webpage, March 4, 2022. As of May 19, 2023: https://www.levada.ru/en/2022/03/04/ukraine-and-donbass.

Levy, Jack S., “Misperception and the Causes of War: Theoretical Linkages and Analytical Problems,” World Politics, Vol. 36, No. 1, 1983, pp: 76–99.

Levy, Jack S., “Declining Power and the Preventive Motivation for War,” *World Politics*, Vol. 40, No. 1, 1987, pp: 82–107.

Leys, Nathan, “Autonomous Weapon Systems and International Crises,” *Strategic Studies Quarterly*, Vol. 12, No. 1, 2018, pp: 60–1.

Lian, Bradley, and John R. Oneal, “Presidents, the use of military force, and public opinion,” *Journal of Conflict Resolution*, Vol. 37, No. 2, 1993, pp: 277–300.

Lombard, Marlize, and Laurel Phillipson, “Indications of Bow and Stone-Tipped Arrow Use 64 000 Years Ago in KwaZulu-Natal, South Africa,” *Antiquity*, Vol. 84, No. 325, 2010, pp: 635–48.

Marcinek, Krystyna, and Eugeniu Han, *Russia's Asymmetric Response to 21st Century Strategic Competition: Robotization of the Armed Forces*, RAND Corporation, RR-A1233-5, 2023.

Massicot, Dara, “Russia’s Repeat Failures,” *Foreign Affairs*, August 15, 2022.

Mastro, Oriana Skylar, “The Taiwan Temptation.” *Foreign Affairs*, June 3, 2021.

Mazarr, Michael J., *Understanding Deterrence*, RAND Corporation, PE-295-RC, April 2018.

Mazarr, Michael J., Arthur Chan, Alyssa Demus, Bryan Frederick, Alireza Nader, Stephanie Pezard, Julia A. Thompson, and Elina Treyger, *What Deters and Why: Exploring Requirements for Effective Deterrence of Interstate Aggression*, RAND Corporation, RR-2451-A, 2018.

Mazarr, Michael J., Joe Cheravitch, Jeffrey W. Hornung, and Stephanie Pezard, *What Deters and Why: Applying a Framework to Assess Deterrence of Gray Zone Aggression*, RAND Corporation, RR-3142-A, 2021.

Mazarr, Michael J., Ashley L. Rhoades, Nathan Beauchamp-Mustafaga, Alexis A. Blanc, Derek Eaton, Katie Feistel, Edward Geist, Timothy R. Heath, Christian Johnson, Krista Langeland, Jasmin Léveillé, Dara Massicot, Samantha McBirney, Stephanie Pezard, Clint Reach, Padmaja Vedula, and Emily Yoder, *Disrupting Deterrence: Examining the Effects of Technologies on Strategic Deterrence in the 21st Century*, RAND Corporation, RR-A595-1, 2022.

Mercer, Jonathan, *Reputation and International Politics*, Cornell University Press, 1996.

Michael Tomz, “Domestic audience costs in international relations: An experimental approach,” *International Organization*, Vol. 61, No. 4, 2007, pp: 821–840.

Molinari, Jim, *Global Hawk Operational Overview*, Presentation, RAV/0032-01, undated. As of June 2, 2023: https://www.faa.gov/about/office_org/headquarters_offices/avs/offices/aam/cami/library/online_libraries/aerospace_medicine/sd/media/Global_Hawk_104Jim_real.pdf).

Morgan, Forrest E., Benjamin Boudreaux, Andrew J. Lohn, Mark Ashby, Christian Curriden, Kelly Klima, and Derek Grossman, *Military Applications of Artificial Intelligence: Ethical Concerns in an Uncertain World*, RAND Corporation, RR-3139-1-AF, 2020.

Mueller, John E., *War, presidents, and public opinion*, New York: Wiley, 1973.

Mullinix, Kevin J., Thomas J. Leeper, James N. Druckman and Jeremy Freese, "The Generalizability of Survey Experiments," *Journal of Experimental Political Science*, Vol. 2, No. 2, 2015, pp: 109–138.

North Atlantic Treaty Organization, Summary of NATO's Autonomy Implementation Plan, webpage, October 12, 2022. As of June 2, 2023: https://www.nato.int/cps/en/natohq/official_texts_208376.htm.

North Atlantic Treaty Organization, Summary of the NATO Artificial Intelligence Strategy, webpage, October 22, 2021. As of June 2, 2023: https://www.nato.int/cps/en/natohq/official_texts_187617.htm.

Nye Jr, Joseph S., *Bound to lead: The changing nature of American power*, New York: Basic books, 1990.

Nye Jr, Joseph S., "Soft Power and American Foreign Policy," *Political Science Quarterly*, Vol. 119, No. 2, 2004, pp: 255–270.

Office of the Director of National Intelligence, Annual Threat Assessment of the U.S. Intelligence Community, PDF, February 6, 2023. As of June 2, 2023: https://www.dni.gov/files/ODNI/documents/assessments/ATA-2023-Unclassified-Report.pdf.

Oimann, Ann-Katrien, "The Responsibility Gap and LAWS: a Critical Mapping of the Debate," *Philosophy and Technology*, Vol. 36, No. 1, 2023, pp: 1–22.

Organski, A.F.K., *World Politics*, New York, NY: Knopf, 1958.

O'Rourke, Ronald, *Navy Large Unmanned Surface and Undersea Vehicles: Background and Issues for Congress*, Congressional Research Service, R45757, April 17, 2023.

President of the Russian Federation, "Open lesson 'Russia Focused on the Future' [Otkrytyj urok «Rossija, ustremljonnaja v budushhee»]", webpage, September 1, 2017. As of April 26, 2023: http://www.kremlin.ru/events/president/transcripts/55493.

Putin, Vladimir, "Extended meeting of the collegium of the Ministry of Defense of the Russian Federation [Rasshirennoe zasedanie kollegii Ministerstva oborony Rossiiskoi Federatsii]," *Army Collection [Armeiskii sbornik]*, No. 1, 2021, p. 8.

Qiao-Franco, Guangyu, and Ingvild Bode, “Weaponised Artificial Intelligence and Chinese Practices of Human–Machine Interaction,” *The Chinese Journal of International Politics*, Vol. 16, 2023, pp: 106–128.

Radin, Andrew, Alyssa Demus, and Krystyna Marcinek, *Understanding Russian Subversion: Patterns, Threats, and Responses*, RAND Corporation, PE-331-A, February 2020.

Rempfer, Kyle, “Automation doesn’t mean soldiers will be out of a job, chief says,” *Army Times*, January 14, 2020. As of June 11, 2023: https://www.armytimes.com/news/your-army/2020/01/14/automation-doesnt-mean-soldiers-will-be-out-of-a-job-chief-says.

Richter, Daniel, and Matthias Krbetschek, “The age of the Lower Paleolithic occupation at Schöningen,” *Journal of human evolution*, Vol. 89, 2015, pp: 46–56.

Rosstat Federal State Statistics Service, “All-Russian population census 2020 [Vserossiiskaya perepis’ naseleniya 2020 goda],” webpage, undated. As of May 23, 2023: https://rosstat.gov.ru/folder/56580.

Russell, Stuart, “AI weapons: Russia’s war in Ukraine shows why the world must enact a ban,” *Nature*, Vol. 614, 2023, pp: 620–623.

Russian Field, “Year of "special military operation" in Ukraine: the attitude of Russians [God «spetsial'noi voennoi operatsii» v Ukraine: otnoshenie rossiyan],” webpage, undated. As of June 19, 2023: https://russianfield.com/godsvo.

Sartori, Anne E., “The Might of the Pen: A Reputational Theory of Communication in International Disputes,” *International Organization*, Vol. 56, No. 1, 2002, pp: 121–49.

Sartori, Anne E., *Deterrence by Diplomacy*, Princeton, N.J.: Princeton University Press, 2005.

Sauer, Frank, and Niklas Schörnig, “Killer Drones – The Silver Bullet of Democratic Warfare?,” *Security Dialogue*, Vol. 43, No. 4, 2012, pp: 363–80.

Sayler, Kelley M., *International Discussions Concerning Lethal Autonomous Weapon Systems*, Congressional Research Service, IF11294, Updated February 14, 2023.

Sayler, Kelley M., *Defense Primer: U.S. Policy on Lethal Autonomous Weapon Systems*, Congressional Research Service, IF11150, May 15, 2023.

Scharre, Paul, *Army of none: Autonomous weapons and the future of war*, WW Norton & Company, 2018.

Schelling, Thomas, *Arms and Influence*, New Haven, Conn.: Yale University Press, 1966.

Schmid, Jon, *The Determinants of Military Technology Innovation and Diffusion*, dissertation, Georgia Institute of Technology, 2018.

Schneider, Jacquelyn, “The capability/vulnerability paradox and military revolutions: Implications for computing, cyber, and the onset of war,” *Journal of Strategic Studies*, Vol. 42, No. 6, pp: 841–863.

Seo, TaeJun, and Yusaku Horiuchi, “Natural Experiments of the Rally ’Round the Flag Effects Using Worldwide Surveys,” *Journal of Conflict Resolution*, online edition, April 23, 2023.

Shipler, David K., “Israeli Jets Destroy Iraqi Atomic Reactor; Attack Condemned by U.S. and Arab Nations,” *The New York Times*, June 9, 1981.

Singer, Peter, *Wired for War. The Robotics Revolution and Conflict in the 21st Century*, Penguin Books, 2009.

Smith, Alexander, “The 'stunning' scale of Russian deaths in Ukraine signals trouble ahead for Putin,” NBC News, May 2, 2023.

Sorokin, Gerald, “Alliance Formation and General Deterrence: A Game-Theoretic Model and the Case of Israel,” *The Journal of Conflict Resolution*, Vol. 38, No. 2, 1994, pp: 298–325.

The Impact of Artificial Intelligence on Strategic Stability and Nuclear Risk Volume I Euro-Atlantic Perspectives, ed. Vincent Boulanin, Stockholm International Peace Research Institute, May 2019.

The Impact of Artificial Intelligence on Strategic Stability and Nuclear Risk Volume II East Asian Perspectives, Lora Saalman, ed., Stockholm International Peace Research Institute, May 2019.

The Impact of Artificial Intelligence on Strategic Stability and Nuclear Risk Volume III South Asian Perspectives, Petr Topychkonov, ed., Stockholm International Peace Research Institute, April 2020.

Thieme, Hartmut, “Lower Palaeolithic hunting spears from Germany,” *Nature*, Vol. 385, No. 6619, 1997, pp: 807–810.

Timoshenko, Denis, “‘Brotherly people’ - the concept of Kyiv scribes, which has become a social doctrine of Russia [«Bratskii narod» – kontsept kievskikh knizhnikov, kotoryi prevratilsya v sotsial'nuyu doktrinu Rossii],” *Radio Freedom [Radio Svoboda]*, September 7, 2020. As of June 19, 2023: https://www.radiosvoboda.org/a/donbass-realii/30364746.html.

Tomz, Michael R., and Jessica L.P. Weeks, “Public opinion and the democratic peace,” *American political science review*, Vol. 107, No. 4, 2013, pp: 849–865.

Tomz, Michael, Jessica L.P. Weeks, and Keren Yarhi-Milo, “Public Opinion and Decisions About Military Force in Democracies,” *International Organization*, Vol. 74, No. 1, 2020, pp: 119–43.

Tullock, Gordon, “Efficient rent-seeking,” in J.M. Buchanan, R.D. Tollison, G. Tullock (Eds.), *Toward a Theory of Rent-Seeking Society*, Texas AM University Press, College Station, 1980, pp: 97–112.

U.S. Army, *Operationalizing Robotic and Autonomous Systems in Support of Multi-Domain Operations*, whitepaper, Fort Eustis, VA: U.S. Army Training and Doctrine Command, 2018.

U.S. Department of Defense Directive 3000.09, “Autonomy in Weapon Systems,” PDF, January 25, 2023, As of April 23, 2023: https://www.esd.whs.mil/portals/54/documents/dd/issuances/dodd/300009p.pdf.

U.S. Department of Defense, “Summary of the 2018 Department of Defense Artificial Intelligence Strategy. Harnessing AI to Advance Our Security and Prosperity,” 2018.

U.S. Department of Defense, *Unmanned Systems Integrated Roadmap FY 2017-2042*, Office of the Secretary of Defense, 2018; U.S. Army, Robotic and Autonomous Systems Strategy, Fort Eustis, VA: U.S. Army Training and Doctrine Command, 2017.

Umbrello, Steven, Phil Torres, and Angelo F. De Bellis, “The future of war: could lethal autonomous weapons make conflict more ethical?,” *AI and Society*, Vol. 35, 2020, pp: 273–282.

United Nations, “Briefing Security Council, US Secretary of States Powell Presents Evidence of Iraq’s Failure to Disarm,” Press Release, SC/7658, February 5, 2003.

Van Creveld, Martin, *The Changing Face of War: Combat from the Marne to Iraq*, New York City, NY: Presidio Pres, 2008.

Van Evera, Stephen, *Causes of War: Power and the Roots of Conflict*, Cornell University Press, 1999.

von Clausewitz, Carl, *On War*, edited and translated by Michael Howard and Peter Paret, Princeton, NJ: Princeton University Press, 1984.

Walsh, James Igoe, and Marcus Schulzke, *Drones and Support for the Use of Force*, E-book, Ann Arbor, MI: University of Michigan Press, 2018.

Weeks, Jessica L., “Autocratic Audience Costs: Regime Type and Signaling Resolve,” *International Organization*, Vol. 62, No. 1, 2008, pp: 35–64.

Wong, Yuna Huh, John Yurchak, Robert W. Button, Aaron B. Frank, Burgess Laird, Osonde A. Osoba, Randall Steeb, Benjamin N. Harris, and Sebastian Joon Bae, *Deterrence in the Age of Thinking Machines*, RAND Corporation, RR-2797-RC, 2020.

Work, Robert O., "AI and Future Warfare: The Rise of Robotic Combat Operations," Mad Scientist Conference, University of Texas at Austin, April 24, 2019. . As of April 26, 2023: https://community.apan.org/wg/tradoc-g2/mad-scientist/m/disruption-and-the-future-operational-environment/274690.

Zabrodskyi, Mykhaylo, Jack Watling, Oleksandr V Danylyuk and Nick Reynolds, "Preliminary Lessons in Conventional Warfighting from Russia's Invasion of Ukraine: February–July 2022," Royal United Services Institute for Defence and Security Studies, 2022.

Zagare, Frank, and Marc Kilgour, "Alignment patterns, crisis bargaining, and extended deterrence: A game-theoretic analysis," *International Studies Quarterly*, Vol. 47, No. 4, 2003, pp: 587–615.

Zegart, Amy, "Cheap fights, credible threats: The future of armed drones and coercion," *Journal of Strategic Studies*, Vol. 43, No. 1, 2020, pp: 6–46.